The Binding Layer: What Stacked Federal and State Climate Policies Buy in U.S. Transportation Fuel

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Abstract

Three climate policies cover U.S. transportation fuel: a federal biofuel mandate, a federal clean-fuel tax credit, and clean fuel standards in four states. I ask what this stack buys and at what cost, using a dynamic equilibrium model of fuel, feedstock and credit markets for 2026–45 in which credit holders' expectations and biofuel investment are estimated on the 2014–25 record. The federal mandate binds: it sets how much biofuel the country uses, so the tax credit and the state standards change emissions only through fuel prices, vehicle choice and where fuel is burned. About half of the emissions cut inside the four states comes from renewable diesel moved there from other states, where fossil diesel replaces it; the rest comes from higher fuel prices and electric vehicles. Where fuel also falls under cap-and-trade, a tonne the standards cut is either an allowance the state no longer sells or one another emitter uses. Either way their cost exceeds the climate value of their abatement at EPA's social cost of carbon; counting the congestion, accident and local air pollution they avoid brings them about to break even at a 2 percent discount rate. The tax credit expires at the end of 2029, and Congress must decide whether to extend it. While the mandate binds, as it does if EPA keeps its percentages at their 2027 level, the extension adds almost no biofuel: it lowers fuel prices, sends up to three quarters of each federal dollar to fuel buyers, and raises emissions by 33 to 67 million tonnes over 2026–45, depending on how much of the credit is capitalized. From 2032 California's standard is not met at its price cap with the fuels and vehicles in the model, and its credit price stays there. Current law abates 1.8 billion tonnes of CO_2 at about $149 a tonne, a fifth more per tonne than a carbon price at the social cost of carbon; at the margin the gap is wider, since a carbon price buys the next 100 million tonnes at $36 a tonne and a larger mandate at $175.

Keywords: overlapping regulation, federalism, tradable credits, banking, expectations, subsidy incidence.

JEL codes: H23, H77, Q48, Q54, Q58.

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1 Introduction

A gallon of diesel sold in California answers to three agencies. EPA's Renewable Fuel Standard (RFS) requires that a share of the country's fuel be renewable and enforces the share with tradable compliance credits, Renewable Identification Numbers or RINs. The Treasury's 45Z is a tax credit, a subsidy per gallon paid to domestic producers of low-carbon fuel. California's Low Carbon Fuel Standard (LCFS), like the clean fuel standards of Oregon, Washington and New Mexico, requires fuel sellers to lower the carbon intensity of what they sell. Much of the fuel that earns its credits is renewable diesel, a diesel substitute made from fats and vegetable oils. The standard is enforced through a market in state credits, which holders may save for later years (the bank) and whose price is capped. The United States has no national carbon price on transportation fuel. At 2026 prices, the RINs, state credits and tax credits this stack creates are worth $45 billion a year, most of it a transfer among fuel buyers, producers and governments.[1] 45Z expires at the end of 2029, and Congress must decide whether to extend it.

Which of these layers does the work depends on how they interact. A mandate binds when it requires more biofuel than the market would use on its own, and a mandate that binds sets the quantity of biofuel. While it binds, a subsidy or a second standard on top of it cannot raise that quantity. It can only change fuel prices, vehicle choice and where the fuel is burned. For one mandate and one subsidy this is a known result (de Gorter and Just, 2009; Lapan and Moschini, 2012). With a federal mandate, a federal tax credit and four state standards, several with banks and price caps, which constraint binds is determined in equilibrium and can change from year to year. I call the constraint that binds in a given market and year the binding layer. It can be capacity, the federal mandate, a state standard below its cap, or a state's price cap. I measure what each other instrument adds, in abatement and in cost, on top of it. In the model the federal mandate is the binding layer in every year from 2026 to 2045. Inside California, the state standard's price cap also binds from 2032.

Two facts support this picture. The first shows the federal mandate binding. Irwin et al. (2020) show that the biodiesel RIN price tracks a fundamental value that falls when the blenders' tax credit, the predecessor of 45Z, is in force. Extending their sample to 2021, I find that each dollar of the credit lowered the RIN price by 41 cents (standard error 13 cents) in the months in which the mandate bound (Appendix G.4). A binding mandate predicts this direction, since a subsidy cannot buy more biofuel and can only lower the cost of complying (Section 6.2). The second fact, noted by Lade and Smith (2025), shows renewable diesel going where the state credits pay for it. California, Oregon, Washington and New Mexico burn 12 percent of the country's on-road diesel but used three quarters of its renewable diesel in 2023–24. In 2025, the year the blenders' tax credit gave

[1] RINs $37.8 billion at observed January–September 2026 prices, state credits $2.8 billion at the states' reported 2026 prices, and the model's 2026 outlay on 45Z, $4.2 billion. New Mexico has no observed 2026 price and is counted at zero.

way to 45Z, they used 97 percent.[2] This is what a fixed national volume being reallocated would look like, and the model measures how much of the states' credited abatement that reallocation accounts for.

Measuring what each layer buys requires a model in which credit banks, price caps and investment decide which layer binds, together with a rule for how credit holders form expectations. I build a dynamic equilibrium model of fuel, feedstock, vehicle and credit markets for 2026–45. In the textbook rule, Hotelling pricing, holders foresee the future and the credit price rises at the interest rate until the bank runs out. This rule does not fit the state markets. Given the bank and the yearly shortfall of credits, Hotelling pricing would put California's credit at $190 to $230 a tonne in 2026. The observed price was $64, which implies that holders expect the bank to last about 43 years, or doubt that the standard will bind.[3]

I estimate how holders form expectations. Holders of state credits look at the size of the bank and form an expectation of the year it will run out, and the credit price follows from that year and the price cap. The rule is estimated on prices and banks in California, Oregon and Washington through 2025. Re-estimated each quarter on earlier data only, it forecasts California's price better than no change when given the realized balance of credits, but not when it must also forecast the balance. For 2026 it predicts $68, against the observed $64. Biofuel capacity follows an investment rule estimated on 46 refinery and plant decisions in 2014–25. Estimated on 2014–19 alone, it predicts the investment of 2020–25. The estimated cost of entry is several times engineering cost. At engineering cost the rule would predict 82 refinery conversions by 2045, against 16 to date. The model also contains the RFS's fuel categories, each with its own RIN and bank, as well as 45Z, the state standards' rules for carrying unmet obligations forward, and the markets for cars and for medium and heavy trucks.[4]

The main result concerns the tax credit. Because the mandate binds, extending 45Z past 2029 pays for biofuel that would have been made anyway, and biofuel use rises by 0.1 percent. The tax credit lowers the price of RINs, and fuel sellers pass the saving into fuel prices. More fossil fuel is burned and fewer electric vehicles are bought, and emissions rise by 67 million tonnes over 2026–45 if all of the credit is capitalized (Figure 7b). Who keeps the money depends on how much of the tax credit is capitalized, that is, how much of it producers, the RIN market and investors treat

[2]State program volumes from the California Air Resources Board (CARB), Oregon DEQ and Washington Ecology, over U.S. consumption in U.S. Energy Information Administration (2026b). New Mexico's program began in April 2026. On-road diesel is from EIA, Adjusted Sales of Distillate Fuel Oil by End Use (on-highway), for 2015–20, the last years of the series, which give 11.7–12.0%.

[3]The bank lasts four years if the end-2025 bank is divided by the observed 2025Q3–2026Q1 shortfall at an annual rate, and ten at the upper end of the projection in Section 2.1. In the model's current-law path the bank runs out in 2031. Prices are $\bar{P}/(1+r)^T$, with the price cap $\bar{P}$ in 2024 dollars and the real holding cost $r = 3.4$ percent measured from forward quotes. The implied horizon is $\ln(\bar{P}/P_t)/\ln(1+r)$, about 29 years at 5 percent.

[4]The fuel and feedstock block extends the single-year model of Wu et al. (2026), and the light-duty vehicle block adapts the demand model of Cole et al. (2023). The medium- and heavy-duty vehicle blocks, the estimation of all three vehicle blocks, the multi-year dynamics, credit banks, the expectation, investment and regulator rules, and all counterfactuals and welfare accounting are new to this paper.

as part of the price. The history of the blenders' tax credit does not pin that share down (Irwin et al., 2020). If all of the credit is capitalized, fuel buyers receive 75 cents of each federal dollar, or 65 cents if the supply of animal fat is kinked. If half is, they receive 38 cents, producers keep the rest of the credit as a windfall, and emissions rise by 33 million tonnes. EPA could turn the tax credit into abatement by raising the mandate, but the two together would still cost more than the abatement is worth. While the RFS binds, the extension lowers fuel prices, and what it buys is set by EPA's percentages.

The same mechanism shapes three further results. First, the stack costs more per tonne than a carbon price, and most of the excess is in the state layer. Relative to a world without these fuel policies, current law abates 1.8 billion tonnes of carbon dioxide over 2026–45 at $149 a tonne.[5] A carbon price set at the social cost of carbon and applied to all transportation fuel in place of these policies would abate more than six times as much, at $125 a tonne on average (Table 3). The federal mandate abates at $132 a tonne and the state standards at more than twice that. Further abatement is also cheaper with a different instrument. On top of current law, a small carbon price buys the next 100 million tonnes at $36 a tonne and a larger RFS at $175. A national clean fuel standard that replaced the RFS and the state standards from 2030 would abate three times as much as current law at two thirds of its cost per tonne. It soon reaches its price cap and then works as a national carbon price, and because it covers every state, fuel cannot meet it by moving across state lines.

Second, about half of what the state standards appear to cut is fuel moved across state lines. The standards cut emissions inside the four states by 1.1 billion tonnes over 2026–45, but national emissions by only half that. The renewable diesel they draw into the four states would otherwise have been burned elsewhere, and fossil diesel replaces it there. The national cut comes from higher fuel prices in the four states and from electric vehicles (Figure 7a). In California and Washington the standards also overlap with cap-and-trade. If the emissions cap does not bind, a tonne the standards cut is an allowance the state no longer sells. Counting that lost revenue, the standards' cost exceeds the value of their abatement by $20 billion in present value. If the cap binds, the allowance the standards free is used by another emitter and the abatement disappears. In both cases, and in any mix of the two, the standards cost more than their climate benefits are worth. The congestion, accident and local air pollution they avoid about make up the difference at a 2 percent discount rate, but not at 3 percent, on either SCC basis (Section 6.1).

Third, California's standard as amended is not met at its price cap from 2032 with the fuels and vehicles in the model. The shortfall at the cap reaches 62 million tonnes in 2045, worth $16.1 billion that year at the cap price, and tightening the standard's targets no longer changes the credit

[5] Costs are discounted to 2025 at 2 percent and divided by the tonnes abated, which are not discounted. The reference also removes the federal carbon-capture credit (45Q) and the state biofuel incentives. The cost counts the resource cost and the fuel-tax and allowance revenue that governments forgo. Allowance revenue is measured on the assumption that the standards' cuts in California and Washington are real.

price. Fuel sellers short of credits then carry unmet obligations forward, which the state may never enforce. Selling credits at the cap price would turn that value into state revenue (Section 8).

How holders form expectations matters mainly for credit prices and for when California's cap binds. The cost of current law stays within $4 a tonne of $149 under the estimated rule and under a bound in which holders keep pricing the standard as they have in recent years (Section 7.2 and Table 4). A regulator that acts on the market's doubt would change the totals. If California relaxes its standard as its past decisions predict, the bank never runs out, the cap never binds, and current law abates 1,471 million tonnes instead of 1,756.

Related literature. The paper's contribution is to measure what each layer of the U.S. fuel-policy stack buys given the layer that binds. The binding mandate and the multi-region fuel market determine the cost of the stack and who receives a federal dollar (Section 6). The credit banks, and how holders price them, determine the year California's cap binds, the obligation that follows and when the 45Z transfer reaches fuel buyers. The dynamic model estimates that pricing. The paper builds on three literatures. The first studies policies stacked on a quantity constraint. On a single binding mandate, a subsidy to biofuel lowers the price of the mandate's credits and leaves output unchanged (de Gorter and Just, 2009; Lapan and Moschini, 2012; Gerveni et al., 2026). The history of the blenders' tax credit supports this. It leaves open how much of the credit was capitalized and shows EPA taking the credit into account when it sets the mandate (Irwin et al., 2020). When two levels of government regulate one externality, abatement by one leaks elsewhere or is cancelled by the other's cap (Goulder and Stavins, 2011; Goulder et al., 2012; Perino et al., 2019; Rosendahl, 2019). The incidence I find spans the range Irwin et al. (2020) leave open, and the counterfactual in which EPA raises the mandate quantifies the response they document. Lade and Smith (2025) describe the state standards as reshuffling renewable diesel with no real effect on emissions. I find that about half of their credited abatement is relocated and trace the rest to fuel prices and electric vehicles. Scott (2024) finds that under a binding cap California's standard raises emissions and the cost of abatement. Here the share of abatement the cap cancels is a parameter (Perino et al., 2019), and the standards fail a cost–benefit test over its whole range, including the low share implied by allowance prices at a floor or ceiling (Borenstein et al., 2019). Larson (2026) shows that when allowance prices sit at a floor or ceiling, linking two markets changes total emissions, in a one-period setting. This paper answers the analogous question for fuel policy over twenty years, with banks, caps and estimated expectations.

The second literature compares instruments. A clean fuel standard taxes high-carbon fuel and subsidizes low-carbon fuel at once, so it can cost far more per tonne than a carbon price and can even raise emissions (Holland et al., 2009). Mandates and clean fuel standards also fall unevenly across regions and incomes (Holland et al., 2015), and how far they trail a carbon price depends on their interaction with existing taxes (Goulder et al., 2016; Parry and Small, 2005; Bento et al.,

2009). Price limits in an intensity standard recover part of the loss (Wang et al., 2021; Lade et al., 2018a; Lade and Lin Lawell, 2021). For electric vehicles, subsidies to charging buy more adoption per dollar than subsidies to purchases, because drivers value a larger charging network (Li et al., 2017; Springel, 2021), and a tax credit buys little when the investment it subsidizes would have happened anyway (Bushnell and Smith, 2024). I compare these instruments in one dynamic model of the whole stack, at equal additional abatement and with and without counting the forgone fuel tax. A carbon price is cheapest, a federal credit for fast charging comes next, on an estimate that depends on how long stations last, and a larger mandate is the most expensive.

The third literature concerns prices in markets for credits that can be saved. Banking and expectations shape the prices of emission allowances (Salant, 2016; Chen, 2018), and allowance prices are likely to sit at a floor or a ceiling (Borenstein et al., 2019). Prices also move on news about the regulator (Lade et al., 2018b), and beliefs about whether a policy will last shape investment (Chen, 2024). Holders need not price a bank with full foresight (Quemin and Trotignon, 2021; Perino, 2024), and pass-through of RIN and state credit prices to fuel prices is incomplete in some markets (Knittel et al., 2017; Mazzone et al., 2022). Forecasts of California's credit balance have projected supply and demand for credits (Bushnell et al., 2020). I instead estimate how holders price the bank, test the rule on years outside its estimation sample, and use it in an equilibrium model. Relative to simulation studies of U.S. climate policy (Bistline et al., 2023; Bushnell and Smith, 2024), the model adds credit banks, the estimated expectation and investment rules and the states' rules for deferring their standards, and reports the main results under alternative descriptions of expectations.

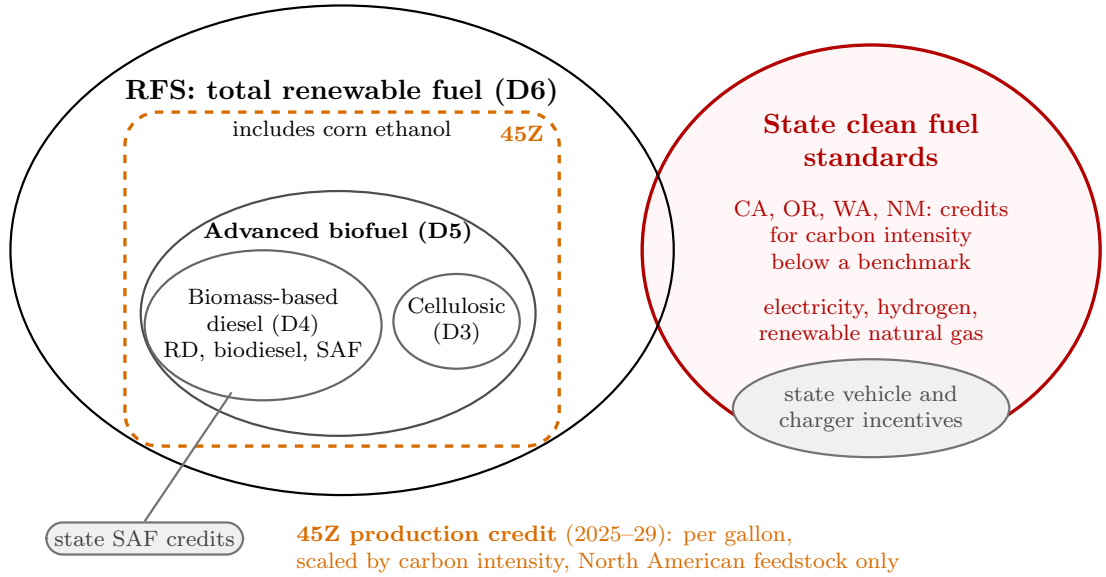
Section 2 describes the policies, Section 3 the mechanism, Sections 4 and 5 the model and its estimation, Sections 6 and 7 the results, and Section 8 concludes.

2 Policy Background

Federal and state instruments regulate the same gallons of U.S. transportation fuel. Figure 1 shows how they fit together. The federal instruments reward gallons of qualifying fuel everywhere, while the state standards reward low carbon intensity and apply only in four states. Because all of them draw on the same feedstocks and the same fuel pools, they interact through prices. Appendix A.1 gives the details.

Two federal instruments apply everywhere. The Renewable Fuel Standard (RFS) requires refiners and importers to retire credits (RINs) in proportion to the gasoline and diesel they sell. The requirements are nested: biomass-based diesel (D4) counts toward the advanced requirement, which counts toward the total. The RIN for the total requirement, mostly earned by corn ethanol, is called D6. Because of the nesting, the D4 price is at least the D6 price, and it exceeds the D6 price only when the biomass-based diesel requirement binds on its own. EPA's 2026–27 rule

(a) The instruments and what they reward



(b) Rules that respond to the state bank

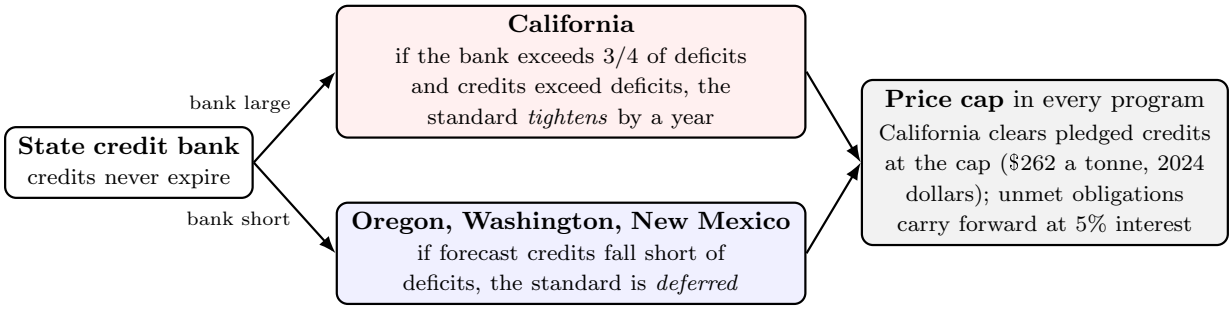


Figure 1: The policy stack for U.S. transportation fuel, October 2026

Notes: (a) The Renewable Fuel Standard's nested requirements, the 45Z tax credit, the four state clean fuel standards and the state credits for sustainable aviation fuel (SAF) and for vehicles and chargers. (b) The rules that make the state standards respond to their own credit banks. Sources: 40 CFR 80; 26 U.S.C. 45Z as amended by U.S. Congress (2025); 17 CCR 95480 ff.; OAR 340-253; RCW 70A.535; 20.2.92 NMAC. Appendix A.1 and Appendix Table 5 give the details.

raised the total requirement from 13.1% of gasoline and diesel in 2025 to 15.8% in 2027 (U.S. Environmental Protection Agency, 2026c). I hold the 2027 percentages for later years, a choice that decides whether the mandate binds after 2027 (Section 6.2). The 45Z credit pays up to $1.00 a gallon, scaled by lifecycle emissions. The 2025 budget law extended it through 2029 and limited it to feedstock grown in North America (U.S. Congress, 2025).

Four states add clean fuel standards. California (since 2011), Oregon (2016), Washington (2023) and New Mexico (2026) set a declining benchmark carbon intensity. Fuel above the benchmark generates deficits, fuel below it generates credits, and credits trade and never expire. Electricity

used in vehicles earns credits. California's 2024 amendments cut the benchmark 30% below 2010 by 2030 and 90% by 2045 (California Air Resources Board, 2025). Each program caps the credit price. California runs a credit clearance market at the cap ($262 a tonne in 2024 dollars), in which holders sell pledged credits (credits they have offered for sale at the cap), and carries obligations still unmet forward at 5% interest. Washington's statute holds its standard at a 20% cut in carbon intensity from 2030 until the Department of Ecology certifies the new biofuel capacity the statute requires. The main specification keeps the 20% limit and treats the path to 45% as a scenario. Renewable diesel was 72% of California's diesel in 2025Q1. Further deliveries are limited by the cost of bringing more of it into the state, since no blend limit binds.

The state programs also respond to their own banks, in opposite directions (Figure 1b). Oregon, Washington and New Mexico must defer their standards when forecast credits fall short of deficits. California instead tightens its standard by one year when the bank exceeds three quarters of the year's deficits and credits exceed deficits. The first test is in May 2027. No deferral has ever been ordered, and these rules work, if at all, through expectations.

2.1 What credit prices say about the statute

While a bank is held, a credit's price should rise at the holding cost and reach the cap when the bank runs out (Cronshaw and Brown-Kruse, 1996; Schennach, 2000). In a market with a price ceiling the price is likely to settle at the ceiling (Borenstein et al., 2019). Each price thus implies a year in which holders expect the bank to run out, $t + \ln(\bar{P}/P_t)/\ln(1+r)$. California's history has two regimes. In 2019–21 the bank covered less than a year of deficits, the price was within about a fifth of the cap, and the horizon it implied was close to a projection of fundamentals. By 2024 the bank had quadrupled and the price had fallen to $64. In 2024–25 the price implied that the bank would last another 43–45 years, while the bank and the current shortfall imply four to ten. The gap widened as the 2024 amendments tightened the standard, and the price did not respond when they took effect in July 2025 or as deficits came to exceed credits. Panel (b) of Figure 2 shows both regimes. Since 2023 the price has been a fifth to a third of the cap while the bank covered one to two years of deficits. A market that believed the statute would have priced credits close to the cap. In both regimes the price moved further than fundamentals, as it would if holders extrapolated the current balance.[6]

The price does respond to news about the regulator. California's price fell 16% in the week the Office of Administrative Law rejected the 2024 amendments and rose 14% in the week it approved them, and Washington's rose 26% (0.23 log points) in the week its Senate passed a bill tightening the standard (HB 1409), while CARB's quarterly data releases move prices little. The price-expectation rule of Section 4 captures the response to the balance but not the response to regulatory news.

[6] Appendix G.2 gives the horizons under alternative holding costs and projections, and a projection-free bound. At the observed 2025Q3–2026Q1 deficit the bank lasts fewer than four years. Oregon shows the same pattern.

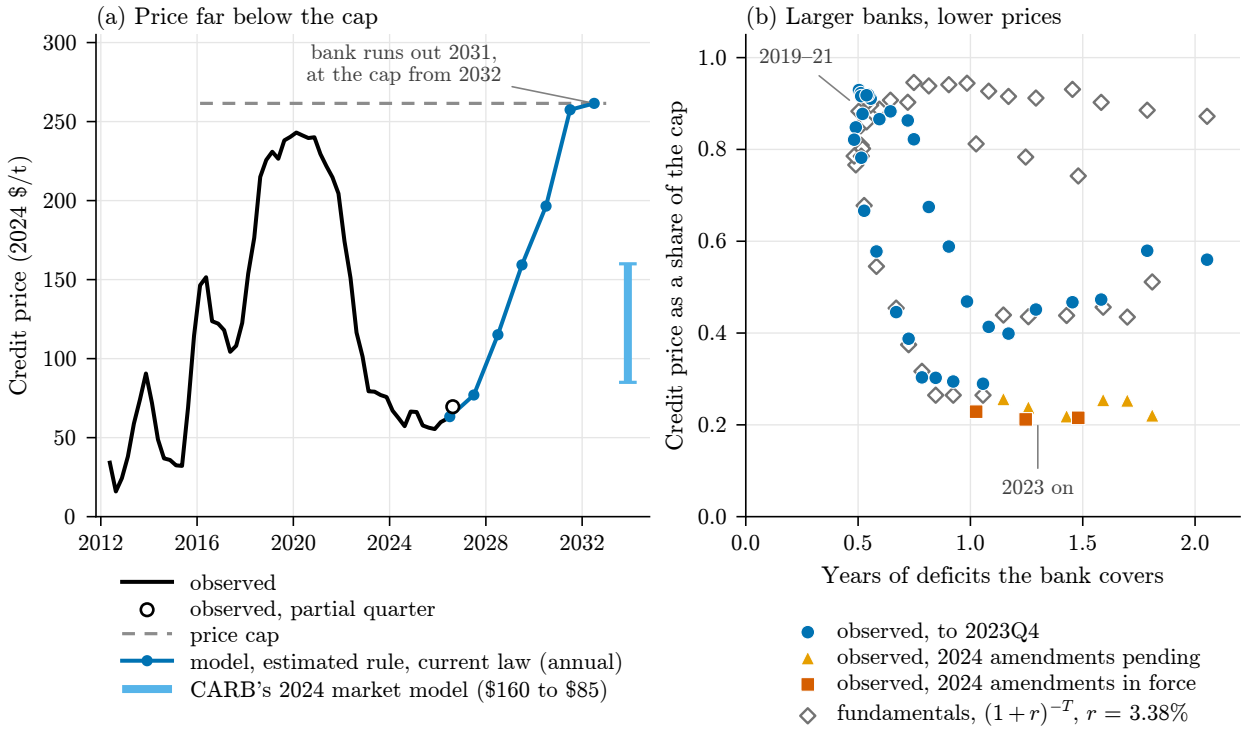


Figure 2: California's credit price sits far below its cap, and larger banks come with lower prices

Notes: (a) Quarterly credit price (2024 $/t), the price cap, the model's current-law path under the estimated rule (annual) and the range of CARB's 2024 market model. (b) Price as a share of the cap against the years of deficits the bank covers (end-of-quarter bank over deficits of the last four quarters), observed and under fundamentals, $(1+r)^{-T}$ with $r = 3.38\%$. Sources: CARB LCFS data and credit transfer log; CPI-U.

3 A Framework: National Support on a Banked State Standard

This section presents a stylized model of a federal fuel subsidy in a country where one region has a bankable credit market. It adapts the analysis of overlapping climate policies in Perino et al. (2019) to an intensity standard with a price cap and a federal mandate, and adds banking. The framework defines the objects that the quantitative model measures and shows which features of the markets determine their size. The central object is what happens to a marginal dollar of national support, that is, whether it becomes a rent, a transfer to fuel buyers or new abatement, and in which region.

3.1 Set-up

There are two regions, $j \in \{S, N\}$. Region S has a carbon intensity standard, and region N has none. Fuel is measured in energy units and sells at price p_j in region j. Demand is $Q_j = D_j(p_j)$, with $D_j' < 0$, and $D = D_S + D_N$ is national demand. Fossil fuel is supplied perfectly elastically at cost m and has carbon intensity c_0. Biofuel is a perfect substitute in use, has intensity $c_1 < c_0$, and is produced by competitive firms with national cost $C(q)$, where $q = q_S + q_N$, $C'' > 0$, and capacity $q \leq H$. Delivering biofuel to S costs $\tau(q_S)$ per unit at the margin, with $\tau' > 0$. Region N

takes biofuel at no extra cost up to a blend or logistics limit $\bar{q}_N$. I call N's room to absorb more biofuel the *outlet*.

Three policies act on this market. The federal mandate requires $q \geq M$. It is met with tradable credits (RINs) at price ρ. Each unit of biofuel generates one RIN, and each unit of fossil fuel carries an obligation $\pi = M/Q^F$, where $Q^F = (Q_S - q_S) + (Q_N - q_N)$ is national fossil fuel use. The federal subsidy pays s per unit of biofuel. The state standard $\sigma \in (c_1, c_0)$ gives each unit of fossil fuel sold in S a deficit of $c_0 - \sigma$ and each unit of biofuel a credit of $\sigma - c_1$. Electric vehicles generate Γ credits. State credits trade at price λ, which cannot exceed the cap $\bar{P}$. When credits fall short at the cap, the state issues the shortfall X^+ at $\bar{P}$, as in the quantitative model (a stand-in for the statute's carry-forward of unmet obligations, Section 6.1).

Fuel prices are

$$p_N = m + \pi\rho, \qquad p_S = m + \pi\rho + (c_0 - \sigma)\lambda, \tag{1}$$

and a unit of biofuel earns $p_N + \rho + s$ if sold in N and $p_S + (\sigma - c_1)\lambda + \rho + s - \tau(q_S) = p_N + (c_0 - c_1)\lambda + \rho + s - \tau(q_S)$ if sold in S. The state's net credit generation, credits less deficits, is

$$b = (\sigma - c_1)\, q_S - (c_0 - \sigma)(Q_S - q_S) + \Gamma = (c_0 - c_1)\, q_S - (c_0 - \sigma)\, Q_S + \Gamma. \tag{2}$$

3.2 Equilibrium

Definition 1. An equilibrium is a set of quantities (q_S, q_N, Q_S, Q_N) with $Q_j = D_j(p_j)$, prices (ρ, λ) satisfying (1), a capacity rent ψ, a shadow value of the outlet ν and issuance X^+ such that

$$C'(q) + \psi + \nu = p_N + \rho + s, \tag{3}$$

$$C'(q) + \psi + \tau(q_S) = p_N + (c_0 - c_1)\lambda + \rho + s, \tag{4}$$

$$0 \leq \psi \perp H - q \geq 0, \qquad 0 \leq \nu \perp \bar{q}_N - q_N \geq 0, \qquad 0 \leq \rho \perp q - M \geq 0, \tag{5}$$

$$b + X^+ \geq 0, \quad \lambda \geq 0, \quad \lambda\,(b + X^+) = 0, \qquad 0 \leq X^+ \perp \bar{P} - \lambda \geq 0. \tag{6}$$

Equations (3) and (4) are the zero-profit conditions for biofuel sold in N and in S (I assume $q_N > 0$). Conditions (5) are the three federal-level constraints: capacity, the outlet and the mandate. Each has a price that is positive only when the constraint binds. Here ψ is a rent to producers, ν is the discount at which the marginal unit is sold in N, and ρ is the RIN price. Condition (6) clears the state credit market, with the cap as a safety valve. Subtracting (3) from (4),

$$\tau(q_S) = (c_0 - c_1)\,\lambda + \nu. \tag{7}$$

When the outlet is slack ($\nu = 0$), biofuel is sold in both regions up to the point where the state credit premium alone covers the cost of delivery to S. The federal subsidy and the RIN are paid

in both regions and so drop out of the condition.

Lemma 1 (The state credit market)**.** *Suppose the outlet is slack. Then* $q_S = \tau^{-1}\big((c_0 - c_1)\lambda\big)$*, and net credit generation is a function* $b(\lambda; \rho)$ *that does not depend on* s*, with*

$$\frac{\partial b}{\partial \lambda} = \frac{(c_0 - c_1)^2}{\tau'} - (c_0 - \sigma)^2 D_S' > 0. \tag{8}$$

The credit price is $\lambda^* = b^{-1}(0)$ *if* $b(\bar{P}) \geq 0$*. Otherwise* $\lambda^* = \bar{P}$ *and the state issues* $X^+ = -b(\bar{P})$*. When the standard binds below the cap, emissions in* S *are*

$$Z_S = c_0(Q_S - q_S) + c_1 q_S = \sigma Q_S + \Gamma. \tag{9}$$

The two terms in (8) are the two ways a higher credit price creates credits. It pulls biofuel into S (first term) and raises the price of fuel in S, which lowers fuel use and deficits (second term). Equation (9) follows from setting $b = 0$ in (2). A binding intensity standard fixes emissions per unit of fuel, so emissions in S change only through fuel use and credits from outside the fuel pool (Holland et al., 2009). Appendix Figure 16 draws the allocation of national biofuel between the regions and the state credit market.

3.3 Where additional national support goes

Consider a marginal increase ds in the subsidy. Where it goes depends on which of the constraints in (5)–(6) binds. Let $\varpi \equiv -(c_0 - \sigma)^2 D_S'/(c_0 - c_1) > 0$, the additional biofuel the state needs to stay in balance when its credit price falls by one dollar.

Proposition 1 (Where additional support goes)**.** *To first order:*

(i) Capacity binds *($q = H$, mandate slack). $d\psi/ds = 1$, so the dollar is a rent to producers, and no quantity or other price changes.*

(ii) The mandate binds *($q = M$, capacity slack), and the outlet is slack or the state is at its cap. $d\rho/ds = -1/(1+\pi)$, and national biofuel does not change. At a given state credit price, the price of every unit of fuel falls by $\pi/(1+\pi)$ and fuel buyers gain $M\,ds$, the whole cost of the subsidy. Below the state's cap, part of the gain to fuel buyers in S is offset by a higher state credit price. This effect is of the order of the demand response and small when demand is inelastic.*

(iii) Neither binds and the outlet is slack. *$dq_N/ds = 1/C''$ and $dq_S = d\lambda = 0$, so the extra biofuel goes to N, and national emissions fall by $(c_0 - c_1)/C''$. The state's credit price, fuel use and emissions do not change, whether or not the state is at its cap.*

(iv) The outlet binds and $\lambda < \bar{P}$.

$$\frac{d\lambda}{ds} = -\frac{1}{(c_0 - c_1) + \varpi\,(C'' + \tau')} < 0, \qquad \frac{dq_S}{ds} = -\varpi\frac{d\lambda}{ds} > 0, \qquad \frac{dZ_S}{ds} = \sigma(c_0 - \sigma)D_S'\frac{d\lambda}{ds} > 0. \tag{10}$$

The dollar lowers the state's credit price, a transfer to its fuel buyers, and the state's emissions rise.

(v) The outlet binds and $\lambda = \bar{P}$.

$$\frac{dq_S}{ds} = \frac{1}{C'' + \tau'}, \qquad \frac{dZ_S}{ds} = \frac{dX^+}{ds} = -\frac{c_0 - c_1}{C'' + \tau'}. \tag{11}$$

Biofuel displaces fossil fuel in S one for one, and the state issues fewer credits.

Figure 3 summarizes the five cases, and Appendix F gives the proofs. Cases (i) and (ii) of Proposition 1 concern the federal constraints. When capacity or the mandate binds, the subsidy does not increase biofuel output, and in case (ii) the whole dollar reaches fuel buyers through a lower RIN price.[7] The level of the biodiesel RIN price has moved with the value of the blenders' tax credit, though by less than one for one (Section 6.2), including in the years the credit lapsed and was restored (Irwin et al., 2020). Because the dollar lowers the price of every unit of fuel, fuel use rises. With biofuel fixed at M, fossil fuel use increases by $-D'(p)\,\pi/(1+\pi)$ per dollar and national emissions rise, so a subsidy on top of a binding mandate subsidizes fossil fuel consumption (de Gorter and Just, 2009; Lapan and Moschini, 2012). Cases (iii)–(v) concern the state, and which of them applies depends on the outlet. When biofuel can still go to N, the arbitrage condition (7) already sends the state the biofuel that its standard pays for, and a subsidy paid in both regions leaves that allocation unchanged. When the outlet is full, the extra biofuel must go to S, and the sign of the interaction depends on the state's regime. Below the cap the standard fixes emissions per unit of fuel, so the extra biofuel lowers the credit price and allows more fossil fuel to be sold in the state. The subsidy becomes a transfer to the state's fuel buyers, and emissions in S rise. This is the waterbed effect of overlapping instruments under a binding target, in which abatement by one instrument is cancelled by the other's fixed target, as pressing on a waterbed raises it elsewhere (Goulder and Stavins, 2011; Perino et al., 2019). At the cap the target is not met. In the terminology of Perino et al. (2019) the waterbed is punctured and the extra biofuel is new abatement. Case (iii) is the fuel-market counterpart of their internal leakage, since the extra biofuel goes to the region where no standard binds. In this framework relocation, the reallocation of a

[7] Case (ii) holds the obligation per unit of fossil fuel, π, fixed, which is exact when fuel demand is inelastic. EPA sets the RFS as percentages of fuel sold, and so does the quantitative model. With a fixed percentage, the lower fuel price raises fossil fuel use, and the required biofuel rises with it by π times that increase. With a fixed volume M, the obligation per unit falls instead. Both corrections are proportional to the demand response to the RIN pass-through and are small when demand is inelastic.

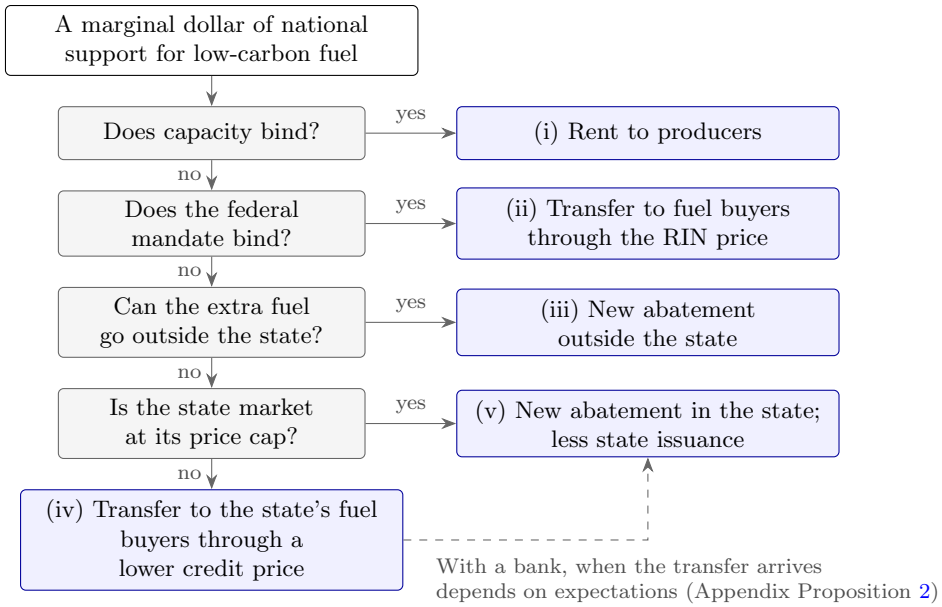


Figure 3: The five cases of Proposition 1. Each question refers to the constraint that binds at the margin. In the quantitative model several apply at once, and Section 6 reports shares.

fixed national volume of biofuel toward S, is caused by the state standard. The subsidy leaves it unchanged. With the mandate binding, tightening σ raises q_S and lowers q_N one for one.

As defined in the introduction, the binding layer is capacity, the federal mandate, a state standard below its cap, or a state's price cap, whichever binds in a given market and year. Proposition 1 says that where an additional dollar goes depends on which constraint binds, and the design of the instrument matters only through that. In the quantitative model the binding layer can be read year by year. Figure 4 shows it for current law. The federal mandate binds in every year, and California moves from a banked market to its price cap.

3.4 When the transfer arrives: banks and expectations

Now let time run, $t = 1, 2, \ldots$, and let the state bank credits. As the standard σ_t tightens, net generation $b_t(\lambda)$ falls over time and eventually becomes negative. The bank evolves as

$$B_t = B_{t-1} + b_t(\lambda_t) + X_t^+, \qquad B_t \geq 0. \tag{12}$$

Credit holders pay a holding cost r. As long as the bank is positive, they hold credits only if the price is expected to rise at rate r. The price is then determined by the year T in which the bank is expected to run out and the price to reach the cap:

$$\lambda_t = \bar{P}\,(1+r)^{-(T-t)}, \qquad t \leq T. \tag{13}$$



Figure 4: Which constraint binds, by policy layer and year, under current law, 2026–45

Notes: Main specification, current law. RFS "bind": the D6 RIN price exceeds \$0.01. The biomass-based diesel requirement never binds on its own. State programs: "bank", price below the cap with a positive bank; "no bank", bank exhausted and price below the cap; "cap", price at the cap; "def.", a step of the standard deferred; "2026 obs.", observed price. Source: model run. Details: Appendix G.2.

This is Hotelling's rule with the cap as a backstop (Hotelling, 1931; Heal, 1976), and it leaves open how holders form their expectation of T. Under perfect foresight, T is the year in which the bank actually runs out given the path (13):

$$B_{t_0-1} + \sum_{t=t_0}^{T} b_t\big(\bar{P}(1+r)^{-(T-t)}\big) = 0. \tag{14}$$

Under the adaptive rule that the paper estimates, holders revise T in proportion to the surprise in the year's net generation,

$$T_t = T_{t-1} + \gamma\big(b_t - E_{t-1}[b_t]\big), \qquad E_{t-1}[b_t] = \bar{b}_{t-1} + \Delta_t^{sched} + \Delta_t^{known}, \tag{15}$$

where $\bar{b}_{t-1}$ is last year's net generation or the average of the last three years (the quantitative model uses the three-year average), Δ_t^{sched} the effect of the scheduled tightening of the standard, and Δ_t^{known} other changes known in advance. The price is (13) with $T = T_t$, solved within the year together with b_t. The quantitative model adds a term that raises the price as the bank runs down, with weight $e^{-Y_t/K}$, where Y_t is the number of years the bank would last at the expected shortfall.

Appendix F.2 works out how a permanent subsidy that raises the state's net generation changes the price path under each description (Proposition 2 and Figure 17). Under perfect foresight the price falls from the first year. In the numerical example, under the adaptive rule it falls by less if the subsidy is a surprise, and not for many years if the subsidy was anticipated, because the part that holders do not treat as news first accumulates in the bank and lowers the price only as the bank runs down. The timing of the transfer depends on what holders count as news. The same accounting, with the policy held fixed, compares any two descriptions of expectations, which gives Corollary 1.

Corollary 1 (Doubt brings scarcity forward)**.** *Take two descriptions of expectations, e and e', that start from the same bank and face the same net generation functions $b_t(\lambda)$, and let T^e and $T^{e'}$ be the first years in which the bank cannot cover the year's shortfall at the cap under each. If the credit price under e is no higher than under e' in every year before* $\min(T^e, T^{e'})$*, then the bank under e is no larger in every such year, and $T^e \leq T^{e'}$.*

Because net generation rises with the price (Lemma 1), a market that expects scarcity to be distant generates fewer credits each year and uses up its bank sooner. Under perfect foresight the price reflects the coming scarcity from the start. A market that extrapolates from a large bank keeps the price low until little of the bank is left, and the price jumps to the cap when the bank runs out. When holders doubt the statute, the bank runs out earlier.

The framework has a single state and a single subsidy paid at the same rate on every gallon, and only one constraint binds at a time. The quantitative model departs from it in three ways that matter for the results. First, 45Z varies with carbon intensity and, from 2026, with feedstock origin. A change in the subsidy then also changes the feedstock mix and the state's credit generation per gallon. Lemma 1 then holds only approximately, and case (iii) can move the state's credit price a little. Second, several cases apply at once, in different markets and years. The RFS can bind nationally while one state is at its cap and another is not. Third, the states' codified responses (acceleration in California, deferral in Oregon, Washington and New Mexico) make the standard itself respond to the bank, which in the framework would make σ a function of B. The quantitative model has to say which constraint binds in each market and year, how steep the slopes C'', τ' and D'_S are, how credit holders map the bank into prices, and how investors value capacity, which determines whether capacity binds. The delivery slope τ', the cost of delivering more renewable diesel to the four states, is the least anchored of these.

4 Quantitative Model

The quantitative model applies the framework to the U.S. fuel and vehicle markets from 2026 to 2045. It has five regions: California, Oregon, Washington and New Mexico, each with its own clean

fuel standard, bank, cap and credit price, and the rest of the country, which has none. Credits do not trade across programs, but fuel moves between states. Each year a fuel market clears fuels, feedstocks, RINs and state credits for given capacity, fleets and credit prices. Banks, holders' expectations, the state rules, capacity and the vehicle stock link each year to the next. The fuel-market block shares its feedstock and crediting structure with Wu et al. (2026). The dynamics, the expectations, investment, vehicles and the four state programs are new. Appendix B states the model in full and summarizes how each block is disciplined by data. Appendix Table 7 sets the model beside the closest models in the literature. It adds multi-year credit banks, estimated expectation and investment rules and four state programs, but it does not model the rest of the economy (Bistline et al., 2023) or treat uncertainty in its inputs (Bushnell and Smith, 2024).

4.1 Credit markets, rules and expectations

In each program j, credits from biofuels, from electricity ($\Gamma^{EV}_{j,t}$, from the vehicle block) and from fixed sources, less deficits from fossil gasoline and diesel, give the year's net credit generation $N_{j,t}$. The market clears as a complementarity problem with a price floor at zero and a ceiling at the cap $\bar{P}_{j,t}$,

$$N_{j,t} + B^{draw}_{j,t} = s^{+}_{j} - s^{-}_{j}, \qquad 0 \le P^{L}_{j} \perp s^{+}_{j} \ge 0, \qquad 0 \le \bar{P}_{j,t} - P^{L}_{j} \perp s^{-}_{j} \ge 0, \tag{16}$$

where B^{draw} is the draw on the bank that holders' expectations allow. Standards follow each program's adopted schedule. Washington's is capped at 20% from 2030 until the Department of Ecology certifies the new biofuel capacity the statute requires. When a program is short at its cap, I close it by issuing the shortfall at the cap price and count the proceeds as state revenue, whereas the statutes carry the unmet obligation forward as a deficit with interest (17 CCR §95485(c)). Section 6.1 states the conditions under which the two treatments give the same prices and quantities.

Credit holders do not know when the bank will run out. They price credits by Hotelling's rule, rising to the cap in their expected exhaustion year at a holding cost $r = 3.38\%$, the real carry measured from forward quotes (Section 5.2),

$$P^{L}_{j,t} = \min\{\bar{P}_{j,t},\ \bar{P}_{j,t}/(1+r)^{\mathrm{gap}_{j,t}}\}, \tag{17}$$

where $\mathrm{gap}_{j,t}$ is the expected number of years to exhaustion. Holders revise it in response to the surprise in the year's net credit generation,

$$G_{j,t}(s) = T_{j,t-1} - t + \gamma_j\big(s - E_{t-1}[s_{j,t}]\big), \tag{18}$$

so a year with more credits than expected moves exhaustion out by γ_j years per Mt and lowers the price. Expected net generation is the average of the last three years plus the changes holders know about in advance: the scheduled tightening of the standard, the cap on credits from crop-based

oils in California's 2024 amendments and the end of the 2026 transitional terms. As the bank runs down, holders also weigh how many years it would last, $Y_{j,t}$:

$$\text{gap}_{j,t} = (1 - w_{j,t})\, G_{j,t} + w_{j,t}\, \min\{G_{j,t}, Y_{j,t}\}, \qquad w_{j,t} = e^{-Y_{j,t}/K}. \tag{19}$$

Equation (17) applies the rule of equation (13) (Heal, 1976; Schennach, 2000). A credit is a claim on the cap at the date the bank runs out, and holders carry it only if the price rises at the carry until then. What the rule adds is how holders arrive at that date. They carry an expected exhaustion year from one year to the next, revise it by γ years for each megatonne by which net credits surprise them (18), and, as the bank shrinks, give increasing weight to the arithmetic of the bank divided by the expected shortfall (19). K sets how close to running out the bank has to be before that arithmetic drives the price. At $K = 2$ it has about a third of the weight when the bank covers two years of expected deficits and a seventh when it covers four. Holders can still expect exhaustion decades away, as the 2024–25 prices imply, and the price discounts over all of that time. The rule is related to the option-pricing model of the biodiesel RIN in Irwin et al. (2020), in which a RIN is a claim on the biodiesel–diesel spread at a known expiry. A state credit never expires, its payoff at exhaustion is the cap, and its exhaustion date depends on the bank and on the price path itself, so their closed form does not apply. The corresponding objects here are the storage model with a price ceiling (Section 7) and the probability that the standard binds implied by prices (Appendix D.2). The rule describes how the credit price has responded to the balance, and it is estimated on that history. Unlike the forecasting rules of Krusell and Smith (1998), it is not required to be accurate along the model's own path, and it is not. Under current law holders expect in 2026 that the bank will last until about 2069, while it runs out at the end of 2031. Their expectation closes the gap only as unexpected deficits accumulate. Corollary 1 describes this mechanism, in which holders who extrapolate a large bank keep the price low and draw the bank down faster. A holder who foresaw the 2031 exhaustion would price California's credit at about $221 in 2026, against the $64 observed (Appendix D.1). Its role is to supply the mapping from credit balances to prices that the counterfactuals need, and it fits the history better than perfect foresight, a regulatory-risk market and a rolling horizon (Appendix D.1).[8] γ and the initial expected exhaustion years are estimated on quarterly prices and banks in California (2016–25, with forward quotes for 2022–25) and in Oregon and Washington (pooled) (Section 5.2). Scheduled and enacted changes to the standards enter holders' expected net credits directly. A counterfactual policy that is announced changes the rule's inputs and leaves its parameters unchanged. The response γ is identified mainly by the 2022–24 surplus, and Section 7 reports the results under other descriptions of expectations.

The other agents form expectations differently, because their markets differ. The RIN market

[8] It has the form of the steady state of a Kalman filter for holders who read each year's net credits as a noisy signal of a persistent balance (Appendix D.4).

is national, its rules are public and set by one agency, and its bank is limited by regulation, so RIN holders store with foresight, the standard description of such a market (Lade et al., 2018b). A deficit can be carried one year but not two in a row. Within the year the biomass-based diesel RIN clears at the marginal cost of the marginal gallon net of 45Z, the linear fundamental price of Irwin et al. (2020) without their option term. Because the D4 price stays above \$0.67 in every year, the option term on its level would be small. The premium of D4 over D6 is zero in the years the biomass-based diesel requirement is slack, where an option model would give a small positive value. Fleets buying trucks see state credit prices on the holders' path, and households see current prices (Anderson et al., 2013). The state credit markets, by contrast, are small, their rules change often, their banks have never run out, and their prices do not follow the statutory path. The model is solved as a temporary equilibrium, in which each year's markets clear given the expectations each group holds. Every agent expects the current policy stack, including the responses written into the statutes, to continue. No agent prices in a scheduled expiry or a proposal that has not been enacted.

The RFS's nested requirements (biomass-based diesel, advanced, total) are EPA's percentages applied to the fossil base. The 2026–27 percentages are EPA's final standards and are held at their 2027 level afterwards. Obligations then fall as fossil demand falls. Every obligated gallon carries the cost of the full obligation, whether its RINs come from this year's production or from the bank. California's acceleration mechanism advances later benchmarks by a year when the bank exceeds three quarters of the year's deficits and credits exceed deficits (17 CCR §95484(b)). Oregon, Washington and New Mexico defer scheduled tightening when the forecast bank falls short, which I model as the smallest temporary standard that keeps the bank non-negative at the cap. Neither rule has been triggered so far. Whether either is triggered in the forward runs is a model outcome that the data have not tested.

4.2 The fuel market in year t

Gasoline, diesel and jet fuel in each region combine a fossil component with biofuels: ethanol in gasoline; biodiesel and renewable diesel in diesel; sustainable aviation fuel (SAF) made from fats and oils (HEFA) or from alcohols (alcohol-to-jet) in jet. Fossil fuel is supplied elastically at a national price plus state differentials and compliance costs. Biofuel producers are price takers, and a pathway produces only if its cost (operating cost, feedstock, utilization cost, delivery) covers its value, which is the fuel it replaces plus its RINs, its state credits at the program's carbon intensity, 45Z and state SAF credits. Lipids form two groups (vegetable oils, and fats and used cooking oil) with upward-sloping supply, and the feedstock mix responds to their relative margins with one estimated response.

The federal–state mechanism of Proposition 1 enters the model through one feature. Renewable

diesel delivered to one of the four states bears a cost that rises with the volume delivered there, which stands in for terminal capacity, logistics and contracts. Without this cost, renewable diesel would move all-or-nothing to whichever program pays most. The slopes are estimated on quarterly deliveries since 2023. They determine whether additional national support raises deliveries to the four states or lowers their credit prices, cases (iii) and (iv) of Proposition 1. They are the least anchored parameters in the fuel market, and I report results at half and at twice California's estimated slope. In California the slope steepens once renewable diesel exceeds 72% of the diesel pool, the highest share in 29 quarters of data.

4.3 Capacity investment

Capacity to make renewable diesel and sustainable aviation fuel (hydroprocessing capacity) is endogenous from 2029. In each year, potential investors (91 refineries that could convert, 19 operating plants that could expand, 20 greenfield entrants, and existing plants that could convert to SAF) decide whether to commit now or wait. Each faces a competitive, single-agent optimal stopping problem with logit shocks, as in Rust (1987), with entry costs estimated from 46 capacity decisions. Investors take prices as given, which is reasonable for a national commodity market whose prices are set by the RFS and the credit markets. Strategic interaction among entrants and differences among firms of the same investor type are not modelled. The value of committing per gallon of capacity is

$$V^{c}_{\ell,t} = \sum_{s=L}^{L+19} \delta^{s} \Big[u^{e}_{t+s} \big(\mathbb{E}_t \, m_{t+s} + F_{\ell}/u_0 \big) - F_{\ell} \Big] - \kappa_{\ell,t}, \tag{20}$$

with a three-year construction lag, a 20-year life, the operating margin m (fuel value plus RINs, state credits and 45Z credits, less feedstock, operating and delivery costs), expected utilization u^{e} from the model's own solution, fixed operating cost F_{ℓ} and entry cost κ_{ℓ}. Because expected utilization falls as capacity is added, each addition lowers the value of the next. Investors hold RIN and feedstock prices at current levels, follow the holders' state credit price path, and expect tax credits due to expire to continue with probability p. Waiting has value because the margin may rise, and a potential investor commits with probability

$$P_{\ell,t}(z) = \Big[1 + \exp\big(-(V^{c}_{\ell,t}(z) - V^{w}_{\ell,t}(z))/\zeta \big) \Big]^{-1}. \tag{21}$$

4.4 Vehicles and charging

The vehicle block extends Cole et al. (2023) to light-, medium- and heavy-duty vehicles in the five regions. The electric share of new sales is logit in the ratio of the costs of ownership and, for light-duty vehicles, in the public charging network. Public stations enter when their margin, including the state credits they earn, covers the cost of building. Station entry is free entry at the

regional level, as in the models of Zhou and Li (2018) and Springel (2021), with the entry elasticities estimated by simulated method of moments (Appendix D.8). The network effects run both ways, from stations to sales and from the electric stock to stations, as in the estimates of Li et al. (2017). Competition among nearby stations is not modelled. Entry on the vehicle supply side, such as a new manufacturer in the electric truck market, is not endogenous either. Vehicle prices follow an exogenous path, with a low-price scenario for electric trucks as a bound. The block returns the electric shares of the fleet, which shift fuel demand, and electricity credits, which enter the state credit balances. Electricity credits are about a fifth of California's credit generation, so the bank after 2030 depends on the pace of electrification.

4.5 Equilibrium and welfare

Each year the fuel, feedstock, RIN and state credit markets clear jointly as a mixed complementarity problem (Appendix B.3). Because holders of state credits do not have rational expectations, the object computed is a temporary equilibrium path (Definition 2). Each year's prices clear that year's markets given expectations formed from the past, the RIN bank solves its storage problem on the path, capacity and fleets follow their rules at the path's prices, and the state rules act on the path's banks. The blocks are coupled by damped iteration over 2026–45 (Appendix B.6). The iteration is a damped fixed-point method and does not search for other equilibria. Every run reported in the main text meets the convergence rule in Appendix H.1. The runs that do not meet it all sit where a policy layer switches between binding and slack.

Welfare is the sum of the surpluses of fuel users (by state), vehicle buyers, biofuel producers, feedstock suppliers, the federal and state governments and owners of capital, less climate damages at EPA's 2023 social cost of carbon (2% discount rate).[9] Transfers cancel in the total. Costs are resource costs, with capital at its capital cost rather than the investment hurdle. California's and Washington's cap-and-trade programs enter as a fixed cost per tonne on road gasoline and diesel sold in those states, at the 2025 auction prices ($27 and $59 a tonne in 2024 dollars, about $0.22 and $0.47 a gallon of gasoline). The cost does not rise after 2025, and jet fuel is not covered. The programs are treated as taxes. The abatement they cover is counted in full, and the allowance revenue that California and Washington lose when covered emissions fall is counted as a cost, like forgone fuel-tax revenue. All values are at a zero offset under the states' caps unless stated. Fuel taxes are a pre-existing wedge, since a gallon not burned loses its tax and avoids the congestion, accident and local air-pollution costs of fuel use (Parry and Small, 2005; Parry et al., 2006; Holland

[9] In January 2025 Executive Order 14154 (Executive Office of the President, 2025) disbanded the Interagency Working Group on the social cost of greenhouse gases, withdrew its estimates and directed agencies to the 2003 version of OMB Circular A-4. I use EPA's 2023 estimates (U.S. Environmental Protection Agency, 2023), the most recent federal estimates, and report the break-even social cost of carbon of each comparison so that readers can apply their own value. The 3% sensitivity uses the Working Group's 2021 interim values at 3% (Interagency Working Group on Social Cost of Greenhouse Gases, 2021), since Circular A-4 (2003) discounts at 3% and EPA's 2023 estimates have no 3% series.

et al., 2016). I report the resource cost, the forgone fuel-tax revenue, the forgone California and Washington allowance revenue and the change in other external costs as separate lines. The cost per tonne in the text includes the first three, and the excise-only value is given beside it in Appendix I. The benchmarks are a carbon price at the social cost of carbon on all fuels and vehicles in the model, and a carbon price that adds 100 Mt of abatement to current law, the common target of Section 6.4. Both are first best only within the model's partial-equilibrium scope. For federal instruments I report outcomes per dollar of federal cost.

The model leaves out EPA's discretion over the percentages, which Section 6.2 and Appendix G.5 vary by raising and cutting them. It also leaves out fuel bought across a state line, a response of California's and Washington's allowance prices to the standards, and renewable natural gas and hydrogen in emissions. Appendix Table 8 lists what the model represents and what it does not.

5 Data, Estimation and Validation

This section covers everything except the feedstock and vehicle blocks, which are estimated in Appendix D.8. The 2026 credit prices and banks are the starting point, four 2026 quantities are calibration targets, and every other 2026 observation is held out for the fit to history. No 2026 observation is used in more than one of these roles. Structural parameters are estimated on data through 2025.

5.1 Data

For the state programs I use the programs' quarterly credit and deficit data by fuel and feedstock and their transfer prices: CARB's LCFS data (2011–2026Q1) and monthly credit transfer reports (California Air Resources Board, 2026c,d), Oregon DEQ's quarterly summaries (2016–2026Q1) (Oregon Department of Environmental Quality, 2026) and Washington Ecology's (2023–2026Q1) (Washington Department of Ecology, 2026). Annual prices are volume-weighted transfer prices. For 2026 the model's prices are pinned at their January–August averages, and the observed 2026 price quoted in the text, \$64 in California, is the average from 1 January to 18 September 2026. New Mexico's program began in April 2026 and has no price history. For the RFS I use data from EPA's Moderated Transaction System (EMTS) on RIN generation, retirements and holdings, and EPA's RIN transaction prices.

Fuel volumes, prices and biofuel production come from EIA (monthly production and capacity of renewable diesel and biodiesel; product supplied; wholesale prices). Feedstock use and prices come from EIA, USDA ERS and farmdoc's monthly series. Feedstock shares by state come from the programs' quarterly data. The investment data are 46 dated decisions to build or convert hydroprocessing capacity in 2014–2025 (16 refinery conversions, 5 expansions, 19 greenfield plants and 6 SAF conversions), each dated at its public announcement, except three, dated at the final

investment decision (Diamond Green Diesel's Port Arthur SAF unit, Calumet's MaxSAF) or at the start of operation (Sinclair), the earliest date recorded.

Data on light-duty electric sales and stocks by state and on public stations from the Alternative Fuels Data Center (AFDC) cover 2020–2025. Truck sales by class use 2025 zero-emission counts from ICCT, the California Energy Commission and Washington's Department of Licensing, with class 3 counts completed for Oregon, New Mexico and the rest of the country (International Council on Clean Transportation, 2026). Incentive funding comes from program records. These are the voucher data of California's Hybrid and Zero-Emission Truck and Bus Voucher Incentive Project (HVIP) as of 31 July 2026, the award list of the Energy Infrastructure Incentives for Zero-Emission Commercial Vehicles project (EnergIIZE), utility make-ready program filings and state budgets (CALSTART / California HVIP, 2026; CALSTART and California Energy Commission, 2026).

Oil prices follow EIA's Short-Term Energy Outlook to 2027 and the Annual Energy Outlook 2025 reference case from 2030, interpolated in between and shifted to the model's price definitions (U.S. Energy Information Administration, 2026b,a). The AEO high- and low-price cases are sensitivities. Fuel demand in 2026–27, electricity prices and inflation follow the same sources.

5.2 Estimation

Four estimated parameters govern the dynamic results, which are credit prices, when California's bank runs out and its cap binds, and the payments at the cap. The cost of the stack and the incidence of a federal dollar move most with capitalization, EPA's percentages and the fuel-demand elasticities, which are not estimated (Section 6.5). The data identify the four estimated parameters unevenly. The response of the credit price to a balance surprise, $\gamma \ln(1 + r)$, is identified from quarterly prices and banks in California and from the forward quotes. The holding cost r is set from the quotes, and γ is reported conditional on it. The bank cover K at which holders price exhaustion rests on three annual shift terms, because no program has approached exhaustion. The investors' continuation probability p is identified mainly from 13 decisions in the 2017–19 lapse and sits at its bound. The delivery slopes are estimated on twelve quarters of deliveries since 2023 and are varied from half to twice California's value. Credit holders behave as if they extrapolate the current balance, and investors face entry costs several times engineering cost. Both rules are tested on years outside their estimation samples. I estimate two sets of expectations, those of credit holders about the end of their bank and those of investors about the policy stack. The credit supply and the margins come from the historical version of the fuel market model, and Appendix D gives the details. Credit holders expect the bank to run out in some year, and the credit price rises toward the cap as that year approaches. Under the price-expectation rule they project net credits forward from the recent past and revise the expected year when the balance surprises them. I estimate the rule by maximum likelihood on California's quarterly prices and banks for 2016–25, together

with forward quotes for 2022–25, and on Oregon's and Washington's quarterly prices and banks, which share one pooled response, with $r = 3.38\%$.[10] A year with a megatonne more credits than expected moves California's expected exhaustion year out by about a year and lowers the price, since $\hat{\gamma}_{CA} = 1.108$ years per Mt (95% interval [0.948, 1.253]).[11] The forward runs set K at 2, inside the 95% interval [1.07, 2.28] estimated with the three annual shift terms and the forward quotes. The main runs use this specification, whose shift terms absorb the years in which the gap between the price and the statutory path opened. Without the shift terms the interval is [2.54, 3.47], which excludes $K = 2$ (Appendix D.1). The value is also in line with other banked credit markets, where prices began to price scarcity one to two years before a bank ran out (Appendix D.1). The closest the sample comes to exhaustion is 2019–21, when the bank covered less than a year of deficits and the price sat within about a fifth of the cap. How early the price begins to rise before a bank runs out is otherwise extrapolated, and K is the parameter that extrapolation rests on. Figure 9 reports other values of K.

The test separates the two things the rule does. The rule is re-estimated each quarter on data up to that quarter and given the net credits that actually occurred after the origin. It then beats a no-change forecast of California's price one, two and three years ahead (Figure 5). Its root mean squared errors are \$40, \$79 and \$108 a tonne. A no-change forecast on the same origins, 2019Q4–2024Q4, has errors of \$48, \$95 and \$138. The rule gets the direction of the change right at 86 to 100 percent of origins (Appendix D.3). This conditional forecast could not have been made at the time. It is the one the model relies on, because in the counterfactuals the balance comes from the fuel market, which is tested on its own (Section 5.4). When the rule must also forecast the balance from the observed state, it does worse than no change at every horizon. The reason is in the rule. With expected net credits at their three-year average it can only predict a rise at the carry, and over these origins the price fell at almost every horizon. The test shows that the rule maps a known balance into prices without forecasting the balance, which is the division of labor the model assumes. In Oregon, whose credit pool is about a tenth of California's, both forecasts do worse than no change. Washington's history is too short for the test. From the end-2025 state it forecast a 2026 price of \$68.0, against \$55.5 from a constant forecast and \$64 observed from 1 January to 18 September 2026. The rule describes how holders revise the expected exhaustion year in response to the balance. It does not explain why the gap of 2023–25 opened. At the bank's fundamentals the 2024–25 price corresponds to holders giving the standard roughly a one-in-three to one-in-two chance of binding as written, against about one in 2019–21 (Appendix D.2). It ignores news about the regulator, which moves prices (Section 2.1). Appendix D.5 estimates that holders priced about

[10] The holding cost is the real carry in forward quotes for adjacent December deliveries, and its 95% interval is [2.94, 3.83]%. The data identify the product $\gamma \ln(1 + r)$, which is what the counterfactuals use. Appendix D gives the forward-quote evidence and the alternative rules.

[11] The smaller programs share a response of 0.159 per percentage point of the obligation, with a 95% interval of [0.115, 0.201].

a sixth of a regulator's announced change. It is the descriptive rule that fits best among those I test (Appendix D). I also report the main results under the price-level bound (Section 7.2).

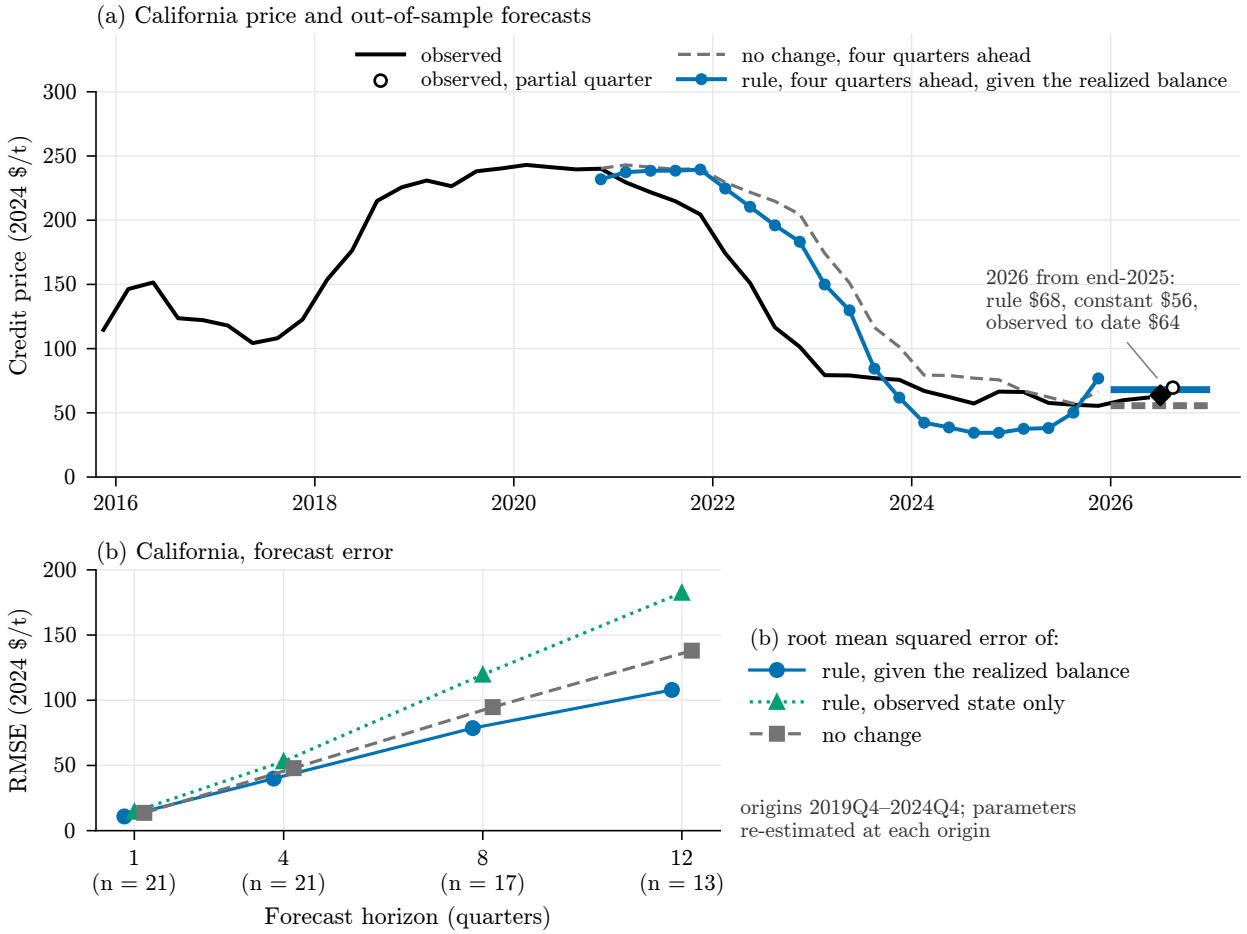


Figure 5: Out of sample, the rule forecasts California's price better than no change when given the realized balance, and worse from the observed state alone

Notes: (a) California's quarterly credit price and the rule's four-quarter-ahead forecast given the realized balance, against no change, at the target quarter. Origins 2019Q4–2024Q4, parameters re-estimated at each origin. Bars: one-step 2026 forecasts from end-2025 (rule $68, constant $56) and the 2026 price to date ($64). (b) Root mean squared error by horizon, on 21, 21, 17 and 13 origins at 1, 4, 8 and 12 quarters. 2024 dollars. Sources: CARB LCFS data and credit transfer log, CPI-U.

The second expectation is investors'. I estimate a competitive, single-agent optimal-stopping logit, as in Rust (1987), on the 46 capacity decisions, with the expected margin of each investor-year computed by the historical model from information public at the time. Entry costs are $20 per gallon of annual capacity for conversions, 10.5 for expansions and new plants and 3.0 for SAF retrofits (Appendix Table 15). For conversions they are about 11 times engineering cost, and for new plants 3 times. The estimate is best read as a hurdle. Besides construction it absorbs permitting, feedstock contracts, the cost of financing first-of-a-kind plants and the risk that the policies change (Appendix D.7 compares it with reported hurdle rates). I do not count the excess over capital cost

as a social cost, and I report capacity fixed at its main path as a bound. Estimated on 2014–19, the rule predicts the investment of 2020–25 better than a constant hazard or a static logit. At engineering cost the rule would predict 82 refinery conversions by 2045, against 16 to date, and a likelihood-ratio test rejects it (Appendix D.7). Because the binding RFS holds biofuel volumes fixed, the extra capacity is mostly idle. Current law then costs $156 a tonne instead of $133 on the excise-only definition, while the results for a 45Z extension do not change.

Investors value a tax credit that is in law as if it continues and a credit that has lapsed almost as if it were gone. The lapses of the blenders' tax credit, which cover 15 of the 46 decisions, separate the two. The estimated probability of continuation is 1 while the credit is in law, with a 95% interval of [0.67, 1.00], and close to zero during lapses. The blenders' tax credit's history fits this reading, since it lapsed four times and was restored retroactively each time. As in Liu et al. (2018), the tax credit is a switching policy, but where that paper sets the switching probabilities, I estimate p from the 46 decisions. Almost all of the identification comes from 13 decisions in the 2017–19 lapse. I also report the counterfactuals with a one-half probability of continuation ($p = 0.5$). A 45Z extension changes a credit that is in law, and investors who already value it as continuing see little news in it.

5.3 Other parameters

The remaining parameters are calibrated, taken from the literature or set by statute. Appendix Table 10 lists the ones that matter most for the results and where each value comes from, and Appendix C gives the details. The levels of the hydroprocessing utilization curve, the feedstock-mix intercepts, the corn oil supply and the RIN bank are set to 2025, the last complete year. The RIN bank is 3.21 billion RINs, from EPA's published bank and 2025 flows. A single year identifies a level but not a slope (Heckelei and Wolff, 2003, pp. 29–30), and slopes come from longer histories. The delivery slopes into the clean fuel states, for example, are estimated on quarterly data since 2023. In 2026 the credit prices are pinned at their observed levels and four 2026 targets identify four terms that fade out by 2028: the renewable diesel delivery terms of California, Oregon and Washington, and the supply of imported fat. RIN prices are outputs of the model.

5.4 Fit to history

The model tracks California's credit price better than a random walk, and six of eight 2026 observations fall within tolerance. The D6 RIN price is the main miss. Run on the history of 2017–25 from start years 2017, 2019, 2021 and 2023, with the oil prices, capacity and standards that occurred, the model tracks California's credit price with a smaller error than a random walk (29.6 against 43.9),

although with nine annual errors the difference is not significant.[12] For Oregon its error is larger than the random walk's. RIN prices and fuel quantities also track history better than a random walk and the Annual Energy Outlook. None of the 2026 observations was used as a target. The two misses are the D6 RIN price, before and (on EPA's September data) after its 31 August decisions on small-refinery exemptions, which lower RIN demand, and the fat share of California's renewable diesel, which fell with the new origin rule and tariffs, a channel the model does not include. The model's 2026 D6 RIN price, $2.29, is above the observed volume-weighted average of $1.54 for 1 January–27 September and $1.92 for September (2024 dollars, Appendix Figure 15). A larger RIN bank or larger exemptions would close the gap. In the run with larger small-refinery exemptions, whose 2026 D6 price of $1.71 is the closest to the observed, the mandate still binds. There the 45Z extension sends 0.756 of each federal dollar to fuel buyers and raises emissions by 67.5 Mt. In the main run the figures are 0.751 and 66.6 Mt (Appendix G.5). Appendix E reports every comparison.

6 Cost and Incidence of the Policy Stack

This section reports what each layer of the stack buys. All results use the estimated expectation rule, cover 2026–45, and value costs at present value at 2% over undiscounted tonnes, including indirect land-use change. The cost per tonne counts the resource cost and the fuel-tax and California and Washington allowance revenue that the policies forgo. The tables call this "E+A", and the version that counts the forgone fuel tax only "E". Appendix Table 24 lists, for each headline number, the parameters that drive it, whether they are estimated, calibrated or a convention, and how far it moves across the alternatives.

6.1 What the stack buys

Current law abates carbon at an average cost a fifth above that of a carbon price at the social cost of carbon, $149 against $125 a tonne. The federal mandate abates more than the state standards, at about half their cost per tonne. Figure 6 shows the main run. The federal mandate binds in every year. California's bank runs out in 2031, and its price reaches the cap in 2032 and stays there (Figure 4). Oregon and New Mexico defer steps of their standards from 2033.[13] Washington keeps its statutory 20% limit. Its standard stops tightening in 2030 while electricity and renewable fuel credits keep growing, and its price falls from about $50 to $6 by 2045 (Figure 6b and Appendix G.1). The D6 price is $2.29–2.45 in 2026–28 and falls to $0.95 in 2045. In the four states the electric share of new light-duty sales rises from 20% to 47%. Elsewhere it rises from 7% to 12%. Appendix G.1 describes the path in more detail.

[12] The rolling replays were computed on the previous version of the model. On the current version, a single replay of 2016–25 at observed capacity gives an error of 38.3 for California's price, against 46.9 for a random walk (Appendix E).

[13] The timing rests on Oregon's rule, which forecasts worse than a random walk one quarter ahead, and on New Mexico, which has no price history.

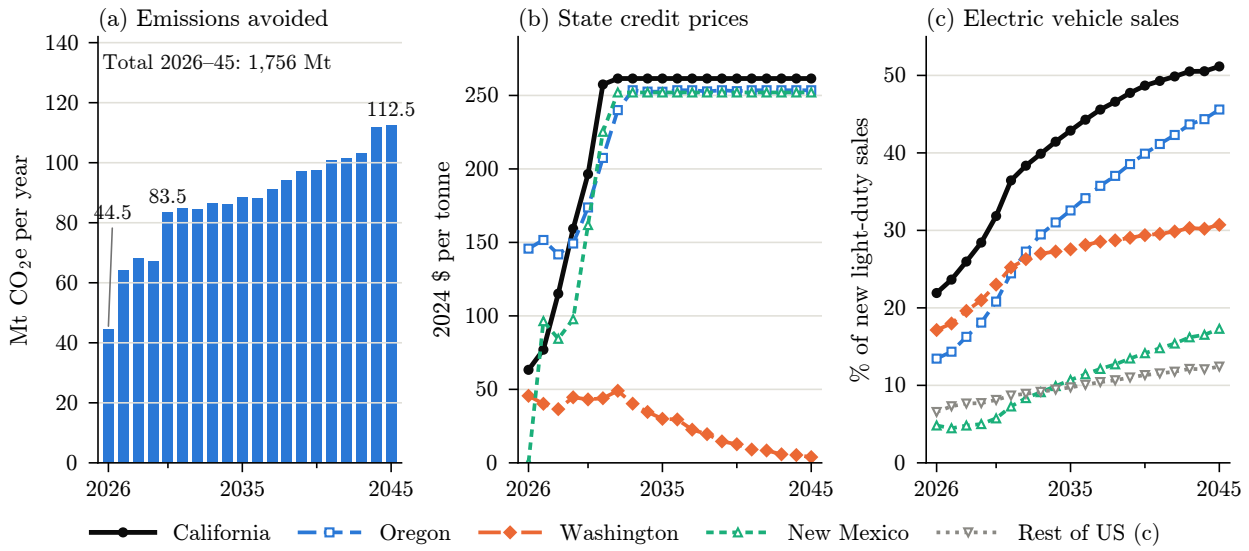


Figure 6: Current law, 2026–45: emissions, state credit prices and light-duty electric vehicle sales

Notes: (a) U.S. lifecycle emissions avoided relative to no fuel policy, including indirect land-use change (Mt CO_2e a year), 1,756 Mt in total. (b) State credit prices, 2026 observed. Washington's price falls because its standard stops tightening at 20% (Appendix G.1). (c) Electric share of new light-duty sales. 2024 dollars. Biofuel volumes: Appendix Figure 18. Source: model runs.

From 2032 the fuels and vehicles in the model fall short of California's standard at the cap price. California's shortfall at its cap grows from 11 Mt in 2032 to 62 Mt in 2045. Valued at the cap, it grows from $2.9 billion to $16.1 billion a year. The compliance options in the model cannot close the gap, because the standard falls to 90% below 2010 by 2045 and from the late 2030s renewable diesel and corn ethanol themselves generate deficits. After 2032, then, each tonne of carbon intensity above the benchmark costs sellers the cap price, $262. The model closes each year by issuing the shortfall as credits at the cap. The statute instead carries an unmet obligation forward with interest and requires it to be retired within five years, which cannot happen when the cap binds every year. The two treatments give the same prices and quantities if the cap binds in every year from 2032 and fuel sellers treat the cost of a deficit as the cap price. The choice then decides only whether the money is held by the state or remains an unpaid obligation of fuel sellers. Appendix Figure 19 shows how much of the gap restoring Advanced Clean Cars II, California's zero-emission vehicle sales rule, or direct air capture under an extended 45Q would close. Under the statute's carry-forward, with no credits issued at the cap, the unpaid obligation of fuel sellers would reach 722 Mt by the end of 2045, against a cumulative shortfall of 536 Mt, because carried deficits accrue interest. Of that obligation, 399 Mt would be past the five-year repayment limit.

I compare current law with a run without fuel policy, which removes the RFS, the four state standards, 45Z, 45Q and the state biofuel incentives from 2026 while keeping existing capacity, banks and vehicle incentives. Over 2026–45 current law abates 1,756 Mt (1.8 Gt). It costs $149 a

tonne.[14] At the ends of the oil-price range, abatement is 898 Mt with high oil prices and 3,176 Mt with low oil prices.

Current law passes a cost–benefit test at 2 percent with EPA's values and fails it at 3 percent with the interagency values. Its break-even social cost of carbon is $169 a tonne in 2030 (2024 dollars), about the same under each discounting convention. EPA's 2023 values for 2030 are $279 at a 2% discount rate and $170 at 2.5% (U.S. Environmental Protection Agency, 2023). The Interagency Working Group's 2021 interim value at 3% is $75 (Interagency Working Group on Social Cost of Greenhouse Gases, 2021). At 2% with EPA's values, current law's net welfare including climate damages is $170.6 billion, and its break-even value is 0.61 times EPA's. At 3% with the interagency values it is −$130.8 billion, or −$141.4 billion if allowance prices follow their floor path. Discounting at 3% while valuing climate on EPA's 2.5% path gives $5.8 billion. On the floor path it gives −$4.8 billion. At 2% with EPA's values, a carbon price at the social cost of carbon would abate at $125 a tonne on average and would achieve about eight times (E+A) the net benefit, mostly because it abates far more.

Against current law without the RFS, the federal mandate abates 627 Mt at $132 a tonne (Appendix Table 32). Against current law without the state standards, the four state standards abate 521 Mt at $284 a tonne. Removed one at a time, the layers do not add up to the total. The federal mandate accounts for 627 Mt, the state standards for 521 Mt and 45Z in 2026–29 for −15 Mt. The remaining 623 Mt comes from the state biofuel incentives and the interaction between the federal and state standards, which the runs do not separate.

Over 2026–45 the state standards cut emissions inside the four states by 1,094 Mt and raise them in the rest of the country by 573 Mt. National emissions fall by 521 Mt (Figure 7a).[15] Renewable diesel and biodiesel drawn from other states account for 600 Mt of the in-state total and contribute nothing nationally. Nationally, 310 Mt comes from higher fuel prices in the four states and 212 Mt from electric vehicles net of charging emissions. With the standards the country uses 1.4% less biofuel, while the four states use more than twice as much (Appendix Table 33). Each in-state tonne the standards record yields about half a national tonne. The relocated share, one minus the ratio of national to in-state abatement, depends on the delivery slopes. These measure the cost of bringing more renewable diesel into a state, and they are the parameters the data identify least precisely. At half and twice California's estimated slope, and at the ends of its absorption kink, the share stays between 49% and 53%. National biofuel is pinned by the mandate and the supply of fats and oils, so almost all of each additional gallon used in the four states comes from the rest

[14] Appendix Table 24 and Appendix I give the cost on the other definitions, from $118 (resource cost only) to $186 (levelized).

[15] The split of the change in fossil fuel use between the price and electric vehicle terms in Figure 7 allocates their interaction by log shares and is approximate. The electric vehicle term includes the growth in electric vehicles induced by fuel and credit prices. Renewable natural gas and hydrogen credits are not modelled in emissions. "Other biofuels" are renewable diesel and biodiesel not relocated, ethanol and SAF, and they fall slightly.

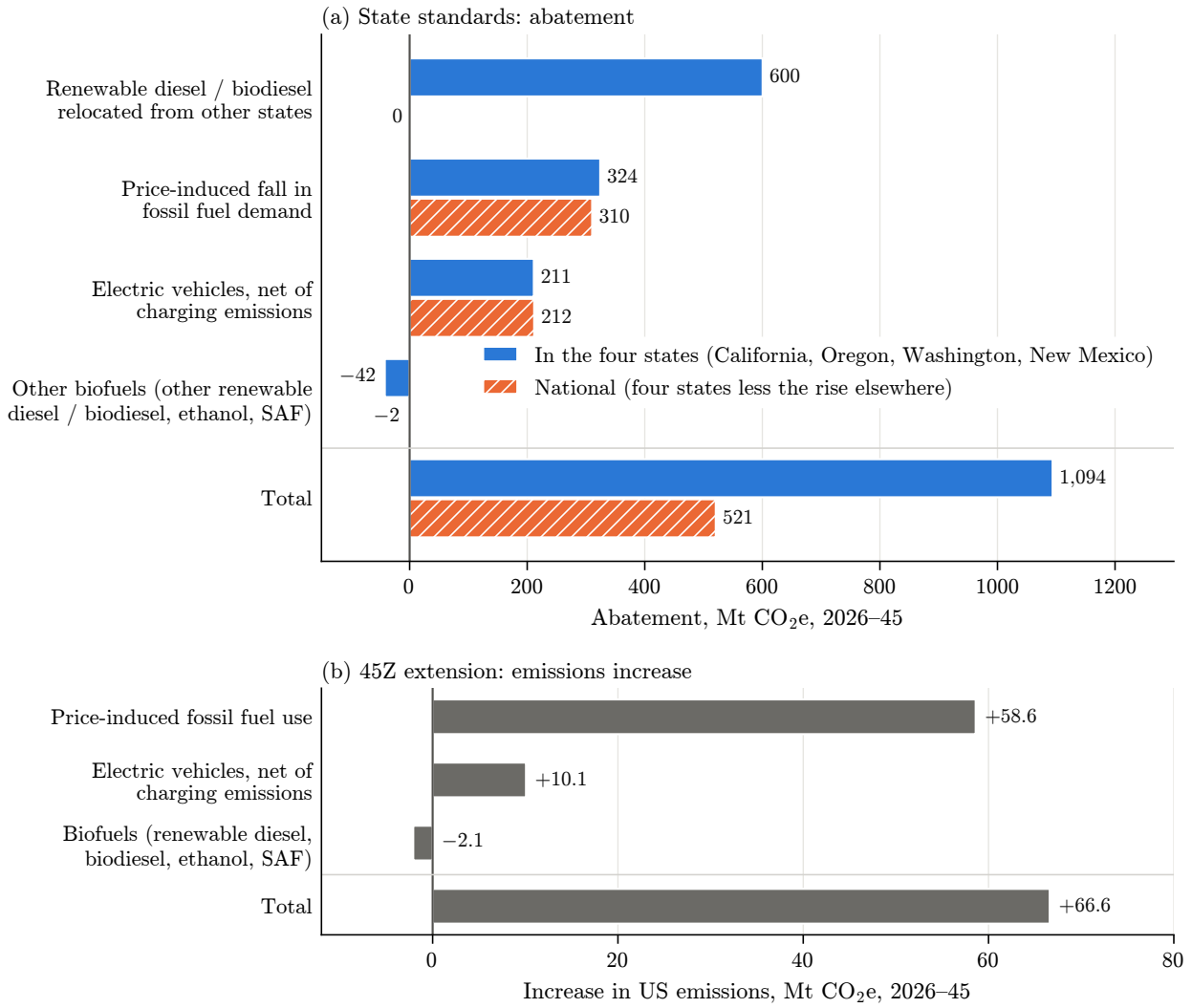


Figure 7: Sources of the emission changes from the state standards and from the 45Z extension, 2026–45

Notes: (a) Current law against current law without the four state standards: emission changes in the four states and nationally (the four states less the rise elsewhere). (b) The 45Z extension against current law: change in U.S. emissions. Mt CO_2e over 2026–45, undiscounted, lifecycle, by consumption state. The split between fuel prices and electric vehicles is approximate (footnote 15). Source: model runs.

of the country. A steeper California slope raises California's credit price before its cap binds, and more credits are issued at the cap after it, while the volume of renewable diesel that moves changes little. California reaches its cap in 2031–32 at every slope. The relocated share depends more on the fuel-demand elasticities, which set the price term. Halving them or raising them by half moves it between 61% and 46% over 2026–45, while the renewable diesel and biodiesel relocated stay at about 600 Mt.

For the nation, the state standards work like a fuel tax in four states combined with a vehicle subsidy. As explained in the introduction, on climate damages alone the standards lose money

whether or not California's and Washington's caps bind. At zero offset the loss is $20.4 billion. The margin is thin at EPA's values, since the state standards' break-even social cost of carbon is $323 a tonne in 2030, 1.16 times EPA's. At a 3% discount rate the loss is $61 billion with EPA's 2.5% path and $101 billion with the interagency values. Under the floor and ceiling paths for allowance prices it is $30 billion and $107 billion at 2%. The climate value of their abatement, $127.3 billion, about covers their resource cost including the forgone fuel tax, $127.7 billion. The loss is the $19.9 billion of allowance revenue that California and Washington forgo when covered fuel use falls, valued at 2025 allowance prices (Appendix Table 29). Under the floor and ceiling price paths the forgone revenue is larger. These figures leave out the congestion, accident and local air pollution costs avoided when fuel use falls in the four states, which the welfare accounts report separately. For the state standards they are $22.9 billion.[16] Counting them, the standards about break even at 2%, with a net benefit of +$2.6 billion and a break-even social cost of carbon 0.98 times EPA's. Across the range of those costs the net benefit is between −$12.1 billion and +$21.1 billion. At 3% it is −$40.8 billion with EPA's 2.5% path and −$80.5 billion with the interagency values.[17] Suppose the cap-and-trade programs cancel a share of the standards' covered abatement (the offset), as a waterbed would. The standards then raise national emissions once the offset exceeds 63% (Perino et al., 2019). Perino et al. (2019) derive an offset of 20% from the estimates of Borenstein et al. (2019). At any offset between zero and one the standards' net benefit stays below zero. Current law as a whole loses its net benefit once the offset exceeds two thirds.

Table 1 shows who gains and who pays. Fuel users in the four states pay $339 billion over 2026–45 for the state standards. Fuel users elsewhere gain $12 billion. Biofuel producers gain $76 billion, credit holders $33 billion and buyers of electric vehicles in the four states $19 billion. Feedstock suppliers lose $12 billion, because the standards move renewable diesel into the states without raising its output. The four states' budgets receive $104 billion from credits issued at the caps in the model. Under the statute this is an obligation of fuel sellers that may never be collected. For current law as a whole, fuel users pay $593 billion, and U.S. farmers gain $86 billion. Every layer that raises fuel prices is regressive as a share of income, and much less so as a share of spending (Appendix G.3).

6.2 A federal dollar inside the binding layer

A 45Z extension adds almost no biofuel. It lowers the price of the mandate's credits and so the price of fuel, and when the credit is fully capitalized 75 cents of each federal dollar reaches fuel buyers. The additional support I study is an extension of 45Z beyond 2029, through 2045, at

[16] Current law against current law without the state standards, 2026–45: present value at 2% to 2025, E+A basis, EPA's 2% social cost of carbon and allowance prices held at 2025. Appendix Table 28 is on the excise-only basis, compares each policy with current law and has no row for the state standards.

[17] With the interagency values the range across those costs is −$93.5 to −$64.1 billion.

Table 1: Who gains and who pays, 2026–2045 ($ billion, present value at 2%)

	State standards	Current law (whole stack)	45Z extension
Fuel users, the four states	−338.7	−374.9	4.6
Fuel users, rest of US	12.3	−218.2	42.5
Vehicle buyers, the four states	18.9	18.4	0.0
Biofuel producers	75.7	118.3	2.2
Feedstock suppliers, US	−11.7	104.8	8.7
of which U.S. farmers	−13.2	86.0	6.9
Credit holders and other credit generators	33.2	54.4	−1.9
Federal budget	−4.1	−18.7	−54.9
State budgets, the four states: credits issued at the caps (model convention)	104.4	104.4	0.9
State budgets, the four states: excise and other lines	−17.9	−15.9	0.2
State budgets, other states	−0.5	−7.2	1.2
State budgets, CA / WA: lost allowance revenue	−19.9	−28.0	−0.1
Other groups (small)	0.7	1.0	0.0
Total: economic welfare (= − net cost)	−147.6	−261.6	3.4
Climate damages avoided (EPA SCC)	127.3	432.2	−16.3
Net welfare including climate	−20.4	170.6	−12.9
Abatement (Mt CO_2e; + = abated)	520.8	1,755.6	−66.6

Notes: Columns: the state standards against current law without them, current law against no fuel policy, and the 45Z extension against current law. Positive: the group gains. Total: economic welfare, minus the net cost including forgone fuel-tax and California and Washington allowance revenue (E+A). U.S. farmers are part of feedstock suppliers. Climate damages at the EPA SCC (2%). Every row and the details: Appendix Table 19.

its current-law rate, measured per dollar of its federal cost. Its federal credit outlays are $54.0 billion. The total RFS requirement binds in every year, which puts the extension in the case of Proposition 1 in which the mandate binds. The mandate binds because the RIN bank is at its floor from 2030 and, without RIN value, biofuel would cost more than the fossil fuel it replaces on the main oil-price path. On the record it bound in 45 of the 50 quarters from 2013Q4 to 2026Q1 that can be classified. The extension raises biofuel output by 0.41 billion gallons over twenty years, 0.11%.

The tax credit replaces RIN value. The RIN value fuel buyers pay falls by slightly more than the federal cost, 1.15 times in the model plus 0.12 for non-road diesel buyers outside it. The main reason is that fuel from imported fat loses the credit and foreign fat suppliers bear part of it. Appendix G.5 gives the decomposition.

The result is conditional on the mandate continuing to bind. It binds in the model because EPA's percentages are held at their 2027 level and agents expect the stack to continue. If EPA lowered the percentages until the mandate were slack, the extension would fall in case (iii) of Proposition 1 and would buy biofuel. I compute that case by lowering EPA's percentages from 2030 until the total requirement is slack. The 20% limit on carrying RINs into the next year is switched off, a simplification that matters only for RINs worth about $0.01 billion drawn in 2044–45. The extension then sends 81 cents of each federal dollar to fuel buyers and raises emissions by 54 Mt over 2026–45. In the main run the figures are 75 cents and 67 Mt. The rise comes from cheaper fuel, as gasoline use rebounds. The additional corn ethanol the extension buys emits about

30% less per gallon than the gasoline it displaces, including land-use change, so on its own it lowers emissions. The saving per additional tonne emitted is $103, against $51 in the main run.

Table 2 divides the federal cost among the groups that receive it. Fuel buyers in the model receive 75 cents of each dollar. Buyers of non-road diesel receive 12 cents. Feedstock suppliers receive 16 cents and biofuel producers 4 cents. Across 184 bootstrap draws of the estimated parameters the share to fuel buyers has a standard error of 0.004. This covers estimation uncertainty only. Capitalization, EPA's response and the calibrated parameters are varied separately below. If the supply of North American animal fat is kinked at the levels that data from USDA's National Agricultural Statistics Service (NASS) imply, the share to fuel buyers falls to 0.65–0.66 and the share to feedstock suppliers rises to 0.28–0.29. The cost of the stack then moves by less than a dollar a tonne.

Table 2: Incidence of a 45Z extension, 2026–2045 (shares of the federal cost; present value at 2%)

	Capitalized share of the credit			Bootstrap
	1 (main)	0.75	0.50	s.e.
Share to fuel buyers in the model	0.751	0.565	0.379	(0.0041)
RIN transfer to non-road diesel buyers	0.122	0.091	0.060	
Feedstock suppliers	0.161	0.126	0.087	(0.0007)
Biofuel producers (net of capital costs)	0.042	0.031	0.017	(0.0030)
Biofuel producers: unpriced credit (windfall)	0.000	0.250	0.500	
Net resource cost	−0.066	−0.055	−0.037	(0.0051)
Federal cost F ($ billion)	54.0	53.4	52.8	(0.01)
Emissions change (Mt)	+66.6	+49.7	+33.2	(0.6)
Net welfare incl. climate (E+A) ($ billion)	−12.9	−9.4	−6.3	(0.2)

Notes: Extension of 45Z from 2030 to 2045 against current law. The Treasury pays the credit in full, and the part not capitalized is a windfall to producers. Net welfare counts forgone fuel-excise and California and Washington allowance revenue (E+A). Bootstrap s.e.: estimated parameters only. Every row, the lower bound 0.083 and investors' extension probability $p = 0.5$: Appendix Table 21.

How much of the dollar reaches fuel buyers depends on how much of the tax credit is capitalized (defined in the introduction). The 75 cents is what a binding mandate implies for a credible, permanent credit in a competitive market, with RIN prices passed fully into fuel prices, as the evidence on wholesale prices supports (Knittel et al., 2017). The record of the blenders' tax credit implies less. Over 2014–21 a dollar of credit moved the D4 price by −0.41 (s.e. 0.13) in the 86 months in which the mandate bound. Two measures of traders' expectations of reinstatement imply that 40–50% of the credit was capitalized. A third, from weekly transaction prices, is consistent with full capitalization (Appendix G.4). The blenders' tax credit lapsed repeatedly and was restored retroactively. Its value was uncertain when fuel was sold. A standing credit on production may be capitalized more fully, and the record cannot settle by how much (Irwin et al., 2020). Table 2 and Appendix Figure 21 report the extension when producers and the RIN market price only part of it, the capitalized share, while the Treasury pays it in full. With half capitalized, 38 cents of each federal dollar reach fuel buyers and emissions rise by 33 Mt, and half the credit stays with

producers as an unpriced windfall. With three quarters capitalized the figures are 57 cents and 50 Mt. The extension adds almost no biofuel at every degree of capitalization, and the size of every other effect scales with the share that is priced.

Capacity barely changes. No capacity constraint binds in any year of the main runs, apart from the small fixed alcohol-to-jet capacity, and biofuel output is held by the mandate. Under current law itself hydroprocessing capacity grows from 4.9 billion gallons in 2026 to 9.3 billion in 2045. Output stays between 4.1 and 4.5 billion gallons, and utilization falls to 0.45. Capacity never binds, and the fixed cost of the idle capacity, $4.6 billion in present value, is counted as a resource cost. Much of the idle capacity comes from the form of the estimated entry rule. The logit entry model keeps a small probability of entry at any utilization, while the value of a gallon of capacity to an investor falls as utilization falls (Appendix B.4). Because capacity never binds, the idle plants change no price or emission result and enter welfare only through their fixed operating cost. With the extension, hydroprocessing capacity rises by 0.06 billion gallons a year on average over 2026–45, 0.8%. Capacity that can make SAF falls by 0.12 billion gallons a year, and capacity that makes only renewable diesel rises, mostly late in the period. Investors who already value the credit as continuing receive little news from the extension. With a one-half probability of continuation, capacity rises by 0.11 billion gallons a year, 1.7%. With none, it rises by 0.20 billion gallons, 3.0%. Because the binding RFS fixes biofuel output in every case, investors' expectations barely move the shares (Appendix Table 21).

Because the extension makes fuel cheaper, fuel use and emissions rise. Over 2026–45 emissions increase by 67 Mt, 62 Mt of it outside the four states (Figure 7b). Fossil fuel burned at lower prices adds 59 Mt and fewer electric vehicles add 10 Mt, while biofuel lowers emissions by 2 Mt. The increase is small, 4 Mt a year, consistent with the small rebound from cheaper blended fuel that the literature expects (Gerveni et al., 2026). The extension saves $3.4 billion on the main measure, $51 for each extra tonne emitted. On resource cost alone it saves $1.5 billion, $23 a tonne. Cheaper imported fat accounts for $3.7 billion, more than the whole saving. This is a terms-of-trade gain. From a global standpoint the extension saves no resources. The saving rests on the calibrated supply elasticity of imported fat. With an elasticity of 2 it is $82 per additional tonne emitted, and with 10 it is $35 (Appendix Table 24). The $1.6 billion of duty the Treasury loses as fat imports fall is a transfer. Valued at the EPA social cost of carbon, the additional emissions cost $16.3 billion, and the extension lowers welfare by $12.9 billion. The state credit markets play a small part. The transfer through the four state credit prices is less than 1% of the federal cost.

Figure 8 shows what happens when EPA raises the requirement alongside the extension. Emissions stop rising at an increase of 2.65%, where fuel buyers receive 40% of the federal cost and feedstock suppliers 29%. The pair lowers welfare at every increase. The extension's own effect stays between −$12.9 and −$13.5 billion at every increase, and the smaller total loss at larger increases comes from the higher requirement. A higher requirement alone reaches the 100 Mt tar-

get at $175 a tonne, while the extension paired with the full offset costs $272 a tonne (Table 3). Appendix G.5 gives the shares at each increase and the case in which EPA cuts the requirement.

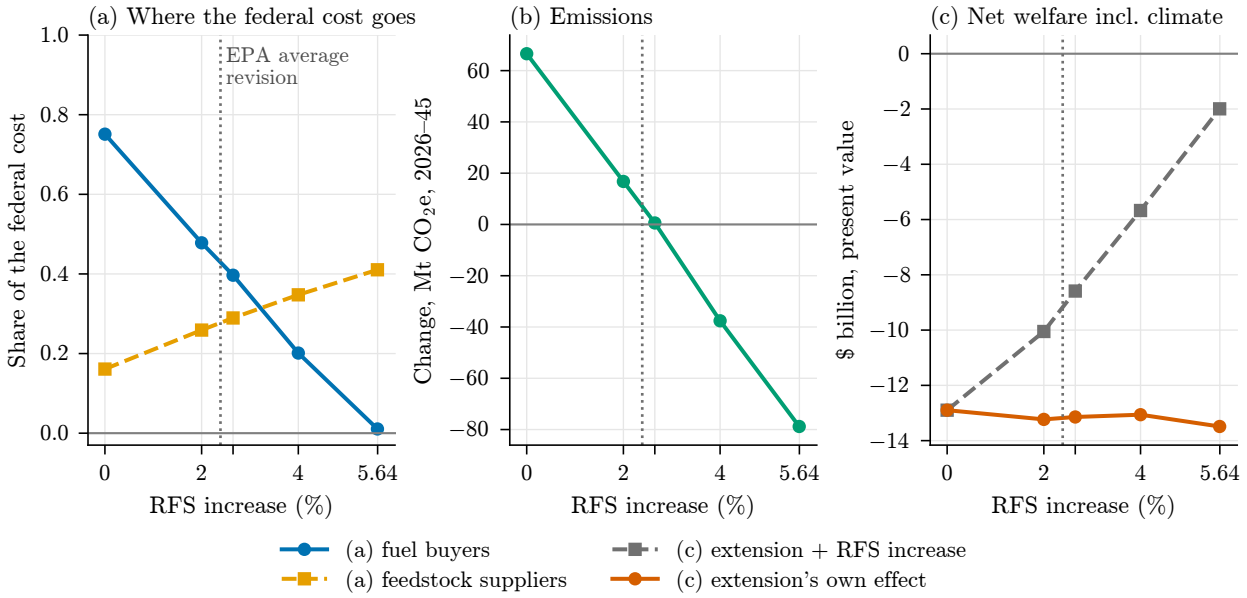


Figure 8: What the 45Z extension does as EPA raises the RFS requirement, 2026–2045

Notes: The 45Z extension paired with an RFS requirement raised by 0, 2, 2.65, 4 and 5.64%, each against current law. At 2.65% (the unlabelled tick) emissions stop rising, and 5.64% returns the average 2030–45 D6 price to its current-law level (full offset). Panels: (a) shares of the federal cost, (b) change in emissions, (c) net welfare including climate damages (E+A, $ billion at 2%). Dotted line: EPA's average revision from proposal to final rule, 2014–20 (2.39%).

6.3 A federal dollar that reaches the state layer

This subsection asks what a federal dollar for electric vehicles buys when it reaches the state credit markets. Every vehicle charged in one of the four states earns credits, so a federal dollar spent on chargers or vehicles adds to the bank. While the bank is large, the dollar lowers the state credit price and adds no abatement in the state, the cases of Proposition 1 in which the state layer is reached. Table 3 reports a credit of 50% of the installed cost of fast chargers and a purchase credit at the former level of the Inflation Reduction Act, $7,500 per light-duty vehicle under Section 30D. Both are paid in 2028–32.

The fast-charging credit abates 172 Mt over 2026–45 at $59 a tonne, for federal credit outlays of $8.2 billion. It rests on two modelling assumptions. When the credit ends in 2032, stations' user cost falls to a floor of 10%, so entry is brought forward into the credit's last year. Stations do not depreciate, and the stations built in 2032 keep raising electric vehicle demand to 2045. With stations depreciating at 10.7% a year the credit abates 121 Mt, and phased down over 2031–32 it abates 150 Mt. With a 5% floor it abates 303 Mt (Appendix H.5). Of its 172 Mt, 159 Mt (92%) is abated outside the four states, almost all through additional electric vehicles. The change in state

credit prices accounts for less than 3 Mt, because California and New Mexico are at their caps from 2032 in both runs. Its rank as the second cheapest instrument in the table rests on the value of a larger charging network to buyers of electric vehicles, most of whom would buy one anyway. That value rests on the estimated network effect and on the price coefficient that converts it into dollars, and both are uncertain. With a weaker network effect the credit costs $157 a tonne and falls behind the purchase credit. Appendix H.1 gives the other specifications. Stations are built ahead of the credit's expiry in 2032 and cannot be withdrawn, and the resource cost includes the losses of the stations that do not recover their cost.

The purchase credit abates 435 Mt at $101 a tonne, for federal credit outlays of $105.3 billion. Seven cents of each federal dollar reach fuel buyers through the lower credit price. Appendix Tables 36 and 37 report the electric share of new sales by class and region under each policy.

6.4 Moving the binding layer

Table 3 compares instruments that relax the binding mandate with instruments that pay inside it, at a common target of 100 Mt of additional abatement over 2026–45. A carbon price of about $2 a tonne on transportation fuel reaches the target at an average cost of $36 a tonne. A higher RFS requirement, raised by 3.94%, reaches it at $175 a tonne. Almost all of the carbon price's cost is the fuel-tax revenue lost on gallons no longer burned. Counting resource cost alone, it costs $7.5 a tonne against $155 for the higher requirement. The ranking of the instruments in Table 3 is the same under every convention for the fuel-tax and allowance wedges and with the change in congestion, accident and local pollution costs counted (Appendix Table 28). It is also the same when the supply of imported fat is twice as elastic.

The existing 45Z credit of 2026–29 falls in the same case as its extension. It raises emissions by 15 Mt and sends 71 cents of each federal dollar to fuel buyers.

The largest move of the binding layer replaces the stack with a single national standard, which in the experiment becomes a national carbon price. In this experiment Congress announces a national low carbon fuel standard in 2028. From 2030 it replaces both the RFS and the four state programs. The standard declines at California's rate and has California's cap, $262 a tonne. It reaches the cap in 2031 and stays there. From then on it too is a $262 carbon price on carbon intensity above the benchmark. Over 2026–45 it abates 5,395 Mt at $98.8 a tonne. Current law abates 1,756 Mt at $149. The national standard's net welfare including climate damages is $784 billion (E+A), against $171 billion (E+A) for current law, a gain of $614 billion (E+A). With jet fuel obligated as well, it abates 5,910 Mt at $95.1 a tonne. Appendix G.7 gives the definition, the path at the cap and the version with jet fuel.

Table 3: The policy menu: abatement, cost and incidence by instrument, 2026–2045

Policy (against current law)	Abatement (Mt)	$ per t abated[b]	Net fiscal cost ($bn)[a]	Federal outlay F ($bn)	Share to fuel buyers
45Z credit as legislated, 2026-29	−14.9	−39	16.8	16.9	0.71
45Z extension	−66.6	−51	53.7	54.0	0.75
45Z extension with EPA offset (RFS +5.64%)	78.8	272	59.5	57.9	0.01
Fast-charging credit (credits 2028-32)	171.8	59	14.2	8.2	0.15
Purchase credit (credits 2028-32)	434.9	101	100.1	105.3	0.07
Higher RFS requirement (+3.94%), 100 Mt target	103.5	175	1.5	0.1	n.a.
Carbon price, 100 Mt target	100.4	36	−54.9	0.0	n.a.
Memo: current law against no fuel policy	1,755.6	149	69.8	17.0	n.a.
Memo: carbon price at the social cost of carbon against no fuel policy	11,640	125			n.a.

Notes: Each instrument against current law, present value at 2% to 2025 over undiscounted tonnes. Cost per tonne: resource cost plus forgone fuel-tax and California and Washington allowance revenue at 2025 allowance prices (E+A). [b]Negative: the resource saving per additional tonne emitted, for rows that raise emissions. F: federal credit outlays. Share to fuel buyers: share of F; n.a.: not applicable. [a]Federal and state budgets, excluding credits issued at the state caps. A national clean fuel standard replacing the RFS and the state standards from 2030 abates 5,395 Mt at an average $98.8 a tonne. Its 3,640 Mt beyond current law cost $74 a tonne (Appendix G.7, Table 23). Fast-charging credit: Appendix H.5. Other definitions: Appendix H.2. Blank: not reported; the carbon price at the social cost of carbon is from Appendix H.2.

6.5 Which inputs move the results

The three headline results move little across the inputs the data identify. They move most with capitalization, EPA's response and the calibrated demand elasticities. Figure 9 sets each input to the end of its range. The bootstrap band over the estimated parameters is narrow, with the 45Z share between 0.741 and 0.754. Halving or raising the fuel-demand elasticities by half moves the extension's extra emissions between 38 Mt and 94 Mt. It moves the cost of the stack between $140 and $161 a tonne. Ends that did not solve are omitted and listed in Appendix H.1. Appendix Table 27 reports every row used in the paper with its residuals, and every failure sits where a layer of policy switches between binding and slack. Appendix G.10 reports further sensitivity analyses, and Appendix G.3 the results at a 3% discount rate, with the interagency 3% values and with EPA's 2.5% path.

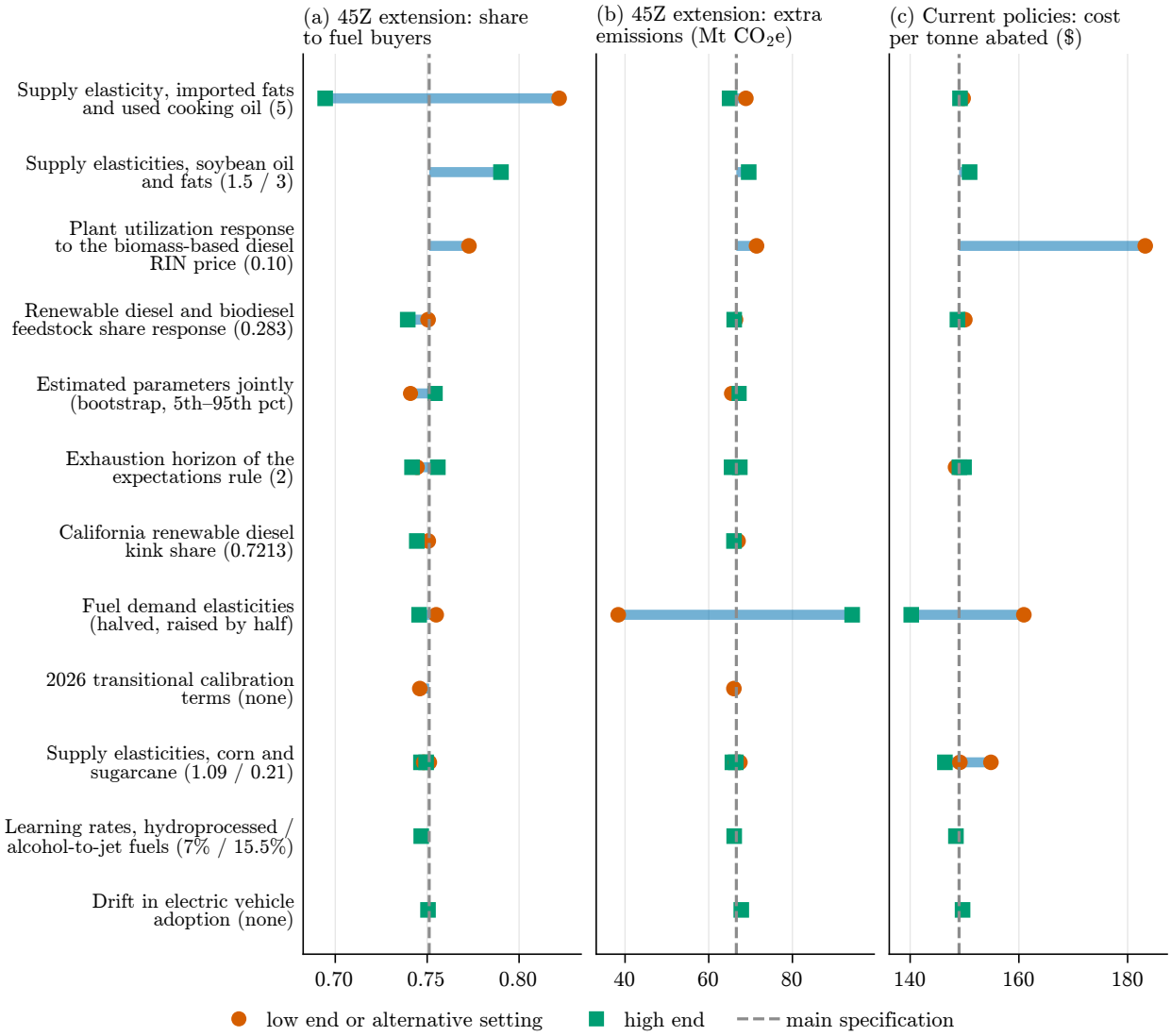


Figure 9: How the main results move with each input, 2026–2045

Notes: Each row sets one input to the end of its range, all else at the main values (dashed line). Panels: (a) share of the 45Z extension's federal cost reaching fuel buyers, (b) the extension's extra emissions (Mt), (c) cost per tonne of current law against no fuel policy (E+A, 2024 $). Orange circles: low end or alternative; green squares: high end. The bootstrap bar spans the 5th–95th percentiles over draws of the estimated parameters. Blank in (c): not run by design. Inputs that move no panel by 0.005, 1 Mt or $1.5: Appendix Figures 23 and 24.

7 Expectations and the Timing of Effects

7.1 Totals and timing

This section asks how much the results depend on how credit holders form expectations. Table 4 reports them under four descriptions of expectations: the estimated rule, a bound in which holders keep the price near its recent level (Section 7.2), and a storage model of California's bank under two regulators. One regulator follows the statute, and the other follows a rule estimated on CARB's past decisions, which shows what happens when the regulator acts on the market's doubt. Under the estimated rule California's bank runs out in 2031 and its price first reaches the cap in 2032. Under the price-level bound both happen a year earlier. Holders who extrapolate a large bank keep the price low, generate fewer credits and draw the bank down faster, as Corollary 1 implies. When the bank runs out, the price jumps to the cap.

Table 4: Results under four descriptions of expectations, 2026–2045

	Current law		45Z extension		California		OR / NM
Expectations	Mt	$/t	share	+Mt	bank out	at cap	deferral
Estimated rule (main)	1,756	149.0	0.751	66.6	2031	2032	2033 / 2033
Storage model, statutory regulator	1,754	148.8	0.749[a]	66.8[a]	2031	2031	2033 / 2033
Storage model, estimated regulator	1,471	125.5	0.710[a]	63.4[a]	never	never	2035 / 2035
Price-level bound	1,747	148.6	0.745	66.1	2030	2031	2033 / 2032

Notes: Current law against the no-fuel-policy run; the 45Z extension against current law under the same expectations. $/t: resource cost plus lost fuel excise and California and Washington allowance revenue (E+A), present value at 2%, per undiscounted tonne. Share: share of the extension's federal cost reaching fuel buyers. [a] 2030–45. Storage model, estimated regulator: mean over 200 paths, all converged. Price-level bound: holders expect a 10 Mt a year relaxation from 2027. Appendix Table 25 gives the ranges, the excise-only cost and the details.

The totals change little across these descriptions. The cost of current law is $149.0 a tonne under the rule and $148.6 under the price-level bound. The share of the 45Z extension reaching fuel buyers is 0.751 and 0.745. Pricing California's bank with a storage model under the statute as written gives the same totals as the rule and reaches the cap in 2031.

One description changes the totals, the one in which holders expect California to relax its standard and the regulator does. When California's regulator follows a policy rule estimated on its past decisions, it relaxes the standard by 2029 in every one of 200 simulated paths, and the bank never runs out. Current law then abates 1,471 Mt. Its cost is $110 a tonne on the excise-only definition and $125.5 with allowance revenue counted, against $149. If the market's doubt is taken at face value, the stack abates less, at a lower cost per tonne, and California's price never reaches the cap.

7.2 Credibility

This subsection asks how much doubt about the statute changes the results. The observed prices lie on the side of doubt, and I compute a bound. Suppose that from 2027 California's holders expect a permanent relaxation of the standard, worth D Mt of extra credits a year, and that the relaxation never comes. I choose D so that California's price in 2027–29 is close to its 2023–25 level of $67 a tonne. The bank, the market, the cap and the codified responses are as in the main run, and only expectations differ. The required misperception is 10.0 Mt a year, which gives an average price of $70.4 in 2027–29. Because holders expect credits that never arrive, they do not build the bank. The bank runs out in 2030 and the price reaches the cap in 2031, a year earlier than in the main run. The totals change little. Current law abates 1,747 Mt at $148.6 a tonne, and in the main run 1,756 Mt at $149.0. The share of the extension reaching fuel buyers is 0.745. The fast-charging credit costs $59 a tonne, as in the main run, and the purchase credit $103, against $101.

8 Policy implications and conclusion

U.S. climate policy for transportation fuel stacks a federal mandate, a federal tax credit and four state standards on the same gallon, each set by a different authority. Over 2026–45 the federal mandate binds every year. The layers above it therefore change emissions only through fuel prices and vehicle choice. Current law abates 1.8 billion tonnes of carbon dioxide at $149 a tonne, a fifth more per tonne than the average cost of a carbon price set at the social cost of carbon. The excess cost has two sources, the state standards and the way further abatement would be bought. At the margin a larger mandate costs about five times as much per additional tonne as a carbon price. Most of the carbon price's cost is fuel-tax revenue forgone, and on resource cost alone the ratio is larger still.

Congress and the Treasury. While the RFS binds, 45Z beyond 2029 is a transfer to fuel buyers or producers. Even if EPA raises the mandate to offset the extra emissions, the combination lowers net welfare. If the goal is abatement, a carbon price is the cheaper instrument. If the goal is farm income or domestic feedstock, which this paper does not evaluate, the tax credit is an expensive way to pursue it. Feedstock suppliers receive one dollar for every $6 of federal outlay, or for every $3.50 if the supply of animal fat is kinked. In present value at 2%, each additional gallon of biofuel the extension buys costs the Treasury about $170, and about $120 if ethanol, whose use falls, is left out. Each cent by which it lowers the average price of gasoline and diesel in 2030–45 costs about $26 billion.

EPA. How much abatement the federal layer delivers is decided by the mandate's percentages, which EPA sets, and the tax credit does not change it. EPA should set the percentages by weighing the RFS's own cost per tonne against the value of its abatement. The Treasury's spending on the

tax credit does not enter that calculation.

California and the other states. A state standard on top of a binding federal mandate draws renewable diesel from other states and raises local fuel prices. In California and Washington, where fuel already falls under cap-and-trade, the standards cost more than their climate benefits are worth whether or not the emissions cap binds. Counting the congestion, accident and local air pollution they avoid, they about break even at a 2 percent discount rate. From 2032 California's binding constraint is its price cap. Once the cap binds, tightening the carbon-intensity targets no longer changes the credit price, and any new source of credits, such as direct air capture, is used only if it costs no more than the cap. In the model, from 2032 California supplies credits at the cap price, and sellers short of credits pay the cap. Those payments are worth $104 billion in present value. In 2045 alone they are $16 billion, about $1.80 for every gallon of gasoline and diesel sold in California. Under the statute no such credits exist. A seller's shortfall is an obligation carried forward, which may never be enforced. Selling credits at the cap, as California's cap-and-trade program does with its price ceiling units (California Air Resources Board, 2026a), would make the cap a hard ceiling (Fell et al., 2012) and turn that value into state revenue, without changing fuel prices or abatement from what the model already assumes.

Congress and the states together. Among the federal alternatives studied, a carbon price buys the next 100 million tonnes most cheaply, at $36 a tonne. A federal credit for fast charging follows at $59. Its abatement depends on how long stations last, and its rank rests on the value drivers place on a larger charging network. A federal credit for electric vehicle purchases costs $101, and a larger mandate $175. The largest change to the stack that this paper studies, short of a carbon price, is a national clean fuel standard in place of the RFS and the four state standards. It would need Congress and the states to act together.

These conclusions have limits. The model is partial equilibrium and ignores the cost of raising public funds through other, distorting taxes. If a dollar of forgone revenue costs society more than a dollar, the forgone fuel-tax and allowance revenue would weigh more (Goulder et al., 2016), and the costs reported here are conservative. Allowance prices in California and Washington are held at their 2025 levels and do not respond to the state standards. Some benefits are not counted. The model includes no learning-by-doing that spills over to other producers, although for the 45Z extension to break even such a spillover would have to be worth about $40 per additional gallon, large for fuels as mature as biodiesel and renewable diesel.[18] The model also stops in 2045, leaving out the value of the electric vehicle stock and biofuel capacity after that year.

Credit holders' expectations carry no regulatory risk, although credit prices move on news about the regulator and the 2024–25 price level is most easily read as doubt that the standard will bind.

[18] Net welfare of the extension, counting forgone fuel-tax and allowance revenue, divided by the extra biofuel it produces, 2026–45, both in present value at 2%. It is $41 per additional gallon of biofuel and $28 excluding ethanol. For biodiesel and renewable diesel alone it is $22.

This doubt matters for the year California's cap binds and for the payments at the cap (Section 7). If the regulator relaxes the standard as its past decisions predict, the cap never binds. California is assumed to supply credits at its cap from 2032, which matches the statute only if fuel sellers expect never to pay more than the cap for their carried obligations. The price channel that carries most of the state standards' national abatement and all of the 45Z extension's emissions rests on fuel-demand elasticities taken from the literature, -0.37 for gasoline and -0.07 for diesel. Halving or raising them by half moves the extension's emissions between 38 and 94 Mt, and it moves the relocated share of the state standards' abatement between 46% and 61%. Drivers in the model cannot buy fuel across a state line, a margin that would shift some of the four states' price effect into leakage. The station entry condition leaves out two features of charging. California's 2025 amendments create capacity-based credits for fast chargers, which the model carries only as an exogenous share of California's credit supply. Light- and medium-duty chargers can apply through 2030 and heavy-duty chargers through 2035. Each earns credits for ten years, within pools capped at 2.5% of deficits. Demand charges are implicit in the per-kWh cost and do not fall with station utilization. Two federal tools that fall in the same case as the 45Z extension, a restored 45Z bonus for SAF and a federal year-round E15 law (which loosens the ethanol blend limit under the same binding total requirement), are left for future work.

A subsidy, a state standard or a larger mandate buys abatement only if it moves the constraint that binds. When different governments set the layers, each one's effect is decided partly by the others. What the Treasury's tax credit buys depends on EPA's percentages, and what a state standard buys depends on the federal mandate and on the state's own cap-and-trade program. In U.S. transportation fuel the binding constraint is EPA's mandate today and, in California, the state's price cap from 2032. Federal and state policies overlap in the same way in renewable portfolio standards under federal tax credits for clean electricity, in state cap-and-trade programs under federal rules, and in zero-emission vehicle credit markets under federal vehicle standards (Goulder and Stavins, 2011; Fischer and Preonas, 2010). In this paper's main case, extending 45Z while the RFS binds raises biofuel output by 0.11% and emissions by 67 million tonnes over 2026–45.

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A Institutional Details

The federal purchase credits for new and commercial clean vehicles (30D, 45W, 25E) ended for vehicles acquired after 30 September 2025, and the refueling property credit (30C) ends for property placed in service after 30 June 2026. Before OBBBA, the Inflation Reduction Act had extended 30C through 2032; the model dates each change by when it became law (Section 4.4).

Credits for residential charging accrue to the distribution utility, which must return at least half of them to drivers and may direct the rest to equity programs and charger grants. Credits from public and depot charging accrue to the equipment owner. The state standards therefore fund electric vehicle and charger subsidies out of fuel-buyer payments, a channel the vehicle block makes explicit.

The largest state program for zero-emission trucks is California's HVIP, a point-of-sale voucher whose amount depends on the vehicle class (California Air Resources Board, 2026b). Vouchers are requested when a truck is ordered and redeemed when it is delivered, so trucks delivered in 2025 were funded mostly by vouchers requested in 2023–24; the funding committed after HVIP reopened in September 2025 pays for trucks delivered in 2026–27. Combined purchase incentives may not exceed 90% of the vehicle's price. Depot charging for trucks is supported by California's EnergIIZE grants (CALSTART and California Energy Commission, 2025), by utility make-ready programs and, through mid-2026, by 30C. Washington exempts zero-emission trucks from sales tax and runs WAZIP; Oregon and New Mexico have smaller rebate and tax-credit programs (New Mexico Legislature, 2024). California's Advanced Clean Trucks sales requirement is not being enforced, and the model removes it. In the model, each incentive is included only to the extent of announced funding (Section 4.4).

Several states outside the clean-fuel bloc offer per-gallon SAF tax credits (Illinois, Minnesota, Nebraska, Iowa and Hawaii), mostly with annual budget caps. Washington has a SAF credit in statute with no certified in-state production.

Figure 10 shows California's market by quarter: the credit price, the bank, and the quarter's deficits and credits, with the dates of the 2018 and 2024 amendments.

A.1 The policy stack in detail

Table 5 summarizes the instruments that the model represents.

Federal instruments

The RFS requires refiners and importers to retire renewable identification numbers (RINs) in proportion to the gasoline and diesel they sell. The obligations are nested: biomass-based diesel (D4) counts toward the advanced requirement, which counts toward the total requirement, so the D4 price is at least the D6 (conventional ethanol) price and exceeds it only when the biomass-based

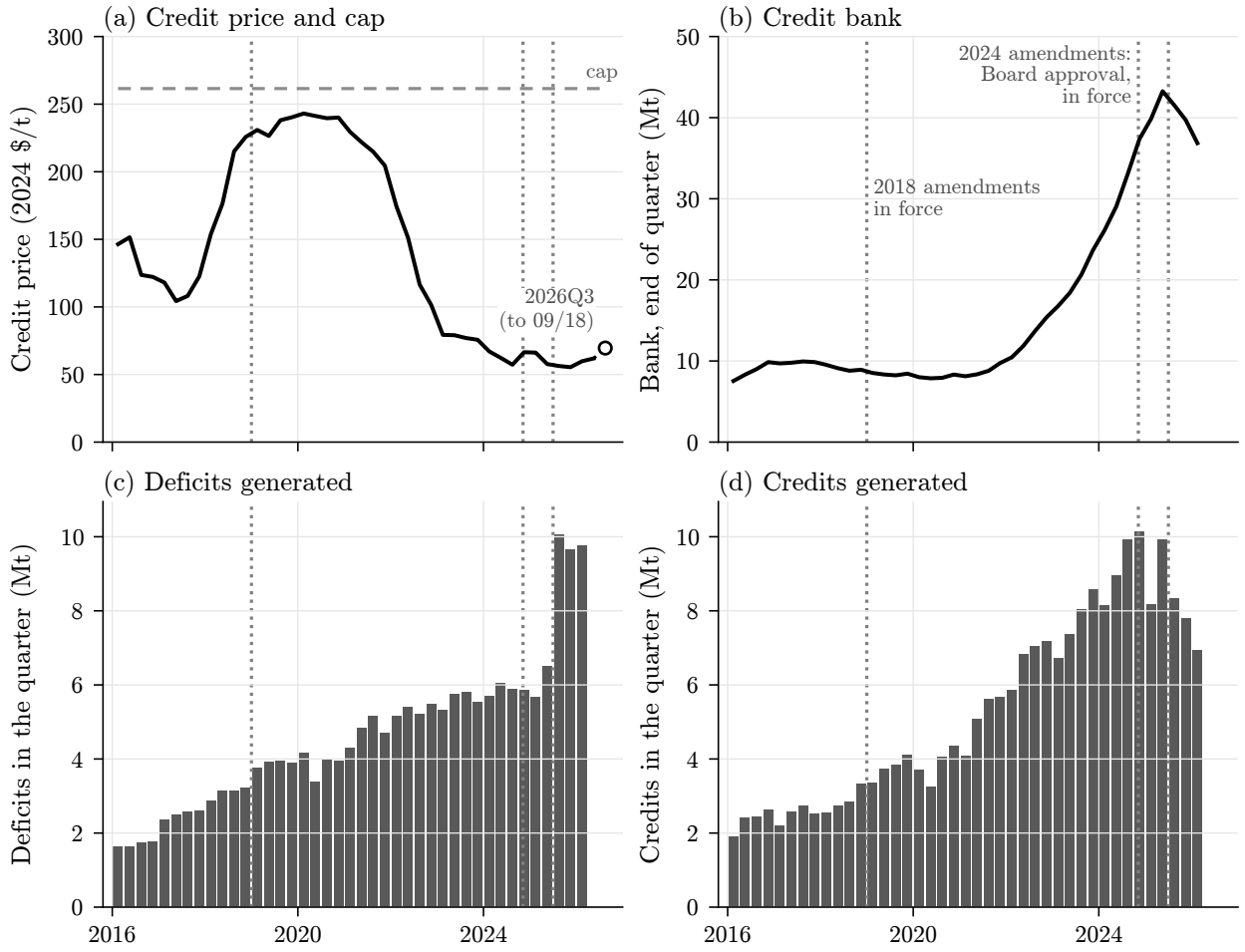


Figure 10: California's LCFS market by quarter, 2016–2026

Notes: (a) Credit price, 2024 dollars, with the cap; hollow point: 2026Q3 to 18 September. (b) Bank at the end of the quarter. (c) Deficits and (d) credits in the quarter. Dotted lines: the 2018 amendments in force, Board approval of the 2024 amendments, and the 2024 amendments in force. Source: CARB LCFS quarterly data summaries (to 2026Q1) and weekly transfer reports.

diesel requirement binds on its own. Each gallon earns RINs according to its equivalence value. EPA's final rule for 2026–2027 (U.S. Environmental Protection Agency, 2026c) raised the biomass-based diesel percentage from 3.15% of gasoline and diesel in 2025 to 5.24% in 2026 and 5.37% in 2027, and the total from 13.13% to 15.50% and 15.78%, and lowered the equivalence value of renewable diesel and sustainable aviation fuel (SAF) from 2027. The RFS has no statutory volumes after 2022; I hold the 2027 percentages for later years. Because the obligations are percentages of fossil gasoline and diesel, the required volume falls as electric vehicles reduce fuel demand. RINs can be used in the year they are generated or the next, carried-over RINs may cover at most 20% of an obligation, and a deficit may be carried forward one year but not two in a row (40 CFR 80.1427).

Since 2025 the federal blenders' credits are replaced by a production credit of up to $1.00

Table 5: The policy stack for U.S. transportation fuels, October 2026

Instrument	What it rewards	Who pays	Bank and price limits	Status
RFS (EPA)	Gallons of renewable fuel, by nested category (RINs)	Obligated refiners and importers, passed to fuel buyers	RINs usable for two years; carry-over $\leq$20%; one-year deficit	Percentages set for 2026–27
45Z (Treasury)	Gallons, scaled by lifecycle intensity; North American feedstock only	Federal taxpayers	—	Expires after 2029
LCFS: CA, OR, WA, NM (state agencies)	Carbon intensity below a declining benchmark, including electricity	Fuel buyers in the state	Unlimited banking; capped prices	Schedules to 2035–2045; CA tightens automatically, OR/WA/NM may defer
Vehicle and charger credits (30D, 45W, 30C)	Electric vehicles and chargers	Federal taxpayers	—	Ended September 2025 (vehicles) and June 2026 (chargers)

per gallon (2022 dollars) scaled by $(50 - \text{ER})/50$, where ER is the fuel's lifecycle emissions in kgCO_2e/MMBtu. The 2025 budget law (U.S. Congress, 2025, OBBBA) extended the credit through 2029, removed the higher SAF rate for fuel produced from 2026, excluded indirect land-use change from the emissions rating, and limited eligibility to fuel made from feedstocks grown in the United States, Mexico or Canada. The origin rule matters for the clean fuel states, whose renewable diesel had come increasingly from imported used cooking oil and tallow. Fuel from facilities claiming the carbon capture credit (45Q) cannot claim 45Z in the same year.

State clean fuel standards

California (LCFS, since 2011), Oregon (Clean Fuels Program, 2016), Washington (Clean Fuel Standard, 2023) and New Mexico (Clean Transportation Fuel Program, April 2026) set an annual benchmark carbon intensity for gasoline and diesel and their substitutes. Fuel above the benchmark generates deficits, fuel below it generates credits, and credits trade. Electricity used in vehicles earns credits, adjusted by an energy economy ratio. Fossil jet fuel carries no obligation, but alternative jet fuel may opt in and earn credits. Credits can be banked without limit and do not expire.

California's 2024 amendments (effective 1 July 2025) cut the benchmark to 22.75% below the 2010 base in 2025, 30% in 2030 and 90% in 2045 (California Air Resources Board, 2025). Oregon targets 20% by 2030 and 37% by 2035, New Mexico 20% by 2030 and 30% by 2040 (against 2018), and Washington, after HB 1409 (signed 17 May 2025), 45% below 2017 by 2038, with annual reductions of 3–4 percentage points in between. The 2028–37 rule is due by 1 January 2028; the model uses the 2025–27 standards in force and the Department of Ecology's draft path for

2028–37, which lies within the statutory range. The statute also bars Ecology from tightening the standard beyond 20% from 2030 until it demonstrates that at least one new or expanded biofuel production facility has received a siting, operating or environmental permit after 1 January 2025 (RCW 70A.535.025). A qualifying county permit appears to have been issued in December 2025, but as of October 2026 Ecology had not made the demonstration, so the main specification caps Washington's standard at 20% from 2030 and treats the statutory path to 45% as a scenario. A further provision letting Ecology relax a year's standard by up to two percentage points to avoid a forecast deferral is not modeled.

Each program limits the credit price, but in different ways. California prohibits any credit transfer above an inflation-indexed maximum price ($261.52 in 2024 dollars; 17 CCR §95487(a)(2)(D)) and runs a credit clearance market at that price; obligations still unmet carry over as an accumulated deficit with 5% compound interest, to be repaid within five years (§95485(c)). Oregon's and Washington's maxima bind sellers in their clearance markets (Oregon Department of Environmental Quality; Washington Department of Ecology), and unmet deficits carry forward with 5% added. New Mexico opens a clearance market at the agency's discretion, with an initial maximum of $270 (New Mexico Environment Department).

Rules that respond to the market

The programs also differ in how the standard itself responds to market conditions, in opposite directions. Oregon, Washington and New Mexico pair their standards with statutory forecast deferral. Each year a state agency forecasts whether banked plus newly generated credits will cover projected deficits; if the forecast falls below 100 percent, the regulator *shall* defer compliance, by temporarily relaxing the standard, holding it at the prior year's level, or suspending deficit accrual (State of Oregon; Oregon Department of Environmental Quality; State of Washington; New Mexico Environment Department, ORS 468A.273; OAR 340-253-2100; RCW 70A.535.110; 20.2.92.601 NMAC). Oregon additionally requires a root-cause review, and permits an emergency deferral, if credit prices rise 100 percent or more over three months, and Washington's 2025 amendments let the regulator loosen annual targets by up to two percentage points to avoid a forecast deferral. California has no deferral provision. Since July 2025 it instead *tightens* its benchmarks automatically: the automatic acceleration mechanism advances the whole benchmark schedule by one year when the bank exceeds three quarters of the year's deficits and credits exceed deficits (17 CCR §95484(b)); the first assessment is in May 2027.

In practice, Oregon's forecast deferral has never been triggered: every annual fuel supply forecast from 2018 to 2027 concluded that banked plus projected credits sufficed (Oregon Office of Economic Analysis), and I found no record of a deferral order in Oregon or Washington. Deferral therefore operates, if at all, through expectations: it bounds how far scarcity can be priced before the

standard itself adjusts. I treat this asymmetry as one possible reason why Oregon's credit prices stayed well below its ceiling in its tighter years, not as an established mechanism; I found no market commentary that links Oregon or Washington prices to deferral. In the model, deferral and acceleration are both applied to the model's own forecasts (Section 4.1).

A.2 EPA's record on the RFS volumes

Table 6 lists each year's total renewable fuel requirement, how EPA set it, and the small refinery exemptions it later granted. From 2010 to 2013 EPA kept the statutory totals and cut only the cellulosic standard. In the rule of December 2015 it cut the 2014–16 totals with its general waiver as well, and the D.C. Circuit vacated part of the 2016 cut in 2017. From 2017 to 2020 it again used only the cellulosic waiver, while exemptions rose to 7–9% of the requirement in 2017–19 before falling to 4% in 2020. From 2023 it sets the volumes itself, and in 2026 it set record volumes and reallocated most of the exempted volume.

Table 6: RFS total renewable fuel requirements and exemptions, 2010–2027 (billion RINs)

Year	Statute	Final	Rule published	Reduction authority	D6 at final ($)	Exemptions (bn RINs)
2010	12.95	12.95	26 Mar 2010	cellulosic waiver	—	—
2011	13.95	13.95	9 Dec 2010	cellulosic waiver	0.03	24 (—)
2012	15.20	15.20	9 Jan 2012	cellulosic waiver	0.02	23 (—)
2013	16.55	16.55	15 Aug 2013	cellulosic waiver	0.94	8 (0.19)
2014	18.15	16.28	14 Dec 2015	cellulosic and general	0.50	8 (0.21)
2015	20.50	16.93	14 Dec 2015	cellulosic and general	0.50	7 (0.29)
2016	22.25	18.11	14 Dec 2015	cellulosic and general[a]	0.50	19 (0.79)
2017	24.00	19.28	12 Dec 2016	cellulosic waiver	0.84	35 (1.82)
2018	26.00	19.29	12 Dec 2017	cellulosic waiver	0.89	31 (1.43)
2019	28.00	19.92	11 Dec 2018	cellulosic waiver	0.19	29 (1.46)
2020	30.00	20.09[b]	6 Feb 2020	cellulosic waiver	0.15	24 (0.82)
2021	33.00	18.84	1 Jul 2022	set after the year	1.32	25 (1.06)
2022	36.00	20.63[c]	1 Jul 2022	statutory table	1.32	22 (0.83)
2023	—	20.94[c]	12 Jul 2023	"set" authority	1.53	26 (0.93)
2024	—	21.54	12 Jul 2023	"set" authority	1.53	28 (1.15)
2025	—	22.33	12 Jul 2023	"set" authority	1.53	29 (1.76)
2026	—	25.82[d]	1 Apr 2026	"set" authority	1.28	—
2027	—	25.98[d]	1 Apr 2026	"set" authority	1.28	—

Notes: Statute: Clean Air Act §211(o)(2)(B); after 2022 EPA sets the volumes. Rule published: Federal Register publication date of the final rule. Reduction authority: the waivers EPA used to reduce the total or advanced requirement below the statute. D6 at final: EMTS volume-weighted price, mean of the five trading days before the announcement (nominal). Exemptions: petitions granted (exempted RINs) by compliance year; 2013–18 counts from U.S. Government Accountability Office (2022), 2017–18 volumes from EPA as reported by EIA (U.S. Energy Information Administration, 2019); other entries from EPA's exemption dashboard as of September 2026 (current status, after later reconsiderations). [a] The general-waiver cut of about 0.5 billion was vacated in U.S. Court of Appeals for the D.C. Circuit (2017). [b] Revised to 17.13 in July 2022. [c] Plus a supplemental 0.25 billion (2022 and 2023) on the remand of U.S. Court of Appeals for the D.C. Circuit (2017). [d] Plus 70% of the 2023–25 exempted volumes, reallocated (26.81 and 27.02) (U.S. Environmental Protection Agency, 2026a). Sources: Federal Register final rules 2010–2026; U.S. Environmental Protection Agency (2015).

B The Quantitative Model

Table 8 lists the policies and rules the model represents and those it leaves out.

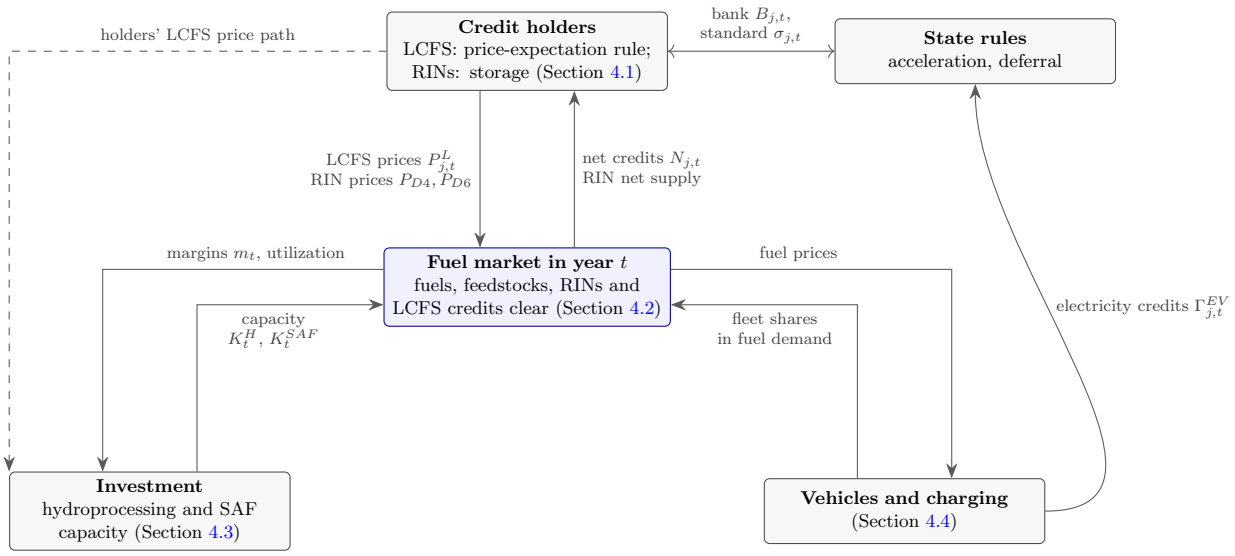


Figure 11: Structure of the quantitative model. Within a year, the fuel market clears given credit prices, capacity and fleets. Across years, credit holders' expected exhaustion years and banks, the state rules, capacity and the vehicle stock carry the state of the economy forward. The blocks are solved jointly over 2026–2045 (Section 4.5).

B.1 Credit markets, rules and expectations

In each program $j \in \mathcal{L}$, credits from biofuels, from electricity ($\Gamma^{EV}_{j,t}$, from the vehicle block) and from fixed sources (biomethane, off-road electricity, refinery projects and infrastructure credits), less deficits from fossil gasoline and diesel, give the year's net credit generation $N_{j,t}$. Infrastructure credits are a share $\iota_{j,t}$ of fossil deficits, set at the observed share in 2026 and rising to the regulatory limit. The balance is

$$N_{j,t} + B^{draw}_{j,t} = s^+_j - s^-_j, \qquad 0 \le P^L_j \perp s^+_j \ge 0, \qquad 0 \le \bar{P}_{j,t} - P^L_j \perp s^-_j \ge 0, \tag{22}$$

where B^{draw} is the draw on the bank allowed by holders' expectations (below) and $\bar{P}_{j,t}$ the program's price cap (2024 dollars: CA \$261.5, OR \$253.5, WA \$245.5; NM \$270 nominal, indexed from 2027). Standards follow each program's adopted schedule; Washington's 2028–37 schedule is the Department of Ecology's draft, within the statutory range, capped at 20% from 2030 until the Department of Ecology certifies the new biofuel capacity the statute requires (Section 2). After a program's last statutory year, the standard keeps falling at the average annual rate of its last five statutory years.

When a program is short at the cap, the state issues the shortfall as credits at the cap price,

Table 7: How this model compares with the closest models in the literature

	Policy layers	Years / dynamics	Credit banking	Expectations	Biofuel capacity	Vehicles, charging	Regions, fuel movement	Welfare wedges	Estimation / tests
Holland et al. (2015)†	C&T, ethanol subsidy, RFS, LCFS, each alone; no EV policy	Static long run, 2022	n/a (static)	n/a	Ethanol plant locations optimized	None; fuel demand elasticity	County supply, aggregated nationally; gains by district	Surplus; permit revenue lump-sum; land-use cost; cost/t vs. SCC	Calibrated; robustness checks; empirical vote model
Bushnell et al. (2020)	CA LCFS incl. infrastructure credits; RFS, C&T not modelled	Quarterly projection 2019–30	Existing bank drawn evenly (assumed)	n/a; credit price not modelled	No; BBD fills residual deficit	Exogenous EV stock (EMFAC); ZEV scenario	California only	None (compliance outlook)	VEC on 1987–2018; 1,000 bootstrapped shock paths
Bistline et al. (2023)	IRA credits incl. biofuel, 45Z, EV, charging (coverage varies by model); RFS and LCFS not described	Multi-year; results to 2035; nine models, economy-wide or power	Not described	Perfect foresight (6 models) or recursive (3)	Energy-system capacity expansion; biofuel not described separately	LDV EV sales (logit in several); commercial vehicles; charging varies by model	1 to 134 regions by model; fuel trade in MARKAL	Net system cost incl. tax credits per t, 2030; SCC benefits separate	Native, unharmonized inputs; no out-of-sample test
Bushnell and Smith (2024)	IRA PTC (wind, solar), ITC (storage); power only	Static, 2030	n/a	n/a; static free entry	n/a (generation entry endogenous)	None	Four WECC regions, transmission limits	Welfare objective; reports capacity, energy, CO_2	Calibrated to 2019; factor-model forecasts give 250 draws of inputs
Scott (2024)†	CA LCFS, C&T with price floor; RFS, ZEV incentives not modelled	Dynamic CGE, annual 2010–30	No LCFS banking	Myopic	Endogenous capital; experience-curve innovation	LDV, HDV services; electricity as fuel	California only	Equivalent variation; C&T revenue rebated; SCC	Calibrated to 2010 SAM; hindcast 2010–22
Gerveni et al. (2026)	RFS + blenders' credit (45Z discussed); LCFS discussed only	Static diagram; history 2011–25	Excludes RIN carryover	n/a	No; upward-sloping supply	None	National	RIN compliance cost; incidence	Parameter-free D4 formula vs. prices, $R^2 = 0.74$
Wu et al. (2026)	RFS, 45Z, CA LCFS, state SAF credits, carbon tax; CA C&T in supplement	Static, 2030	Fixed LCFS bank draw; no RIN bank	n/a	No capacity block	Exogenous EV share	CA and rest of U.S.	Fuel cost + spending − carbon-tax revenue, per t	Calibrated; no validation reported
This paper	RFS (nested), 45Z, LCFS in 4 states, CA/WA C&T (fixed $/t), 30C, truck incentives	Annual path 2026–45 (temporary equilibrium)	RIN bank; four LCFS banks with caps	RINs: foresight; LCFS: estimated rule (perfect foresight as alternative)	Hydroprocessing from 2029: estimated optimal stopping	LDV, MDV, HDV logit; L2/DC station entry	CA, OR, WA, NM, rest of U.S.; rising delivery cost	Surpluses − SCC damages; fuel-tax, allowance revenue separate; partial eq.	Rolling forecasts of LCFS price; investment 2014–19 → 2020–25; parameter bootstrap

Notes: Entries describe each model as used in the cited paper. "Not described": the cited paper does not describe the feature. C&T: cap-and-trade; BBD: biomass-based diesel; LDV/MDV/HDV: light-, medium-, heavy-duty vehicles; SCC: social cost of carbon. † Rows for Holland et al. and Scott describe the working-paper versions (Energy Institute at Haas WP 220R, 2011; 2023 draft). Bistline et al. compare nine models, so their row gives the range. Bushnell and Smith model input uncertainty, and Bistline et al. cover the whole economy; this paper does neither.

Table 8: Policies and rules in the model

Layer	Represented	Not represented
RFS	Nested biomass-based diesel, advanced and total requirements on a fossil base; RIN banking with the two-year life, 20% carry-over and one-year deficit; small-refinery exemptions as three scenarios; EPA's 2026–27 percentages and reallocation	EPA's discretion as a choice; RIN market power
45Z	Credit scaled by tax carbon intensity; North American feedstock only from 2026; expiry after 2029 (in law; agents treat it as continuing); inflation adjustment	Prevailing-wage conditions; transferability
State standards	CA, OR, WA, NM schedules; unlimited banking; price caps and clearance (CA deficit carry-forward replaced by issuance at the cap); California's acceleration mechanism and crop-oil cap; Washington's 20% limit from 2030 (the 45% path as a scenario); Oregon, Washington and New Mexico forecast deferral; electricity and infrastructure credits	Washington's 2-point deferral relief; New Mexico's program review
Other incentives	State SAF and biodiesel credits as budget-capped steps; 45Q; truck purchase incentives (HVIP stacking)	Local and utility programs beyond those listed
Markets	Gasoline, diesel and jet pools in five regions; ethanol, biodiesel, renewable diesel, HEFA-SAF, alcohol-to-jet; two lipid groups and corn; hydroprocessing investment; electric vehicles and chargers by class	World feedstock prices, indirect land use, refinery closures

the bank ends at zero, and the revenue goes to the state. The statutes instead carry an unmet obligation forward as a deficit with interest (17 CCR §95485(c) in California); when the cap binds year after year that deficit is never repaid, so issuance at the cap is the safety valve a regulator would have to adopt. The statutory ledger is reported separately.

LCFS credit holders in program j do not know when the bank will run out and the price reach the cap. Their expectation of that year, $T_{j,t}$ (the expected exhaustion year), prices credits by Hotelling to the cap at the holding cost $r = 3.38\%$, the real carry measured from forward quotes,

$$P^{L}_{j,t} = \min\{\bar{P}_{j,t},\ \bar{P}_{j,t}/(1+r)^{\mathrm{gap}_{j,t}}\}, \tag{23}$$

where $\mathrm{gap}_{j,t}$ is the expected number of years to exhaustion. Holders update it by the surprise in the year's net credit generation $s_{j,t}$ relative to what they expected,

$$G_{j,t}(s) = T_{j,t-1} - t + \gamma_j\big(s - E_{t-1}[s_{j,t}]\big). \tag{24}$$

A year with more credits than expected moves the exhaustion date out by γ_j years per Mt and lowers the price.[19]

[19] One interpretation is the steady state of a Kalman filter for holders who read each year's net credits as a noisy

Expected net generation has two parts: a baseline and the changes holders know about in advance,

$$E_{t-1}[s_{j,t}] = \bar{s}_{j,t-1} + \Delta^{sched}_{j,t} + \Delta^{known}_{j,t}, \tag{25}$$

where Δ^{sched} is the mechanical effect of this year's change in standards at last year's quantities (including any acceleration or deferral already decided) and Δ^{known} the effect of other changes known in advance: the crop-oil cap, scheduled import volumes and the fading out of the 2026 transitional terms (Section 5.3). Scheduled expiries of federal credits are not among them: like investors, holders treat 45Z as continuing. Counting its expiry after 2029 as a known change would move California's 2030 price by about \$2/t. Expected changes use the information available at the time; in forward years there are no rule-change jumps, so changes in the rules enter only through Δ^{sched} and Δ^{known}. I consider two baselines $\bar{s}_{j,t-1}$:

- *Rule A (last year)*: $\bar{s}_{j,t-1} = s_{j,t-1}$;
- *Rule B (three-year average)*: $\bar{s}_{j,t-1} = \frac{1}{3}(s_{j,t-1} + s_{j,t-2} + s_{j,t-3})$, adaptive expectations in the sense of Nerlove (1958), with the available years at the start of the sample.

Holders also look at how many years the bank would last at the expected rate of shortfall, $Y_{j,t} = \max(B_{j,t-1}, 0)/(-E_{t-1}[s_{j,t}])$ (and $Y_{j,t} = \infty$, so $w_{j,t} = 0$, when $E_{t-1}[s_{j,t}] \geq 0$), and put more weight on it as it shrinks:

$$\text{gap}_{j,t} = (1 - w_{j,t})\, G_{j,t} + w_{j,t}\, \min\{G_{j,t}, Y_{j,t}\}, \qquad w_{j,t} = e^{-Y_{j,t}/K}. \tag{26}$$

K is the horizon, in years of bank left, over which holders start to price exhaustion. With a large bank the price follows the rule (18); as the bank runs down, the price rises toward the cap even without new surprises. The blend can only raise the price. Within a year, the price and the net generation it induces are a fixed point, and the bank evolves as $B_{j,t} = B_{j,t-1} + s_{j,t} + s^{-}_{j,t} - s^{+}_{j,t}$, so a year that clears at the cap ends with an empty bank. If the expected price would overdraw the bank, the year clears instead: the whole bank is available, equation (16) sets the price, and the expected exhaustion year is reset to match that price.

γ_j, K and the initial expected exhaustion years are estimated on California's and Oregon's prices and banks for 2016–25. Oregon, Washington and New Mexico share one response γ per unit of their obligation, $\gamma_m = 100\, g/O_m$ with O_m the program's 2025 obligation in Mt, and the change in T is limited to 20 years per year in Oregon and Washington and 5 in New Mexico. The main specification uses Rule B with $K = 2$ in all four programs; Rule A is the alternative.[20]

signal of a persistent component of the balance (Appendix D.4).

[20] Under Rule A, the bank-years blend (19) is used in California only, because Oregon's bank never fell in the sample and the blend is not identified there; a version with the blend in all four programs is a sensitivity. Under Rule B the blend applies to all four programs.

The RIN bank is held in the conventional (D6) market. Holders are competitive and store RINs when the expected return covers the interest rate, which is the solution of a planner's problem that minimizes the discounted cost of meeting the obligations,

$$\min_{\{B_t,z_t\}} \sum_{t=2026}^{2045} d_t\big[C_t(B_t-B_{t-1})+\Psi_t(B_t)\big]+d_{2046}\,C_{2046}(-B_{2045}) \quad \text{s.t.} \quad B_t \geq -k\,R_t\,z_t,\ z_t+z_{t-1}\leq 1, \tag{27}$$

where $C_t(n)$ is the area under the year's inverse net-supply curve of RINs (built from the year's solves of the fuel market), $d_t=(1+r)^{-(t-2025)}$ with the holding cost $r=3.38\%$, Ψ_t a soft penalty for carrying more than 20% of the next year's obligation, and z_t marks a year in deficit. A deficit can be carried forward one year, not two in a row (40 CFR 80.1427(b)), and is limited to 10% of the obligation, applied to the industry as a whole. The 20% carry-over limit applies to each obligated party; the model's aggregate bank stays below 10% of the next year's obligation in 2026–30, so the limit does not bind in aggregate. At an interior bank, prices rise at 3.38% a year. The terminal year 2046 is a static copy of 2045 in which the bank is drawn down. A version that does not allow deficits is a sensitivity.

RIN markets are national and nested. Obligations are EPA's percentage standards applied to the obligated base,

$$\text{Base}_t=\textstyle\sum_j Q_{G0,j,t}+(1-f_{D,t})\big(\sum_j Q_{D0,j,t}+X_t\big), \tag{28}$$

where Q_{F0} is fossil gasoline and diesel, X_t is obligated non-road, locomotive and marine diesel outside the model, and $f_{D,t}$ the share of diesel exempted for small refineries in EPA's 2026–27 rule (U.S. Environmental Protection Agency, 2026c). The total requirement (net of cellulosic, which has no supply in the model), the advanced requirement and the biomass-based diesel (BBD) requirement each bind as a complementarity with its price: P_{D6}, the advanced premium μ_t^{ADV}, and the biomass-based diesel premium $P_{D4}-P_{D6}-\mu_t^{ADV}$. The 2026–27 percentages are EPA's final standards, including the reallocated exempted volumes; from 2028 the reallocation is removed and the percentages are held at their 2027 level (net of reallocation) and applied to each year's base, so obligations fall as fossil demand falls. RIN equivalence values are 1.5 for BD and 1.6 for RD and SAF (RD 1.7 in 2026). Every obligated gallon carries the cost of the full obligation,

$$w_t^{RFS}=\pi_t^{BBD}P_{D4}+(\pi_t^{RF}-\pi_t^{BBD})P_{D6}+(\pi_t^{ADV}-\pi_t^{BBD})\mu_t^{ADV}, \tag{29}$$

whether the RINs come from this year's production or from the bank. Exempt refiners keep this cost as rent.

California's 2024 amendments advance all later benchmarks by one year, effective the following January, when the bank exceeds three quarters of the year's deficits and credits exceed deficits in the same year (17 CCR §95484(b)). The model applies the rule to its own bank and flows from

2027 on. Whether the mechanism triggers depends on small differences in the 2027 balance, so I do not treat a trigger, or its absence, as a finding.

The other three programs must defer scheduled tightening when the agency forecasts that banked and new credits will fall short of the obligation, by adopting a temporary standard that better reflects the forecast supply of credits. I model the temporary standard as a weighted average of this year's scheduled standard and the one in force last year,

$$\sigma_{j,t}(d) = (1-d)\,\sigma_{j,t}^{sched} + d\,\sigma_{j,t-1}^{\text{in force}}, \qquad d \in [0,1], \tag{30}$$

where d is the smallest deferral that makes the bank plus this year's net credits non-negative when the credit price is at the cap. Deferral is considered only when the bank is expected to run short at the scheduled standard, with expected net credits that include the changes already known, and holders already expect the cap price. The scheduled path resumes the next year. Two approximations are involved: the statutes' forecasts do not condition on price, so "supply at the cap price" stands in for potential supply; and very short deferrals can be implemented only through a temporary standard, not by keeping last year's standard for a full compliance period. A rule that allows only whole-year deferral has no stable solution when a program sits at its cap, because a full deferral pulls the price below the cap and the next year's forecast then calls for none.

B.2 The fuel market in year t

Fuel pools are gasoline G, diesel D and jet fuel J. In each region, each pool combines a fossil component with biofuel pathways $i = (f,k,j)$, indexed by fuel f, feedstock group k and region j: corn and sugarcane ethanol in gasoline; biodiesel (BD) and renewable diesel (RD) in diesel; and sustainable aviation fuel (SAF) made by hydroprocessing of the same lipids (HEFA-SAF) or from ethanol (alcohol-to-jet, ATJ, with or without carbon capture). Lipids form two groups: soybean and other vegetable oils, and fats (animal fat, used cooking oil and distillers' corn oil). The year's market is a mixed complementarity problem. Time subscripts are dropped within a year where unambiguous.

Producers are price takers. A pathway produces in region j only if its cost covers its value:

$$0 \le Q_i \;\perp\; c_{f,t}^{op} + \theta_{i,t}P_k + \phi_g(u_g) + \rho_g + \nu_{f,j} + \tau_i + w_i^{A1} + \mathbf{1}\{i \in \mathcal{H}\}\big(\psi_t + \rho_t^{SAF}\big) + \mu_i^{cc} - \big[\varphi_f P_{F(f),j}^{conv} + V_i\big] \;\ge 0, \tag{31}$$

where c^{op} is the variable operating cost per gallon (falling with cumulative output for hydroprocessing), θ_i the feedstock yield (8.2 lb of lipids per gallon of RD, 9.35 per gallon of HEFA-SAF, 7.55 per gallon of BD), P_k the price of feedstock group k, ϕ_g and ρ_g the utilization cost and capacity rent of technology group g, ν the shadow price of a blend wall, τ_i the cost of delivering RD to an LCFS state, w_i^{A1} the marginal cost of departing from the observed feedstock mix (below), $\mathcal{H}$ the

HEFA-SAF pathways, ψ_t and ρ_t^{SAF} the marginal cost of shifting hydroprocessing output toward jet and the rent on SAF-capable capacity (below), μ_i^{cc} the rent of California's cap on crop-oil-based fuels (from 2028), and φ_f the value of a gallon relative to the fossil fuel it replaces. The policy value of a gallon is

$$V_i = \mathrm{EV}_i\big(P_{D4},\ P_{D6} + \mu_t^{ADV} \text{ or } P_{D6}\big) + P_j^L\,(\sigma_{F,j,t} - CI_i^L)\,ED_i \cdot 10^{-6} + z_{i,t}^{45Z} + z_{i,t}^{state}, \tag{32}$$

the RINs it generates (EV_i per gallon), its LCFS credits or deficits at the program's carbon intensity for that fuel and feedstock group CI_i^L, the federal clean fuel production credit and state credits.

Renewable diesel delivered to a clean fuel state bears a cost that rises with the volume delivered there,

$$\begin{gathered}\tau_j = c_j + d_j X_j + \mathbf{1}\{j = CA\}\, e\, E_{CA}, \qquad X_j = \textstyle\sum_{k \in RD_j} Q_k + IMP_j^{RD}, \\ 0 \le E_{CA} \ \perp\ E_{CA} - X_{CA} + \bar{s}\, Q_{CA}^D \ge 0,\end{gathered} \tag{33}$$

which stands in for terminal capacity, logistics and contracts that the model does not represent. Without it, RD would move all-or-nothing to whichever program pays most. The slopes d_j are estimated on quarterly deliveries, 2023Q1–2025Q4 for California and Oregon and 2024Q3–2025Q4 for Washington. In California the slope steepens, by $e = \$2.396$/gal per billion gallons, once RD exceeds $\bar{s} = 72.13\%$ of the state's diesel pool Q_{CA}^D, the highest share in 29 quarters of CARB data (2025Q1). I interpret the kink as competition from other outlets for renewable diesel (Canada, Europe, Oregon and Washington) and logistics into the state, not as a technical blend limit: renewable diesel can replace fossil diesel entirely. Without the kink, the model fills California's diesel pool with RD from 2027 and prices California diesel below the rest of the country, which the state has never seen; a run without the kink is a sensitivity. The slopes d_j are the least-anchored parameters in the fuel market. Calibrated to one year's deliveries, d_j is close to that year's LCFS premium spread over that year's RD volume, and it decides whether additional national support raises deliveries to the clean fuel states or lowers their credit prices (cases (iii) and (iv) of Proposition 1). Results are therefore also reported at half and at twice California's estimated slope.

Fossil gasoline, diesel and jet fuel are supplied perfectly elastically at the national wholesale price $PC_{F,t}$, which follows a single rule that links observed 2026 prices, EIA's Short-Term Energy Outlook and the Annual Energy Outlook. The rack price of fossil gasoline and diesel adds a state price differential and the RFS and LCFS compliance costs per gallon,

$$P_{F,j}^{conv} = PC_{F,t} + \Delta_{F,j}^{fos} + w_t^{RFS} + \mathbf{1}\{j \in \mathcal{L}\}\, P_j^L\,(CI_{F0,j} - \sigma_{F,j,t})\, ED_F \cdot 10^{-6}, \qquad F \in \{G, D\}, \tag{34}$$

where $\mathcal{L} = \{CA, OR, WA, NM\}$, P_j^L is program j's credit price (\$ per t CO_2e), $CI_{F0,j}$ the fossil carbon intensity and $\sigma_{F,j,t}$ the standard (gCO_2e/MJ), ED_F the energy density (MJ/gal) and w_t^{RFS} the RFS compliance cost per gallon (equation (29)). The differential $\Delta_{D,j}^{fos}$ is the observed whole-

sale premium of fossil diesel in each state over the national benchmark ($0.070/gal in California, $0.031 in Oregon and Washington, $0.099 in New Mexico, zero elsewhere); it is a real logistics and specification cost, so it enters welfare as a resource cost. The gasoline differential is zero. Fossil jet fuel carries neither obligation. In the four LCFS states fossil diesel enters as a complementarity, so renewable diesel can in principle fill the whole diesel pool.

Capital is sunk: producers pay operating cost, not full cost, on existing capacity, and the return to capital is the capacity rent $\rho_g \geq 0$, paired with the capacity limit as a complementarity. Running a plant harder costs more as utilization rises,

$$\phi_g(u_g) = a_g \left(\frac{u_g - u_{0,g,t}}{u_{ref,g,t} - u_{0,g,t}} \right)^{+}, \tag{35}$$

for hydroprocessing (RD and HEFA-SAF together, exports included), and a two-sided version for biodiesel that lets plants run below their reference rate while covering only variable cost. The curves are fitted to monthly U.S. production, and a fixed operating cost is charged on nameplate capacity whether or not it runs.

Each plant can in principle use either lipid group, and with constant costs the cheapest group would take a whole state's production, switching from year to year between corners. Observed feedstock mixes move slowly. I therefore add a quadratic cost of departing from the observed 2025 mix, in the spirit of positive mathematical programming (Howitt, 1995; Heckelei and Wolff, 2003). For each nest $n = (f, j)$ of a fuel $f \in \{\text{BD}, \text{RD}, \text{HEFA-SAF}\}$ in region j, with volumes Q_k of the two groups, total $Q_n = \sum_k Q_k$ and observed 2025 shares s_k,

$$C_n = \sum_k a_k^{A1} Q_k + \frac{g_n}{2} \sum_k \left(Q_k - s_k Q_n\right)^2, \qquad g_n = \frac{1}{2\, c\, \bar{Q}_n}, \qquad \textstyle\sum_k s_k a_k^{A1} = 0, \tag{36}$$

and $w_k^{A1} = \partial C_n / \partial Q_k$. The intercepts a_k^{A1} reproduce the observed 2025 shares, and $\bar{Q}_n$ is the nest's observed 2025 volume, fixed in every year and counterfactual. The cost penalizes changes in the mix, not in scale: total volume in a state is still set by the delivery cost and the utilization curve. With two groups the fat share responds to the gap between the margins of the two groups as $\partial s / \partial(\text{gap}) = c\, \bar{Q}_n / Q_n$, so at reference volumes every state's share moves by c per $1/gal of margin gap. One response c is shared by RD and BD and estimated on the history (Appendix D.8). Within each group the composition (for example soybean against canola oil) is held at its 2025 level, and the group's credited carbon intensity is backed out of the programs' quarterly data. C_n is a resource cost; payments in excess of it are producer rent.

Hydroprocessing plants can shift output between RD and jet only if they have SAF-capable equipment. National HEFA-SAF output is limited by SAF-capable capacity,

$$\textstyle\sum_{i \in \mathcal{H}} Q_i + \text{slack}_t^{SAF} = K_t^{SAF}, \qquad 0 \leq \rho_t^{SAF} \perp \text{slack}_t^{SAF} \geq 0, \tag{37}$$

which is 0.46 billion gallons in 2025 (U.S. Energy Information Administration, 2025) and grows with announced projects to 2028 and with the investment rule afterwards. Within that capacity, raising the jet share of a plant's output has a rising cost, $\psi_t = (h/H_t)\sum_{i\in\mathcal{H}} Q_i$, with H_t hydroprocessing nameplate capacity and h = \$0.34/gal; it represents the hydrogen cost of the jet-heavy configuration, since yield losses are already in θ_i. A product-slate limit keeps the jet share at or below that of the jet-maximizing HEFA configuration in Baldwin et al. (2022, Fig. 1), where jet is about 55% and renewable diesel about 25% of total product.

Demand for gasoline, diesel and jet fuel is isoelastic in the retail price,

$$Q_{F,j,t} = \xi_{F,t} A_{F,j,t} B(s_{j,t}) (\bar{P}_{F,j,t} + W_{F,j,t})^{\beta_F}, \tag{38}$$

where P^{dem} is the energy-weighted blend of fossil and biofuel prices in the pool, W a fixed real wedge for taxes, retail margins and state premiums (so compliance costs pass through fully to retail prices), and $B_F(\cdot)$ the share of the fleet using fuel F that still burns liquid fuel, evaluated at the electric shares from the vehicle block. $A_{F,j,t}$ is fuel use if every vehicle burned liquid fuel, rebuilt each year from AEO vehicle miles divided by AEO conventional-vehicle fuel economy; $B(s_{j,t})$ is the share of that use left when the electric share of the fleet is $s_{j,t}$, taken from the vehicle model, and the calibration share is removed through $B(s^{calib}_{j,t})$ so electrification is not counted twice. The elasticities are -0.37 for gasoline and -0.07 for diesel (Coglianese et al., 2017; Dahl, 2012), and -0.26 for jet fuel on the rack price. They are long-run utilization responses with the fleet held fixed; changes in fleet composition come through B_F. Applied to the oil-price spike of 2026, a long-run elasticity would cut 2026 gasoline demand by about a tenth, which did not happen. The national multiplier $\xi_{F,t}$ therefore sets 2026–27 demand at the levels of EIA's Short-Term Energy Outlook and returns linearly to one by 2030; it does not vary across counterfactuals.

Ethanol's blend value falls steeply above a kink in its share e_j of the gasoline pool, $P_{E,j} = \varphi P^{conv}_{G,j}\big[1 - \mathbf{1}\{e_j > k_j\}(2^{(e_j-k_j)/w} - 1)\big]$, with $k_j = 0.10 + 0.05a_j$, where a_j is the E15 share of gasoline and $w = 0.0036$ is calibrated to the observed blend share; a floor keeps the share at or above 10%. Biodiesel is limited to a share of each region's diesel (7.5% in California, 12% in the other LCFS states, 20% elsewhere). State biodiesel mandates and incentives outside the LCFS states are included as step complementarities.

Feedstock markets are national. Supply of soybean and other vegetable oil to fuel is isoelastic with elasticity 1.5; North American animal fat has elasticity 3 and grows 0.8% a year to 2035 and 0.5% after; distillers' corn oil adds supply to the fat market with elasticity 0.5. Imported fats and used cooking oil are a separate residual supply with elasticity 5. Since 2026, fuel made from imported feedstock earns no 45Z credit, so imported fat trades at a discount to North American fat equal to the credit it forgoes. Price floors at pre-RFS real levels (\$0.35/lb for soybean oil, \$0.28/lb for fat) keep feedstock prices from falling below what other uses would pay. Corn and imported

sugarcane ethanol have their own supply curves.

Imports of RD, BD and SAF enter as fixed volumes: they count toward pools, blend walls, RIN supply and LCFS balances, but use no U.S. feedstock or capacity. Exports of RD (mostly to Canada) and BD are exogenous and do use U.S. feedstock and capacity.

The 45Z clean fuel production credit pays $z_{i,t}^{45Z} = \bar{z}_t \max\{0, (47.39 - CI_i^{Tax})/47.39\}$ per gallon for fuel made from North American feedstock, with $\bar{z}_t \approx \$1.03$ (2024 dollars) and carbon intensities from 45ZCF-GREET without indirect land use change, as the One Big Beautiful Bill Act (OBBBA) provides from 2026. Under current law the credit ends after 2029, there is no SAF premium, and fuel from imported feedstock and sugarcane ethanol does not qualify. ATJ with carbon capture can claim 45Q instead. State SAF credits (Illinois, Minnesota, Nebraska, Iowa, Hawaii) are step schedules with budget caps, solved as complementarities on each step.

B.3 The year-t complementarity problem

Table 9 lists the conditions of the year-t fuel market and the variable paired with each. Given the state at the start of the year and the credit prices (LCFS prices from holders' rule, or from (16) when the year clears; RIN prices from the storage problem), the problem is a mixed complementarity problem in the listed variables.

Table 9: Conditions and paired variables of the year-t market

Condition	Variable	Equation
Zero profit of pathway i	output $Q_i \geq 0$	(31), (32)
Pool balance, fuel F, region j	retail price $P_{F,j}^{dem}$	(46), (38)
Fossil supply at the rack price	fossil volume $Q_{F0,j} \geq 0$	(34)
Feedstock balance, group k	price $P_k \geq \underline{P}_k$	(47)
Capacity of technology group g	rent $\rho_g \geq 0$	Section 4.2
SAF-capable capacity	rent $\rho_t^{SAF} \geq 0$	(37)
Utilization cost	ϕ_g	(35)
Feedstock-mix cost	w_i^{A1}	(36)
Delivery cost and California kink	τ_j, $E_{CA} \geq 0$	(33)
Blend walls and biodiesel share limits	shadow price $\nu_{f,j} \geq 0$	Appendix B.2
Crop-oil cap (California, from 2028)	rent $\mu^{cc} \geq 0$	Section 4.2
RFS requirements: BBD, advanced, total	$P_{D4} - P_{D6} - \mu^{ADV}$, μ^{ADV}, P_{D6}	(48)
LCFS balance, program j	P_j^L, surplus s_j^+, issuance s_j^-	(16), (49)
State SAF credit steps	step prices	Appendix B.2

B.4 Capacity investment

Hydroprocessing capacity (RD and HEFA-SAF) is endogenous from 2029; earlier capacity is under construction. Four types of investment add or convert capacity: (a) conversion of a petroleum refinery, (b) expansion of an operating biofuel plant, (c) a greenfield plant, and (d) conversion of

operating hydroprocessing capacity to produce SAF. In each year, each potential investor of type τ decides whether to commit now or wait. This is competitive, single-agent optimal stopping with logit shocks of scale σ, as in Rust (1987), with entry costs estimated from 46 capacity decisions. Investors take prices as given, which fits a national commodity market whose prices are set by the RFS and the credit markets; strategic interaction among entrants is not modelled.

The value of committing per gallon of annual capacity is

$$V^{c}_{\ell,t} = \sum_{s=L}^{L+19} \delta^{s} \Big[u^{e}_{t+s} \big(\mathbb{E}_t \, m_{t+s} + F_\ell / u_0 \big) - F_\ell \Big] - \kappa_{\ell,t}, \qquad \ell \in \{a, b, c\}, \tag{39}$$

with a construction lag $L = 3$, a 20-year life, $\delta = 1/1.07$ and κ_ℓ the entry cost (capital cost plus the hurdle the data imply, falling with capital learning). $F_\ell = \varphi \cdot capex_\ell$ is the fixed operating cost, charged on nameplate capacity at $\varphi = 4\%$ of capital cost a year (National Renewable Energy Laboratory, 2024; Davis et al., 2018); it is written so that the value is unchanged at the reference utilization $u_0 = 0.70$, where the history has the plants, and falls when expected utilization is lower. The margin m_t is the RD operating margin (fuel value plus RIN, LCFS and production credits, less feedstock, operating and delivery cost and the feedstock-mix cost w^{A1}) averaged over destinations with weights equal to each region's share of RD deliveries, net of the 2026–28 transitional terms and of the gap between the 2026–28 utilization ramp and the long-run utilization curve. Expected utilization u^{e}_{t} is the discounted average of the utilization the model solves for over the plant's life, so additional capacity lowers the value of further capacity. RIN and feedstock prices are expected to stay at current values; the LCFS component follows the path the rule implies and the scheduled standards; statutory credits that are due to expire are expected to continue with probability p, so $\mathbb{E}_t z_{t+k} = z^{law}_{t+k} + p\,(z_t - z^{law}_{t+k})$. The main specification sets $p = 1$ while a credit is in law and 0.18 during a lapse; estimated, they are 1 [0.67, 1.00] and 0 [0, 0.31] (Section 5.2).

Waiting has value because the margin may rise. Write $V^{c}_{\ell,t}(z)$ for the value of committing as a function of the margin index $z = m_t$. The value of waiting solves

$$V^{w}_{\ell,t}(z) = \delta \, \mathbb{E}\Big[\zeta \ln\Big(e^{V^{c}_{\ell,t+1}(z')/\zeta} + e^{V^{w}_{\ell,t+1}(z')/\zeta} \Big) \Big], \qquad z' = z + \Delta_{t+1} + s_z \, \varepsilon, \;\; \varepsilon \sim N(0,1), \tag{40}$$

where Δ_{t+1} is the expected drift of the margin from the scheduled LCFS standards, holders' LCFS price path and the statutory credits, and s_z is the standard deviation of the yearly change in the margin index in the history (the rest of the margin is a random walk). Once the drifts end, V^{w} is stationary. The logit shocks have mean zero, so κ_ℓ is the entry cost including the hurdle the data imply. A potential investor commits with probability

$$P_{\ell,t}(z) = \Big[1 + \exp\big(-(V^{c}_{\ell,t}(z) - V^{w}_{\ell,t}(z))/\zeta \big) \Big]^{-1}, \tag{41}$$

and expected commitments in year t are the sum over the pools of potential investors of the commitment probabilities times the expected capacity per commitment. The forward runs start from 91 refinery sites that could convert (the 106 sites in EIA's 2026 list less 15 with a conversion or co-processing project), 19 renewable diesel plants that could expand (the 2025 risk set) and 20 potential greenfield entrants a year (a normalisation). A site leaves the refinery pool when it announces a conversion and a plant joins the plant pool when built, so by 2045 the pools are 85.1 and 26.2. Refinery closures are not modelled.

The gain from converting a gallon of hydroprocessing capacity to SAF is the margin of SAF over RD actually earned on SAF sales, net of the RD output given up,

$$g_t = \max\big(0,\ S_t - \vartheta R_t\big), \qquad V_{saf,t} = u_{saf} \sum_s \delta^s\, \mathbb{E}_t g_{t+s} - \kappa_{saf,t}, \tag{42}$$

where S_t is the SAF-over-RD margin weighted by realized HEFA-SAF sales (zero when no SAF is sold), R_t the RD margin, $\vartheta = 1.125$ gallons of RD capacity lost per gallon of SAF capacity (Gerveni et al., 2025) and $u_{saf} = 0.70$. Every operating hydroprocessing plant is a potential converter, the conversion takes two years, and the decision follows the same logit rule. Conversion and new capacity share σ.

New capacity is not chosen to be SAF-capable in the model. Half of the refinery conversions of 2017–23 were built SAF-capable at a time when SAF earned nothing over RD, which no margin index can rationalize; in the data the choice looks like a nearly costless design option. Future refinery conversions are therefore SAF-capable in the historical proportion (0.4535 of conversion capacity), with $h_b = 0.2$ of their nameplate able to produce jet, the capacity-weighted share of Rodeo, Great Falls and Kapolei. SAF-capable capacity then evolves as

$$K_t^{SAF} = \min\Big(K_t^{H},\ K_{t-1}^{SAF} + A_{t-2}^{saf} + h_b \cdot 0.4535 \cdot A_{t-3}^{a} + \Delta_t^{proj}\Big), \tag{43}$$

where A^{saf} and A^{a} are conversion and refinery-conversion commitments, Δ_t^{proj} projects already announced (2026–28) and K_t^{H} hydroprocessing nameplate capacity.

Why hydroprocessing capacity keeps growing. An entrant's expected margin per gallon of output splits exactly into deliveries to the four LCFS states and everything else (national RINs, 45Z, and fossil-fuel value net of cost). From 2030 the second part is about zero: competition holds it at cost. The first part is \$1.00 a gallon in 2029, \$1.28 in 2030 and \$1.48–1.62 in 2031–45, and from 2030 99–100 percent of domestic renewable diesel goes to the four states. This margin is not the value of the LCFS credit: in California the credit term for renewable diesel is negative after 2036 (−\$1.69 a gallon in 2045). It appears instead in the gap between the value of fossil diesel in California and elsewhere, \$0.87 a gallon in 2030 and \$3.28 in 2045. Because capacity never

binds and costs rise only above utilization of about 0.86, added capacity changes no price; but an investor's return is the margin times utilization less the fixed cost of idle capacity, and the value of a gallon of capacity falls from $7.47 in 2031 to about $6.0 in 2040–45. Entry continues because the estimated logit rule includes a random taste term: at utilization near zero each potential entrant still enters with probability of about 2.8 percent a year. Of the 5.5 percent entry probability in 2045, about half comes from this floor; the rest comes from the rising margin, learning-driven falls in capital cost, and a larger pool of potential entrants. These are accounting decompositions of the main run. Re-solving current law with hydroprocessing capacity frozen at its 2029 level changes no credit price by more than $0.72 a tonne and no RIN price by more than $0.004, within the solver's tolerance, and net emissions over 2026–45 by 0.1 Mt; it lowers the present value of idle fixed costs from $4.6 billion to $2.0 billion. The model has not been re-solved without the floor.

B.5 Vehicles and charging

The vehicle block extends Cole et al. (2023) to light-, medium- and heavy-duty vehicles, five regions and the channels through which LCFS revenue reaches vehicle buyers. The electric share of new sales of class v in region j is logit in the ratio of total costs of ownership and, for light-duty vehicles, in the charging network:

$$\ln \frac{x_{jvt}}{1-x_{jvt}} = \alpha_{jv} - A^{ACT}_{jvt} + \beta_{1v} \ln \frac{TCO^{EV}_{jvt}}{TCO^{ICE}_{jvt}} + \mathbf{1}[v = LDV]\Big(\beta_2 \ln \frac{N^{L2}_{j,t-1}}{S^{EV}_{j,t-1}} + \beta_3 \ln N^{DC}_{j,t-1}\Big), \qquad (44)$$

Vehicle classes follow FHWA gross-weight classes: light-duty is classes 1–2 (including class 2b vans, with light-duty parameters), medium-duty classes 3–6, and heavy-duty classes 7–8, with heavy-duty costs dominated by class 8 tractors. Medium-duty costs, subsidies, sales, shares and mileage weight class 3 and classes 4–6 by sales. The charging block and the electricity credits use the same classes. For given prices each agent's choice has a closed form or a scalar root.

In equation (44), S^{EV} is the electric stock and N^{L2}, N^{DC} the Level-2 and DC fast public stations. For light-duty vehicles, β_1 and β_3 are estimated with $\beta_2 = \beta_3/2$ (Section D.8). For trucks, $\beta_{1v} = \eta_v / w_{p,v}$ maps the literature price elasticity η_v into the TCO coefficient, where $w_{p,v}$ is the purchase price's share of TCO: the elasticity of the electric truck's TCO with respect to its purchase price, computed by the model at the 2025 anchor (0.779 for medium-duty and 0.752 for heavy-duty trucks). The implied purchase-price elasticity is therefore η_v. The medium-duty η of -2.5 is the approximate median of light-duty estimates chosen by Cole et al. (2023, p. 318 and fn. 2), applied to medium-duty trucks; the heavy-duty η of -2.65 is the midpoint of the range of -2.5 to -2.8 in Weber (2024, p. 16). A^{ACT} removes the component of truck sales in California, Oregon and Washington attributable to the Advanced Clean Trucks rule, which is no longer enforced. The TCO includes the purchase price net of incentives and resale value, discounted fuel or electricity

cost, maintenance and registration fees. Households discount at 15% over 15 years, an assumption; 7% and 25% are sensitivities. Fleets discount at 7% over a five-year holding period, as in Zhao et al. (2026), who take these values from Ledna et al. (2024, Table 1); Basma et al. (2023, p. 7) use the same rate and period for class 8 trucks. For heavy-duty trucks the five years are Ledna et al.'s horizon for classes 7–8. For medium-duty trucks they are an assumption: Ledna et al. (2024) use three years for class 3 and four for classes 4–6. Residual values after five years are, for heavy-duty trucks, 35% for diesel trucks and, for battery-electric trucks, 49% for the battery (the model-year 2030 value of Basma et al., 2023) and 35% for the rest of the truck (Zhao et al., 2026, Supplementary Table 11). For medium-duty trucks they are 55% for diesel trucks, derived from the depreciation parameters of Burnham et al. (2021, Table 3.11), and 40% for battery-electric trucks, an assumption. Federal purchase credits end after September 2025. Rebates funded by utilities' residential LCFS credits lower the purchase price. State purchase incentives (California's HVIP, Washington's sales-tax exemption and WAZIP, Oregon's rebates, New Mexico's tax credit) enter only to the extent of announced funding and are zero after the last funded year. A truck's combined purchase incentives are capped at 90% of the electric truck's price, the HVIP stacking rule; credits the owner earns directly are outside the cap. The intercepts α_{jv} match 2025 sales by region and class with the incentives in force in 2025 and are held fixed in every counterfactual.

Public Level-2 and DC fast stations enter freely at the regional level, as in Zhou and Li (2018) and Springel (2021), until a station's annual margin, including the LCFS credits it earns, equals the user cost of building now rather than next year,

$$\frac{D^k_{j,t}}{N^k_{j,t}+X^k_{j,t}}\left[\mu^k_j (N^k_{j,t})^{-\eta_k} + P^L_{j,t} e_{j,t}\right] = \left[(C^k_t - g^k_{j,t}) - \frac{(1-\varrho)(C^k_{t+1} - g^k_{j,t+1})}{1.07}\right], \qquad \eta_k = \frac{1}{\upsilon_k} - 1, \quad (45)$$

where D^k is public charging demand, e credits per kWh, C^k the installed cost, g^k grants and tax credits, ϱ depreciation and X^k stations built outside this market. Grants and the federal 30C credit enter through the installed cost net of grants. The user cost is floored at 10% of the installed cost, a floor that binds in many region-years. $\varrho = 0.1072$ is the annual depreciation in the user cost; the station stock itself does not depreciate. Next year's cost is evaluated under the law known in year t: the federal charger credit (30C) is expected to continue in 2025 and earlier years, as the Inflation Reduction Act then provided, and to end in mid-2026 from 2026 on, after OBBBA. The entry elasticities υ_k are estimated by simulated method of moments (Section D.8). Depot chargers for trucks have no entry decision; their capital cost is reduced by announced grants (30C through mid-2026, California's EnergIIZE and utility make-ready programs, Oregon's program).

The electric stock accumulates sales by vintage. The block returns the electric shares of the fleet, which enter fuel demand (38), and electricity credits $\Gamma^{EV}_{j,t}$, computed per kWh as $(\sigma^*_{j,t} - CI^{el}_{j,t}/EER_v)\, EER_v \cdot 3.6 \cdot 10^{-6}$ at the standard in force (after acceleration or deferral), which enter the LCFS balance (16). Electricity credits are about a fifth of California's credit generation, so the

bank after 2030 depends on the pace of electrification.

In the setting of Section 3 with one region, suppose electric vehicles generate E credits. If $\lambda = \bar{P}$, the fuel price, biofuel and fuel use are set by $\bar{P}$ alone, and a marginal change in E changes only the shortfall issued at the cap: $\partial X^{+}/\partial E = -1$ and $\partial Z/\partial E = 0$. Below the cap, $\partial\lambda/\partial E < 0$ and, by equation (9),

$$\frac{\partial Z}{\partial E} = 1 + \sigma\, D'\left(c_0 - \sigma\right)\frac{\partial \lambda}{\partial E} \;>\; 1.$$

Holding the electric fleet's direct effect on fuel demand fixed, each credit from electricity permits at least one more unit of fuel-cycle emissions below the cap, because it also lowers the credit price and the fuel price; at the cap it has no effect on fuel emissions.

B.6 Equilibrium, solution and welfare

Given the state of the economy at the start of year t (capacity K_t^H and K_t^{SAF}, vehicle fleets, LCFS banks and expected exhaustion years, the RIN bank and the standards in force), the fuel market clears when the zero-profit conditions (31) hold for every pathway, fuel demand is (38), and the following balances hold, each paired with its price. For each fuel pool F and region j,

$$Q_{F0,j} + \textstyle\sum_{i\in F,j} \varphi_{f(i)}\, Q_i = Q_{F,j}^{dem}. \tag{46}$$

For each feedstock group k, use equals supply, with the floor price $\underline{P}_k$ (Appendix B.2),

$$0 \le P_k - \underline{P}_k \;\perp\; S_k(P_k) - \textstyle\sum_{i:\,k(i)=k} \theta_i Q_i - \theta X_k^{exp} \;\ge 0. \tag{47}$$

For each technology group g, output is within capacity, $0 \le \rho_g \perp K_{g,t} - \sum_{i\in g} Q_i - X_g^{exp} \ge 0$. The three nested RFS requirements clear with the year's draw on the RIN bank, $\Delta B_t = B_{t-1} - B_t$, which counts toward the total requirement:

$$\begin{aligned} 0 \le P_{D4} - P_{D6} - \mu_t^{ADV} \;&\perp\; \textstyle\sum_{i\in BBD} \kappa_i Q_i - \pi_t^{BBD}\mathrm{Base}_t \ge 0, \\ 0 \le \mu_t^{ADV} \;&\perp\; \textstyle\sum_{i\in ADV} \kappa_i Q_i - \pi_t^{ADV}\mathrm{Base}_t \ge 0, \\ 0 \le P_{D6} \;&\perp\; \textstyle\sum_{i} \kappa_i Q_i + \Delta B_t - \pi_t^{RF}\mathrm{Base}_t \ge 0, \end{aligned} \tag{48}$$

with RIN generation net of exports and $BBD \subset ADV$. Each LCFS program clears by (16), with net credit generation

$$\begin{aligned} N_{j,t} = \;& \textstyle\sum_{i\in j}(\sigma_{F(i),j,t} - CI_i^L)\, ED_i\, Q_i \cdot 10^{-6} \\ & - (1-\iota_{j,t}) \textstyle\sum_{F\in\{G,D\}} (CI_{F0,j} - \sigma_{F,j,t})\, ED_F\, Q_{F0,j} \cdot 10^{-6} + \Gamma_{j,t}^{EV} + \Gamma_{j,t}^{fix}, \end{aligned} \tag{49}$$

where Γ^{fix} are credits from fixed sources. Appendix B.3 lists the full problem, variable by variable.

Because LCFS holders do not have rational expectations, the object I compute is a temporary equilibrium path: each year's prices clear the year's markets given expectations formed from the past, rather than a rational-expectations path.

Definition 2. Given the 2025 state and the exogenous paths (oil prices, standards as scheduled, statutory credits, exemption scenario), a temporary equilibrium path is a sequence of annual allocations, prices, expected exhaustion years, banks, capacity and fleets for 2026–2045 such that

(i) in every year, the fuel market clears as above, given that year's capacity, fleets, banks and standards;

(ii) from 2027, each LCFS price satisfies holders' rule (17)–(19) given their expected exhaustion years, and these and the banks are updated from the realized net credits and any issuance or surplus, $B_{j,t} = B_{j,t-1} + N_{j,t} + s^{-}_{j,t} - s^{+}_{j,t}$;

(iii) the RIN bank path solves the storage problem (27) on the net-supply curves of the path;

(iv) capacity follows commitments (21) evaluated on the path, with the construction lag;

(v) fleets and stations follow (44) and the entry conditions at the path's prices; and

(vi) acceleration and deferral decisions (30) are those the rules imply on the path.

Each year's fuel market is a mixed complementarity problem, solved with PATH (Dirkse and Ferris, 1995); Gurobi then checks the solution against every constraint of the original formulation. The feedstock-mix cost (36) keeps the problem linear with a symmetric positive semidefinite Jacobian, and removes the ties in feedstock choice that made solutions depend on the starting point. Within a year, each LCFS price is a one-dimensional fixed point found with the price pinned, and the four programs are swept in turn. The blocks are coupled by damped iteration over the whole 2026–45 path: prices from the previous round give vehicle sales and credits, learning effects on cost, and capacity; the RIN storage problem is re-solved on the updated supply curves; and every year is re-solved with the new RIN path. The iteration stops when, between rounds, LCFS prices move by less than \$0.5/t, rack prices by less than \$0.005/gal, capacity by less than 0.02 billion gallons, deferral amounts by less than 0.02 years and the RIN storage conditions hold within \$0.05 per RIN. Acceleration and deferral decisions are then re-checked on the converged path; if one changes, the iteration resumes. Prices are unique in every check; some quantities are not (when two uses of a fuel are equally valued), so I report aggregates.

C Data and Parameters

Table 10 lists the parameters that matter most for the results, ordered by my judgment of their influence on the headline numbers (based on the sensitivity runs). For each it gives where the value

comes from: estimated on the history, calibrated to a single year, exactly identified by 2026 targets, borrowed from the literature, set by statute, or assumed. Appendix C describes the data sources.

Table 10: Parameters: sources and identification

Parameter	Value	Origin	Data and precision	Sensitivity
Expectations				
Exhaustion horizon K	2 (Rule B)	Estimated, weakly	CA and OR, quarterly 2016–25 ([1.07, 2.28] with shift terms and forward quotes, $r = 3.38\%$); annual data do not identify it under Rule A (Section 5.2)	$K = 1.07$, 2.28 (interval ends); $K = 3$ without shift terms
Response to balance surprises γ_{CA}; pooled g (OR, WA, NM)	1.108; 0.159	Estimated	Same	parameter bootstrap (r, γ, g)
Investors' extension probability p; lapse value	1; 0.18	Set; estimated as 1 [0.67, 1.00] and 0 [0, 0.31]	46 decisions, 2014–25 (15 in lapses); lapse value from RIN prices	$p = 0.5$; $p = 0$
Fuel market				
California RD kink	share 0.7213, slope 2.396	Observed maximum; fitted	CARB quarterly 2019Q1–2026Q1; slope rests on 2 quarters (s.e. about 1.0)	No kink; slope $\times\frac{1}{2}$, $\times 2$; kink share as a parameter with an interval
2026 transitional terms (CA, OR, WA absorption; imported fat supply)	CA 0.131, OR 7.52, WA c 0.058	Exactly identified	Four 2026 targets	Kept permanently; faded to 2023–25 values
End-2025 RIN bank	3.21 bn (D4 1.40)	Calibrated	EMTS (21 Sept 2026); EPA has not published the bank	2.27–3.42; 6.0
Feedstock share response c (RD and BD)	0.283	Estimated (chosen on a grid; no standard error)	CA, WA, NC fat shares 2016–25 (about 39 observations)	RD response at the ends of its two-parameter interval
Hydroprocessing and BD utilization curves	HEFA a 1.381; BD a 2.761, u_0 0.655, kink 0.83	Estimated; level calibrated	HEFA slope: EIA monthly 2022–26 with year effects; BD level: 2016–24	HEFA slope 0.10; BD slope 0.285 and 0.017
RD delivery slopes d_j (CA, OR, WA)	0.100; 3.83; 0.68	Estimated	Quarterly 2023Q1–2025Q4 (WA 2024Q3–2025Q4), intercepts fixed	2025 single-year values
Feedstock supply elasticities	soy 1.5, fat 3, imports 5, corn oil 0.5	Set (soy, fat); assumed (imports, corn oil)	Soy / veg oil 1.5: set; out-of-sample price fit favours 1.5–2. North American fat 3: set; out-of-sample price fit gives 1.5–3 depending on the price series. Imported fat / UCO 5: assumed (residual-supply formula, Baker and Bresnahan, 1988). Distillers' corn oil 0.5: assumed (by-product)	Soy 0.75, 2.25; fat 1.5, 4.5; imports 2, 10; corn oil 0.25, 1.0
Fuel demand elasticities	G -0.37, D -0.07, J -0.26	Literature	Coglianese et al. (2017); Dahl (2012)	Retail -0.20
Oil price path	STEO–AEO rule	Data, EIA	—	AEO high and low price; Monte Carlo

The estimated investment parameters in force in every reported run are the joint estimates with the expansion and greenfield entry costs pooled ($\zeta = 2.79$, $\kappa_a = 20.25$, $\kappa_b = \kappa_c = 10.5$, $\kappa_{saf} = 3.0$); the SAF conversion cost falls with capital learning, $\kappa_{saf,t} = 3.0 - 1.38\,(1 - \ell_t)$, to 2.75 in 2045. The two engineering-cost rows of Appendix G.10 replace them with capital costs (1.9, 3.44, 3.4, 1.38). The capital costs are from announced project costs, in 2024 dollars per gallon of annual nameplate capacity: refinery conversions, the median of six U.S. conversions; expansions, Diamond Green

Table 10: Parameters: sources and identification (continued)

Parameter	Value	Origin	Data and precision	Sensitivity
Investment				
Logit scale ζ; entry costs κ_a, κ_{bc}	2.79; 20.25; 10.5	Estimated	1,492 decision-years, 46 decisions	Capacity fixed; $p = 0$
SAF conversion cost κ_{saf}	3.0 [−1.35, 9.70]	Estimated	6 conversion events; bootstrap relative s.e. about 100%	Engineering cost (about $1.38 plus lost RD capacity)
Fixed O&M share φ	0.04	Literature	TEA, 2.7–6.5% (National Renewable Energy Laboratory, 2024; Davis et al., 2018); not identified by history	0.03; 0.05
SAF-capable share of new conversions h_b; conversion loss ϑ	0.2; 1.125	Calibrated; literature	Rodeo, Great Falls, Kapolei; Gerveni et al. (2025)	$h_b = 0.5$; $\vartheta = 1$
Vehicles				
LDV price and network coefficients b_1, b_3	−4.92 (0.35); 0.31 (0.10)	Estimated (SMM)	2021–25 sales shares and station counts	Price elasticity −2.8
Truck price elasticities	MDV −2.5, HDV −2.65	Literature	MDV: light-duty median, Cole et al. (2023, p. 318); HDV: midpoint of −2.5 to −2.8, Weber (2024, p. 16)	−6; −10
Statutory parameters (45Z, 45Q, RIN values, RFS percentages, LCFS schedules, caps, AAM)	Section 2	Statute	Federal Register; state codes	45Z, 45Q extensions are counterfactuals

Diesel's 400-million-gallon Norco expansion (about $1.1 billion in 2018); new plants, the cost per gallon of Artesia ($350 million for 125 million gallons) and of Diamond Green Diesel's Port Arthur plant ($1.43–1.45 billion for 470 million gallons), rounded to $3.4; SAF retrofits, Diamond Green Diesel's Port Arthur SAF project ($315 million for 235 million gallons).

Four sets of parameters are calibrated to 2025, the last complete year, by matching observed quantities at observed prices: the level of the hydroprocessing utilization curve, the feedstock-mix intercepts a^{A1}, the corn oil supply anchor and the 2025 RIN bank. A single year identifies a level, not a slope; in a calibrated model with quadratic costs, one year's first-order conditions are an underdetermined system, and different slope choices that fit the base year equally well imply different simulation behavior (Heckelei and Wolff, 2003, pp. 29–30). I therefore checked each one-year parameter against its own history, and report the multi-year value as a sensitivity where the history is informative.

The RD delivery slopes have a structural break around 2022. California's annual slope was $2.1–3.5 per gallon per billion gallons in 2016–21 and $0.02–0.18 in 2023–25; Oregon's fell from 67–89 to 2.5–7.1. The annual slopes move with the year's credit price: the elasticity of the log slope with respect to the log of credit value per gallon delivered is close to or above one in the annual data. A slope calibrated to a single year is therefore largely that year's LCFS premium spread over that year's RD volume, a residual rather than a fixed transport cost. The main specification instead estimates the slopes on quarterly data after the break (California 0.100 (0.022), Oregon 3.83 (0.78), Washington 0.68 (0.09; six quarters), Newey-West), and the 2025 single-year values are a sensitivity. On quarterly data the slope still co-moves with credit value per gallon (elasticity 1.29

in California, R^2 0.986), so when California's price rises toward its cap a fixed slope understates the marginal cost of absorbing more RD there; in the forward runs the kink in equation (33) carries this.

2025 is not a representative year for biodiesel. It was the first year of 45Z, which replaced the \$1 blenders' credit and excluded imported feedstock, and biodiesel utilization fell to 3.6 standard deviations below its 2016–24 mean. The level of the biodiesel utilization curve is therefore set from 2016–24, and 2025 is treated as a one-off shock year. The 2025 feedstock-mix intercepts for biodiesel also break from their 2016–24 values; their 2021–24 means are a sensitivity. The other one-year anchors (corn oil, the California RD feedstock intercept) are in line with their recent history.

The 2026 LCFS prices of California, Oregon and Washington are pinned (Section 4.1). Four 2026-specific terms are then exactly identified by four 2026 targets: the RD delivery terms of California, Oregon and Washington and the supply level of imported fat, which absorb the one-off shocks of 2026 (the oil price spike, OBBBA's feedstock-origin rule and tariffs, California's July 2025 amendments). They fade out by half in 2027 and fully in 2028. Holders' and investors' expectations include the fade-out as a known change. The other 2026 observations remain out of sample (Section 5.4). The RFS is not pinned: the D4 and D6 prices of 2026 are outcomes, and the RIN storage problem iterates with the fuel market until the 2026 D6 price used in calibration and the price the run produces agree within \$0.05. I have not found a study in this literature that treats a shock year with terms that exist only in that year; the closest analogues start from an observed state and let the model take over (Borenstein et al., 2019; Bushnell et al., 2020; Fowlie et al., 2016).

EPA has not published the RIN bank after 2024. EPA's final 2026–27 rule uses an effective 2024 carryover of 3.60 billion RINs, which it assumes carries unchanged into 2026–27 (U.S. Environmental Protection Agency, 2026b, Table 1.8.2-1). The main specification uses 3.21 billion, with 2.27–3.42 and 6.0 as sensitivities; the 2026 D6 price is sensitive to this number. Expected small-refinery exemptions for decisions in 2025 are 0.78 billion RINs, from the September 2025 supplemental proposal. From 2026 the main specification uses the 2023–25 average exempted volume, 10.07 billion gallons; the final rule's 1.11 billion RINs (7.55 billion gallons) is a sensitivity (Appendix G.6) (U.S. Environmental Protection Agency, 2025, 2026c).

A few inputs have no public source and are labelled as assumptions with sensitivities: the share of new chargers in census tracts eligible for 30C (0.60; for reference, about two-thirds of the population lives in eligible tracts, U.S. Department of the Treasury, 2024), the split of 2025 HVIP redemptions by truck class (the 2023–24 request mix; public data give the request year, not the redemption date), the shares of depot-charger grants going to electric trucks, and the class 3 counts outside California and Washington.

D Estimation

D.1 The price-expectation rule

From 2022 California's credits exceeded deficits every year and the bank quadrupled, yet the price stayed positive and then fell. A bank growing at a positive price is what Hotelling predicts when holders expect the standard to bind later, so the question is quantitative. Read through the cap and a holding cost, the price implies an expected exhaustion year, $t + \ln(\bar{P}/P_t)/\ln(1+r)$, and that year moved by decades over 2016–26 (Section 2.1). None of the three standard descriptions of expectations reproduces this history. A static market is rejected by inspection: the bank is positive and the price is not zero. Figure 12 makes the comparison quarter by quarter: it plots the price at which California's historical fuel market would have generated no net credits in the quarter, the static price λ^* of Section 3, against the observed price. In 2019–21 the two are close (an average of \$221 against \$231); in 2022–24 the observed price is above it (\$92 against \$42), the difference being the value of expected future scarcity; and from 2025Q3, after the amendments, the static price is above the observed price (about \$60), which is the doubt Section 2.1 describes. The static price is censored at zero or at the cap in 16 of 41 quarters, and its 2025–26 level depends on the slope of the credit supply above \$60.

Perfect foresight fails whether or not it is given hindsight: its quarterly price error over 2016–25 is 84.5 when holders know only the rules adopted by then and 80.9 with the rules as finally adopted, against 34.0 for the rule without its three shift terms (estimated at $r = 5\%$) below. With the rules of each year it implies prices of \$34 in 2024–25, against \$63 and \$59; once the 2024 amendments are known it implies \$177 for 2025 and \$186 for 2026, against about \$63. Before the amendments holders priced more scarcity than the rules of the day implied, and after them much less than the amended rules imply. The third description, a market that prices the chance that the regulator relaxes or tightens the standard, does not fit, and the reason is itself informative (below).

I therefore describe holders' expectations with the rule (18)–(19), and I treat it as a description of how the credit price responds to the balance, estimated on the history, and not as a model of how holders form beliefs: the rule maps the credit balance and the scheduled standards into the price holders expect, it is judged by whether it gets the direction and size of price responses to the balance right, and the equilibrium path is computed with it. Unlike the forecasting rules of Krusell and Smith (1998), it is not required to be accurate along the model's own path, and it is not: under current law holders expect in 2026 that the bank will last until about 2069, while it runs out at the end of 2031, and their expectation closes the gap only as unexpected deficits accumulate. Corollary 1 is this mechanism: holders who extrapolate a large bank keep the price low and draw the bank down faster. A holder who foresaw the 2031 exhaustion would price California's credit at about \$221 in 2026, against the \$64 observed. Its parameters (γ_{CA}, the pooled small-program response g, K and the initial expected exhaustion years) are estimated by maximum likelihood on

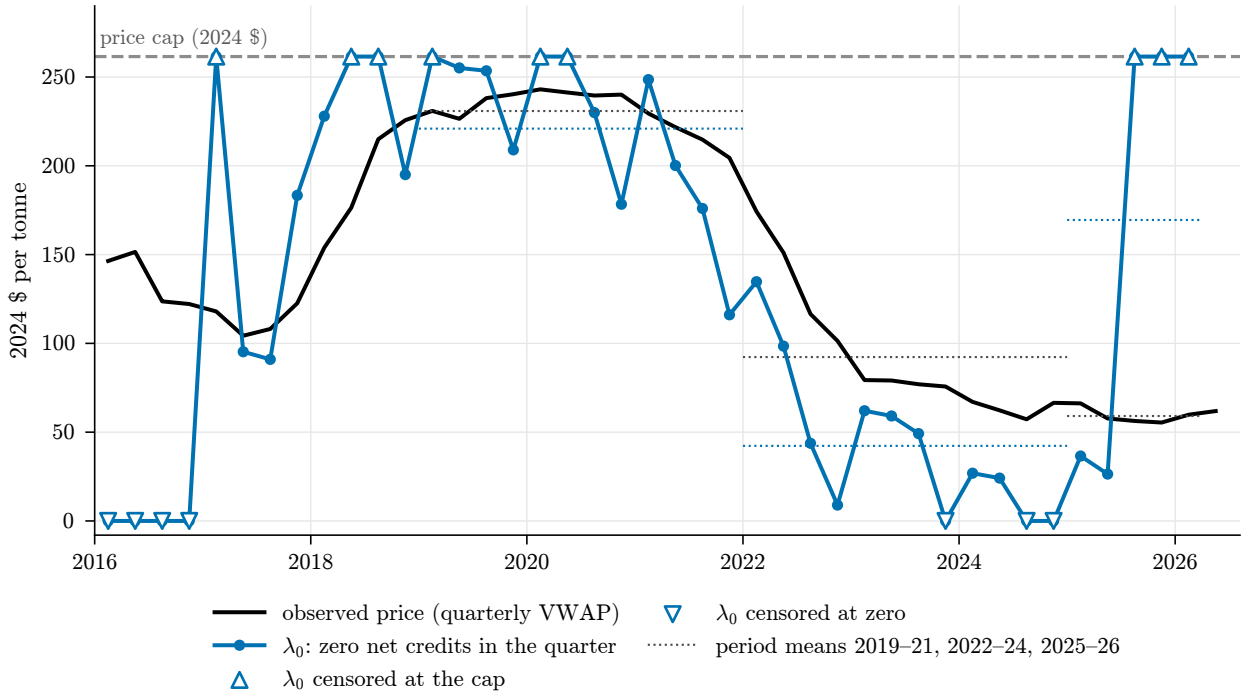


Figure 12: California's observed credit price and the static price at which the quarter's net credits are zero, 2016–26

Notes: Observed quarterly price and the static price λ^* (λ_0 in the figure) from California's historical fuel market; open triangles mark quarters in which the static price is censored at zero or at the cap; dotted lines are period means (2019–21, 2022–24, 2025–26). Because of the censoring, the 2019–21 and 2025–26 means are lower bounds and the 2022–24 mean is an upper bound. 2024 dollars. Sources: CARB LCFS data; model's historical fuel market.

the observed prices and banks of California and Oregon, 2016–25, with normal errors on log prices and on banks. For a parameter vector ω, the historical model gives each program's price $P^L_{j,t}(\omega)$ and bank $B_{j,t}(\omega)$, and

$$\ln L(\omega) = \sum_{j\in\{CA,OR\}} \sum_t \Big[\ln\phi\Big(\frac{\ln(P^{obs}_{j,t}+10) - \ln(P^L_{j,t}(\omega)+10)}{s_P}\Big) + \ln\phi\Big(\frac{B^{obs}_{j,t} - B_{j,t}(\omega)}{s_{B,j}}\Big) - \ln s_P - \ln s_{B,j}\Big], \tag{50}$$

with ϕ the standard normal density and the error scales estimated with ω. The shift of \$10 a tonne keeps the log errors well behaved at the low prices of 2016–17 and 2025; forward quotes enter the same way. The credit supply comes from the historical fuel market solved at each candidate price.

Three annual shift terms (2018, 2023 and 2024) enter the history only; they are not used in forward years. They fall in rulemaking years, but no dated announcement moved prices by the amounts they imply (below), and they are not the rational response to the changes announced in those years: the 2018 term moves the expected exhaustion year by −34.9 years, while the announced change moves the perfect-foresight exhaustion year by +0.18. I describe them as the years in which the gap between the price and the statutory path opened. The forward runs, which give the paper's

results, do not use them; the annual replay of the full model in Appendix E includes them in the history years (Table 16). The sample ends in 2025.

For the horizon K, on quarterly data, 2016–25 (40 observations for California), Rule B, with the forward quotes in the likelihood, gives a 95% interval for K of [1.07, 2.28] (best grid value 1.5; on spot prices alone 2.29 [1.61, 3.04]). The interval rests on extrapolation: no program in the sample came close to exhausting its bank. Without the three annual shift terms the estimate is [2.54, 3.47] (best grid value 3; [2.81, 4.42] on spot prices alone), which excludes the $K = 2$ used in the forward runs; that value lies within the interval of the estimate with the shift terms. The holding cost r is set, not estimated. Across r from 0.25% to 10%, $\hat{K}$ stays between 2.26 and 2.84, but $\hat{\gamma}$ moves roughly in proportion to $1/r$ (0.76 at 5%, 1.08 at 3.38%, 1.30 at 2.8%, 0.50 at 8%), and the likelihood rises without an interior maximum as r falls, so the data do not separate r from γ. What they pin down is close to the product $\gamma \ln(1 + r)$, the response of the log price to a balance surprise, which is what the counterfactuals use. I set r at 3.38%, the real carry measured from forward quotes (below), and report γ conditional on it. Evidence from other bankable-credit markets points to a short horizon. Prices are forward-looking and react to news within a day or two, but large rises come as jumps once scarcity falls within a limited horizon: in the 2013 RFS episode, D6 RIN prices jumped about one to two years before the RIN bank was projected to run out (Irwin and Good, 2013; Lade et al., 2018b); Acid Rain Program prices moved about five years ahead of the Clean Air Interstate Rule (Schmalensee and Stavins, 2013); rolling-horizon models of the EU ETS fit with horizons of about 10–13 years (Quemin and Trotignon, 2021; Sitarz et al., 2024); and RECLAIM, with no banking, is the $K = 0$ limit (South Coast Air Quality Management District Governing Board, 2001).

The main specification is Rule B (expected net generation as the three-year average, Nerlove, 1958) with $K = 2$, applied to all four programs. Rule A is the alternative; on the quarterly data Rule B fits markedly better (AIC higher by 22.1 for Rule A). Oregon, Washington and New Mexico share one response per unit of obligation. Oregon's bank never fell in the sample; the rule turns small surprises in a small credit pool into large revisions, and its backtest is worse than a random walk. I treat Oregon's price path as the least reliable of the four. Per percentage point of the obligation, the responses estimated separately for each program are similar: 0.192 in California (spot prices only, like Oregon and Washington; 0.197 in the main specification), 0.167 in Oregon and 0.154 in Washington (pooled over Oregon and Washington, 0.159).

The holding cost is set at 3.38%; the data identify the product $\gamma \ln(1 + r)$, which is what the counterfactuals use. OPIS quotes for adjacent December deliveries imply a nominal carry of 5.7% a year ([5.3, 6.1], 2022–26). Net of expected inflation over the same days (the five-year breakeven rate, 2.3%) the real carry is 3.4% ([2.9, 3.8]); with the Cleveland Fed's one-year expectation it is 3.0%. The first is the rate used here, the real carry measured from forward quotes; the 5% of an earlier version lies above both. The spread is a one-year horizon, and the breakeven rate is a five-year

rate that includes liquidity and risk premia. Because the rule enters the counterfactuals through $\gamma \ln(1+r)$, a lower r with the same product leaves the results under the rule unchanged; it changes the perfect-foresight paths and RIN storage, and puts the price-implied exhaustion year later. From the forward quotes it is 2057–2076 at a real carry of 3.4% (from CARB's daily transaction prices, 2073 in 2025) and 2047–2060 at 5%, against a first year at the cap of 2032 in the model. On spot prices alone the rule at $r = 3.38\%$ gives $\gamma_{CA} = 1.08$ and $\hat{K} = 2.29$; with the forward quotes (the main specification) $\gamma_{CA} = 1.108$ at $K = 2$, and $\gamma_{CA} \ln(1+r)$ is 0.037 at both rates. Moving from $r = 3.38\%$ to $r = 5\%$ with $\gamma_{CA} = 0.753$ (the same product) in the rule, in perfect foresight and in RIN storage changes the cost per tonne of current law by 0.1%, the share of the 45Z extension reaching fuel buyers by 0.007 and its emissions by 0.2 Mt; the saving per additional tonne emitted under the extension rises from \$51 to \$58.

Table 11 compares the descriptions of expectations by the error of rolling forecasts of California's price from each quarter's observed state, at horizons of one to twelve quarters, against a no-change forecast. As usual for asset prices, none beats no-change at one quarter in a way the data can detect (Meese and Rogoff, 1983; Alquist et al., 2013; Irwin et al., 2020). Given the net credits that actually occurred, however, the rule forecasts prices far better than no-change (errors of 34, 46 and 50 at four, eight and twelve quarters, against 50, 87 and 117, with the full-sample parameters; Figure 14). Re-estimated at each origin, it still beats no-change, by less (Appendix D.3). It maps the credit balance into prices well; its weakness is forecasting the balance. From the observed state its forecasts are biased upward (mean errors of \$41 and \$108 at four and eight quarters, against \$35 and \$81 for no change), because from a state with expected net credits at their three-year average the rule can only predict a rise; given the realized balance the bias is downward with full-sample parameters (\$15 and \$20 at four and eight quarters) and upward with re-estimated parameters (\$16 and \$46), the difference coming from the small response estimated at origins before 2022Q3. The one-step forecast of 2026 from the end of 2025 is \$68 against \$64 observed to 18 September. The counterfactuals need a mapping from the balance to prices, which a no-change forecast cannot supply, and the upward bias is one reason every result is also reported under perfect foresight and with the response at its model-free value.

The rolling horizon of Quemin and Trotignon (2021), in which holders price the bank over a finite horizon H, is not identified at the measured carry: the 95% set for H is not an interval ($\{6.5\text{–}7.5\} \cup \{8.25\text{–}8.75\}$ years), the version with a zero price beyond the horizon goes to the 40-year bound, and its AIC exceeds Rule B's by 91 with the forward quotes. Re-estimated at each origin it is significantly worse than no change one quarter ahead, and it forecasts \$93 for California in 2026.

Table 11: Rolling forecasts of California's credit price from the observed state

	No change	Rule B (main)	Rule A	Regulatory risk ($r = 2.8\%$)	Rolling horizon
1 quarter	14.8	13.5	14.3	67.7	59.1
4 quarters	49.9	43.0	44.6	69.1	59.3
8 quarters	87.4	89.1	85.4	77.9	70.6
12 quarters	117	132	127	110	104.3

Notes: Root mean squared error, \$/t, 2016–25, forecasts from each quarter's observed price and bank. Regulatory risk at $r = 2.8\%$; rolling horizon at $r = 3.38\%$. Rule B does not differ significantly from no change in California (Diebold–Mariano); in Oregon and Washington it is worse at one quarter ($p = 0.03$). The static model and the perfect-foresight benchmarks, which are solved as paths rather than rolled forward, are compared with the history in the text above. Path errors of the full replay are in Section 5.4.

D.2 The probability that the standard binds

Read through the bank's fundamentals rather than through an expected exhaustion year, the price implies a probability that the standard binds as written: $q_t = P_t(1 + r)^{T_t^f - t}/\bar{P}_t$, with T_t^f the exhaustion year implied by the bank and the current shortfall, or by the fundamentals projection of Table 14 with and without its trend term. This is the pricing of a permit as the discounted penalty times the chance of a shortage, applied with the cap as the penalty (Seifert et al., 2008; Borenstein et al., 2019). In California q is about one in 2019–21 and falls to between 0.26 and 0.53 in 2024–25 across the three fundamentals measures and holding costs of 2.9–3.8%; Oregon falls from 0.69–0.74 to 0.33–0.40, and Washington from 0.53 in 2023 to 0.16 in 2025 (Figure 13). Weekly changes in q in the weeks of regulatory events are not distinguishable from ordinary weeks (California, permutation $p = 0.25$). Allowing the exhaustion year to be uncertain, with the spread the rule's own updates imply, raises the expected discount factor by a third in 2026 and lowers California's 2024–25 q only slightly (0.48 to 0.43 on the projection with trend). The storage rows of Table 4 are the equilibrium counterpart of this reading.

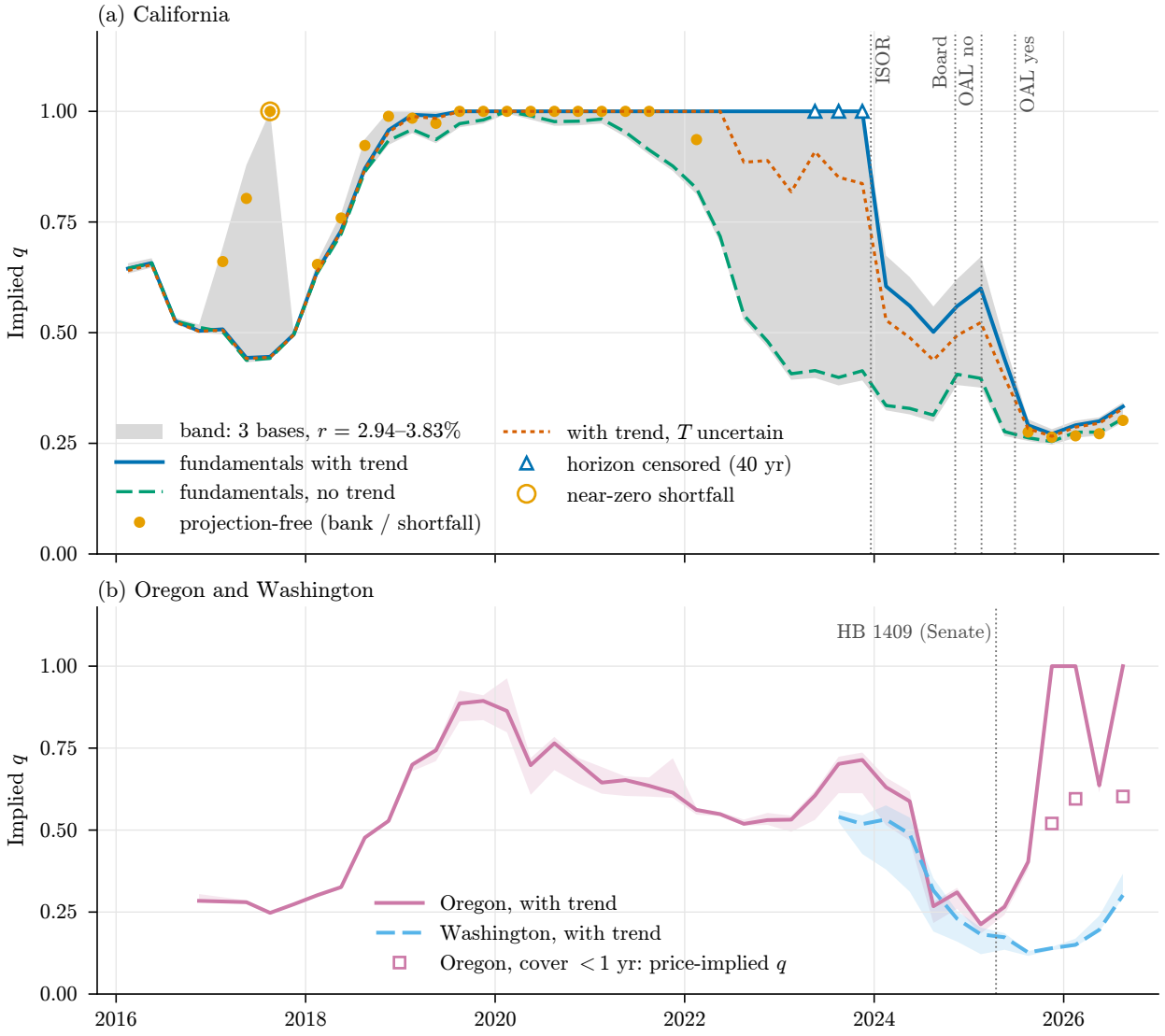


Figure 13: The probability that the standard binds implied by credit prices, California, Oregon and Washington

Notes: $q_t = P_t(1+r)^{T_t^f - t}/\bar{P}_t$, capped at one, with T_t^f from the bank and the current shortfall or from the fundamentals projection with and without its trend term; band over the three measures and holding costs of 2.94–3.83%. Quarters with less than a year of bank cover keep $q = P/\bar{P}$. The 2017Q3 point (open circle) is capped at one: the bank then covered about 35 years of a shortfall that had nearly vanished. Three 2023 quarters reach the 40-year horizon of the fundamentals projection; at 3.38% their q is one in any case. Sources: CARB, Oregon DEQ and Washington Ecology credit data.

D.3 Out-of-sample test of the expectation rule

The test was specified before it was run. At each quarter from 2019Q4 to 2024Q4, the rule's parameters are re-estimated with the same likelihood, data and holding cost ($r = 3.38\%$) as the main estimate, using only information dated at or before that quarter, including forward quotes and regulatory events. The rule then forecasts the price one to twelve quarters ahead in two ways. The first starts from the observed state and must also forecast the balance of credits; it contains no estimated parameter, so re-estimation does not change it. The second is given the net credits that actually occurred after the origin. It is a conditional forecast, not one that could have been made at the time, and it tests the part of the rule the model relies on: how the price responds to the balance. Both are compared with a no-change forecast on the same origins, by root mean squared error and by Diebold–Mariano tests with a HAC variance and the Harvey, Leybourne and Newbold correction. The test statistic is not computed twelve quarters ahead, where there are fewer than twice as many origins as quarters. Washington has fewer than eight quarters of history before every origin and is not tested.

In California the forecast given the realized balance has a lower error than no change at every horizon, and the difference is significant eight quarters ahead ($p = 0.001$); it gets the direction of the price change right at 86 to 100 percent of origins (Table 12). The forecast from the observed state alone has a higher error than no change at every horizon. In Oregon both forecasts have higher errors than no change at every horizon.

Figure 14 repeats the California exercise with the parameters fixed at the full-sample estimate, on all origins from 2016. The given-balance forecast's errors are then \$34, \$46 and \$50 four, eight and twelve quarters ahead, against \$50, \$87 and \$117 for no change, and it gets the direction right at 81 to 96 percent of origins at these horizons. Because these parameters use data from after each origin, the figure shows fit, not out-of-sample performance. On the out-of-sample origins, the full-sample parameters give errors of \$24, \$27 and \$21 (Table 12, panel B); the gap comes from the small response to the balance estimated at origins before 2022Q3.

Table 12: Out-of-sample test of the expectation rule: forecast errors against a no-change forecast

Panel A. Parameters re-estimated at each origin; origins 2019Q4–2024Q4. RMSE, 2024 $/t; DM p-value vs no change in parentheses.

Market	Quarters ahead	n	Given realized balance	Observed state only	No change	Direction, given balance
CA	1	21	10.9 (0.117)	14.5 (0.002)	13.7	86%
CA	4	21	39.9 (0.400)	53.2 (0.113)	47.9	95%
CA	8	17	78.6 (0.001)	119.6 (0.019)	94.7	100%
CA	12	13	107.9 (–)	182.4 (–)	138.0	100%
OR	1	21	28.5 (0.022)	29.4 (0.024)	18.7	33%
OR	4	21	73.5 (0.024)	85.1 (0.086)	40.9	33%
OR	8	17	103.1 (<0.001)	142.0 (0.026)	43.9	0%
OR	12	13	138.0 (–)	156.5 (–)	56.8	0%

Panel B. California, parameters fixed at the full-sample estimate.

Origins	Quarters ahead	n	Given realized balance	Observed state only	No change	Direction, given balance
2019Q4–2024Q4	1/4/8/12	21/21/17/13	10.5 / 24.3 / 27.1 / 21.1	14.5 / 53.2 / 119.6 / 182.4	13.7 / 47.9 / 94.7 / 138.0	86 / 95 / 100 / 100%
2016–2024	1/4/8/12	39/36/32/28	13.0 / 33.8 / 46.5 / 50.3	13.5 / 43.0 / 89.1 / 131.7	14.8 / 49.9 / 87.4 / 117.2	67 / 81 / 91 / 96%

Notes: The given-balance forecast uses the net credits realized after the origin and is a conditional forecast. The observed-state forecast contains no estimated parameter. Diebold–Mariano tests use a HAC variance with $h-1$ lags and the Harvey–Leybourne–Newbold correction; – = not computed (fewer than $2h$ origins). Direction: share of origins at which the forecast change from the origin price has the sign of the realized change. Washington is not tested (fewer than eight quarters before every origin). Sources: CARB and Oregon DEQ credit data; CPI-U.

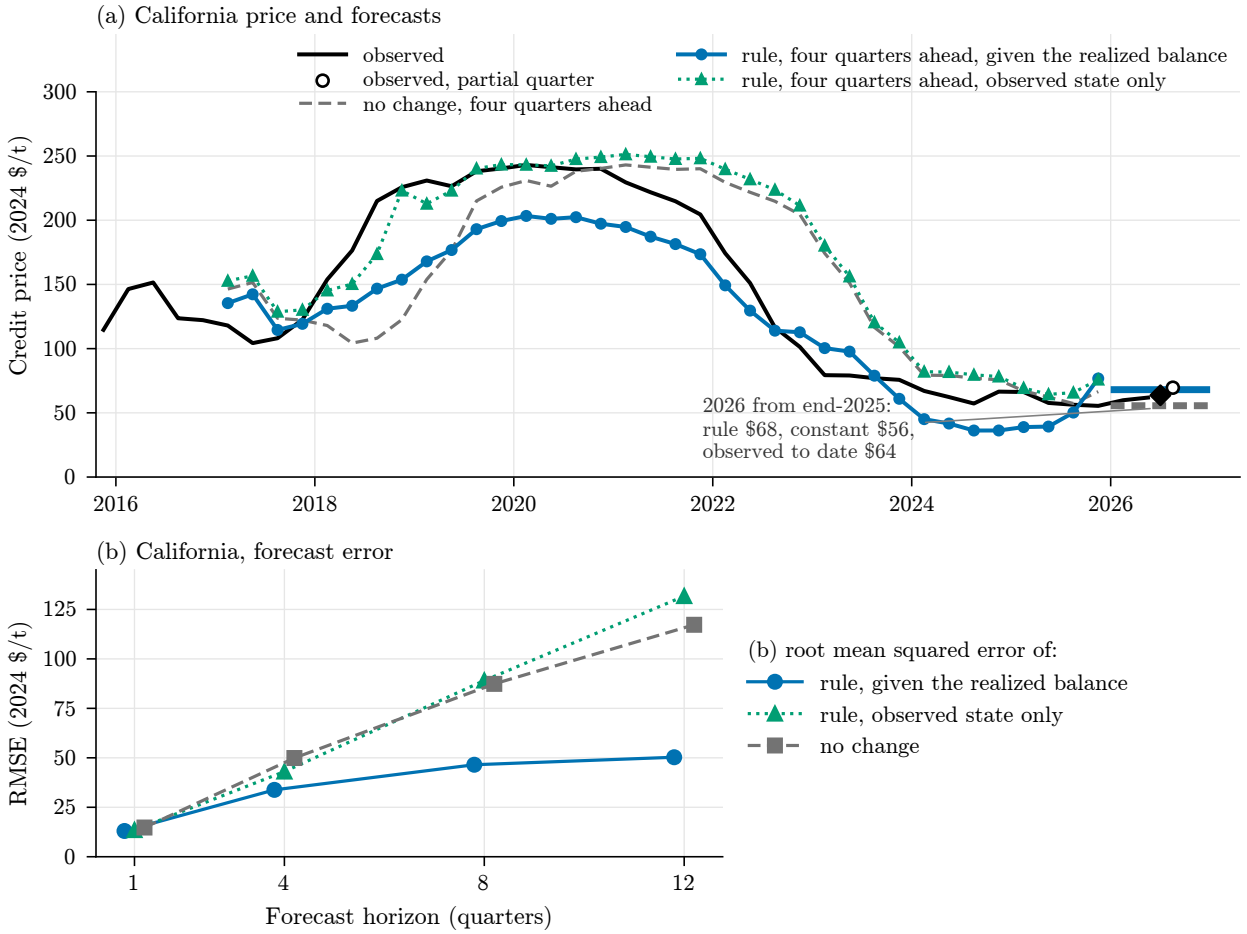


Figure 14: With full-sample parameters, the rule given the realized balance tracks California's price closely; this shows fit, not out-of-sample performance

Notes: Parameters fixed at the full-sample estimate (Rule B, $r = 3.38\%$); origins 2016Q1–2024Q4, targets to 2025Q4; 2024 dollars. Out-of-sample results are in Table 12.

D.4 The rule as a constant-gain filter

This appendix shows that the rule has the form of a steady-state Kalman filter. The equivalence is formal: it says which updating problem the rule would solve, not that holders solve it, and nothing in the main text depends on it.

Let T_t^* be the year in which the bank would run out. It is not known and receives persistent shocks: $T_t^* = T_{t-1}^* + \varepsilon_t$, $\varepsilon_t \sim N(0, q)$. Holders read the year's net surplus as a noisy measure of how far the scarcity date implied by this year's market is from their previous expectation, $y_t \equiv T_{t-1} + \lambda\big(s_t - E_{t-1}[s_t]\big) = T_t^* + v_t$, $v_t \sim N(0, \varsigma)$, with λ years per Mt. This is a local-level model. The Kalman filter updates $T_t = T_{t-1} + G_t(y_t - T_{t-1}) = T_{t-1} + G_t\lambda\big(s_t - E_{t-1}[s_t]\big)$, and its gain converges to

$$G = \frac{\Sigma}{\Sigma + \varsigma}, \qquad \Sigma = \tfrac{1}{2}\Big(q + \sqrt{q^2 + 4q\varsigma}\Big),$$

where Σ is the steady-state prior variance. In the steady state the update is the estimated rule with $\gamma = G\lambda$; only the product is identified. The gain rises with the signal-to-noise ratio q/ς, and the data show no sign of a gain that declines over time or rises with the bank.

D.5 What moves prices

The θ results in this subsection are estimated at $r = 5\%$.

To see what the rule captures and what it misses, I collected 59 dated LCFS events and 71 RIN events for 2016–26, each checked against its primary source, and 43 CARB quarterly data releases, and classified them as scheduled changes in the standards, news about the regulator's ability to tighten, or data on the credit balance. News about the regulator moves prices most. California's price fell 16% in the week of the Office of Administrative Law's rejection of the amendments (18 February 2025) and rose 14% in the week of their approval (27 June 2025); RIN prices rose 4.6% on average over ten trading days after such news (95% interval [1.9, 7.3]). Scheduled changes move prices when they carry new information (absolute daily LCFS returns 4.7 times those of ordinary days, $p = 0.02$) and not when they are procedural steps already expected; the largest is the Washington Senate's passage of HB 1409, which raised Washington's price by 26% (0.23 log points) within a week. Quarterly data releases move prices little: the response of California's price to the surprise in a release is -0.013 per Mt (s.e. 0.018).

Proposition 2(c) holds when announced changes do not move the expected exhaustion year. Allowing an announcement to move it by a share θ of its effect on the perfect-foresight exhaustion year, the quarterly estimate for California is $\hat{\theta} = 0.15$, 95% interval $[-0.30, 0.59]$. California's proposals did not move prices (model-free θ near zero), while Washington's statute did (near one, from a single trade in a thin market). I keep $\theta = 0$ and read Proposition 2(c) as one end of a range.

The model gives a check on the size of these responses (Table 13). If holders priced the whole announced change in the statutory path ($\theta = 1$), the rule would overstate the price moves on the

two Office of Administrative Law rulings six- to thirteenfold, and perfect foresight would overstate them by a similar order of magnitude. With $\theta = 0.15$, the value estimated from California's other scheduled announcements, the rule reproduces the rejection and overstates the approval (0.26 against 0.14). For Washington's statute the reverse holds: full pass-through matches the observed move and $\theta = 0.15$ does not. On these three events, holders priced about a sixth of a regulator's action and nearly all of a legislature's. In the full model, a 45Z extension beyond 2029 announced in 2026 does not move the state credit prices in 2027 under any description of expectations and moves the D4 price only from 2030, which is consistent with the absence of state price responses on federal dates.

Table 13: Observed and model-implied price responses to three events (change in log price)

	Observed	Rule, $\theta = 1$	Rule, $\theta = 0.15$	Perfect foresight
OAL rejects the 2024 amendments (CA)	−0.16	−1.03	−0.16	−1.35
OAL approves the amendments (CA)	+0.14	+1.76	+0.26	+1.45
Senate passes HB 1409 (WA)	+0.23	+0.27	+0.04	+0.63

Notes: θ is the share of the announced change in the perfect-foresight exhaustion year that enters the expected exhaustion year; $\theta = 0.15$ is estimated from California's other scheduled announcements, not from these events. Perfect foresight is an offline approximation.

D.6 A market that prices regulatory risk

A natural alternative is a rational market that prices regulatory risk: holders expect the standard to be tightened when the bank is large and relaxed or deferred when it is short, and the credit price is the probability-weighted value of the Hotelling paths to the cap under each regime. Disciplined by the record (the shape of the hazard from the 2011–26 regulatory record, the chance that a proposal becomes law from the two 2025 rulings, the holding cost from forward quotes, and only the level of the hazard from prices), such a market cannot reach the observed price level. Its quarterly price error is 69.9 in California against 34.0 for the rule without its three shift terms (estimated at $r = 5\%$), it predicts 222 for California in 2026 (earlier estimate at $r = 2.8\%$) against 63 observed, and the estimated level of the hazard is zero. The reason is informative. The tightenings and relaxations in the record roughly offset, so a hazard disciplined by the record cannot lower prices this far; only a holding cost well above market rates, or the expectation of a permanent relaxation larger than any in the record, could (Salant, 2016). Table 14 compares the price-implied years to exhaustion with the fundamentals projection by program; Washington's market is thin and new, and its rows are illustrative.

D.7 The investment rule

I observe 46 investment decisions in 2014–2025: 16 refinery conversion or co-processing announcements (8 of them SAF-capable) among 1,268 refinery-years at 124 refinery sites (refineries in the

Table 14: Price-implied years to exhaustion, $\ln(\bar{P}/P)/\ln(1+r)$, and the fundamentals projection

Programme	Year	P	$r = 0.4\%$	$r = 2.8\%$	$r = 3.38\%$	$r = 5\%$	Fundamentals ($r = 2.8\%$)
CA	2016	135.9	163.9	23.7	19.7	13.4	3.3
CA	2017	113.2	209.7	30.3	25.2	17.2	2.3
CA	2018	192.8	76.4	11.0	9.2	6.3	2.1
CA	2019	234.0	27.9	4.0	3.3	2.3	3.2
CA	2020	241.0	20.5	3.0	2.5	1.7	4.9
CA	2021	217.6	46.0	6.7	5.6	3.8	7.2
CA	2022	135.8	164.1	23.7	19.7	13.4	21.0
CA	2023	77.8	303.8	43.9	36.5	24.9	$\geq$38.8
CA	2024	63.3	355.5	51.4	42.7	29.1	24.5
CA	2025	58.9	373.4	54.0	44.8	30.5	15.3
OR	2016	67.1	333.2	48.2	40.1	27.3	2.0
OR	2017	60.8	357.7	51.7	43.0	29.3	3.1
OR	2018	95.4	244.9	35.4	29.4	20.0	2.4
OR	2019	181.0	84.5	12.2	10.1	6.9	3.4
OR	2020	159.0	116.8	16.9	14.1	9.6	5.6
OR	2021	144.7	140.4	20.3	16.9	11.5	3.1
OR	2022	127.8	171.7	24.8	20.5	14.0	1.9
OR	2023	133.7	160.4	23.2	19.2	13.1	5.4
OR	2024	82.6	280.8	40.6	33.8	23.0	9.1
OR	2025	79.6	290.3	42.0	34.8	23.7	3.2
WA	2023	95.1	237.7	34.4	28.5	19.4	9.4
WA	2024	49.3	402.3	58.2	48.3	32.9	20.1
WA	2025	26.1	561.2	81.1	67.4	45.9	10.8

Notes: Price-implied columns: $\ln(\bar{P}/P)/\ln(1+r)$ at the year's mean price, in years from the end of the calendar year. Fundamentals: the regulatory-risk model's projection under the schedule in force, starting from the reported end-of-quarter bank (final-vintage data through 2026Q1), measured from each quarter's end (on average 0.375 years earlier) and averaged over the year's quarters. Projected net credits are the mean of the last four quarters' deseasonalized net credits at the coming standard, plus the deficits added by the scheduled tightening at fixed fuel energy, plus the three-year trend in net credits extrapolated for three years, plus a supply response to the price along the Hotelling path; California's acceleration mechanism and the Oregon and Washington deferral apply as written. While an amendment is pending (CA 2018Q2–Q4 and 2024Q1–2025Q2, OR 2020Q2–2022Q3, WA 2025Q2) the date is the mean of the dates under the proposal and under the law. r enters the fundamentals column only through the supply response; at $r = 5\%$ it changes by at most 1.7 years. Without the trend term, California's column is 4.6, 8.3, 10.0 and 7.3 in 2022–25. $\geq$: three of the four quarters reach the 40-year horizon.

same state and city count as one site; 92 to 113 are at risk in a year, against 129 to 142 operable refineries in EIA's count); 5 expansions and 6 SAF retrofits at existing renewable diesel plants, both among the same 106 plant-years at 21 plants; and 19 greenfield announcements over 12 years, each year with a set number of potential entrants. All four types enter one likelihood and share ζ. Before 2023 the SAF margin is zero in every row, so κ_{saf} is identified mainly by the 56 plant-years and 5 retrofits of 2023–25. For each investor-year the margin index m of equation (20) is computed from the historical model at observed LCFS prices, with LCFS expectations on the path the estimated rule implies and statutory credits valued as they stood at the decision date. Information is dated at the time it became public, including the LCFS standards (by their adoption date) and the announcements of SAF projects. The four types share the logit scale ζ and are estimated jointly

by maximum likelihood,

$$\ln L(\zeta, \kappa_a, \kappa_b, \kappa_c, \kappa_{saf}, p) = \sum_r \big[y_r \ln P_r + (n_r - y_r) \ln(1 - P_r) \big], \tag{51}$$

where P_r is the probability that investor-year r commits under the optimal stopping rule (Rust, 1987) and $\delta = 1/1.07$ is fixed.

Table 15 reports the estimates. Equality of the expansion and greenfield entry costs is not rejected (LR 0.83, $p = 0.36$), so they are pooled. The entry costs are well above engineering capital cost: about 11 times capital cost for refinery conversions and 3 times for new plants. The excess over capital cost reflects risk, option value and permitting, and it is not counted as a social cost. Estimated on 2014–19, the rule predicts 2020–25 better than a constant hazard (log predictive score +13.3) and than a static logit on current margins (+6.9); most of the gain comes from SAF conversions, the rule alone does about as well as the static logit for renewable diesel, and it under-predicts the 2020–21 conversion wave by about half.

The data reject entry costs at engineering capital cost: with κ fixed there, the likelihood rises without bound in ζ, toward investment that ignores margins, and the likelihood-ratio statistic against the estimates exceeds 283 (three restrictions) at every ζ; at $\zeta = 80$ the rule would predict 165 refinery conversions against 16. The estimated entry costs correspond to real hurdle rates of about 51% for conversions, 22% for expansions and new plants and 18% for SAF retrofits (constant margins, 20-year life, three-year construction, two for SAF), which include option value and permitting and policy risk as well as the cost of capital. For comparison, U.S. firms report real hurdle rates averaging 12.2% (Poterba and Summers, 1995), and firms' discount rates average about twice their cost of capital (Jagannathan et al., 2016). The implied hurdles for new plants, expansions and SAF retrofits are above that average but of the same order; 51% for refinery conversions is high even by that standard, consistent with conversion-specific risk: margins that depend on policy, and first-of-a-kind plants. The SAF conversion cost rests on six events and is imprecise (bootstrap relative standard error about 100%); the engineering cost of a conversion lies inside its confidence interval, and a run with κ_{saf} at engineering cost is a sensitivity.

The investment estimates were fitted with investors expecting the California credit price to rise at 5% a year toward its cap; the forward runs use the holders' 3.38%. At the same current price this lowers the value of expected credit revenue by about a tenth. Re-estimating the investment rule at 3.38% is left to the next version.

The main specification sets the probability at 1 while a credit is in law and at 0.18 during lapses, the restoration probability implied by RIN prices. Letting the probability differ in the months when the blenders' credit had lapsed (15 decisions, 609 decision-months) and when it was in law (31, 883) estimates both: 1 in law (profile interval [0.67, 1.00], [0.74, 1.00] adjusted for the boundary) and 0 in lapses ([0, 0.31]). The data reject equal values and do not reject the main specification's

Table 15: Investment rule, maximum likelihood, 2014–2025

	Estimate	95% profile interval
Logit scale ζ	2.788	[2.01, 4.11]
Entry cost, refinery conversion κ_a (\$/gal-yr)	20.25	[16.25, 27.50]
Entry cost, expansion and greenfield κ_{bc}	10.5	[7.75, 15.00]
Entry cost, SAF conversion κ_{saf}	3.0	[−1.35, 9.70]
Extension probability p	1	at the upper bound
Log likelihood	−179.75	
Decision-years; decisions	1,492; 46	

Notes: Expansion and greenfield entry costs pooled (LR test of equality 0.83, $p = 0.36$). Parametric bootstrap 90% intervals: ζ [2.01, 3.73], κ_a [16.50, 25.50], κ_{bc} [7.75, 13.76], κ_{saf} [−0.90, 8.53] (200 replications). Profile intervals: likelihood within 1.92 of its maximum. Decision-years: 1,268 refinery-years, 106 plant-years for expansions and the same 106 for SAF retrofits, and 12 greenfield years; decisions 16 conversions, 5 expansions, 19 new plants and 6 SAF retrofits. Fixed: conversion loss 1.125; jet share of SAF-capable capacity $h_b = 0.2$. The estimate of p is at its upper bound of 1 (profile interval [0.67, 1.00]). Estimated on 2014–19 alone (ζ 3.09, κ_a 25.0, κ_{bc} 11.25, κ_{saf} 4.15), the rule beats the benchmarks by +13.3 and +6.9 in log predictive score over 2020–25; with the price-expectation rule also refitted on data before 2019, by +9.5 and +3.0.

pair. Freeing the lapse value moves the other estimates little: σ from 2.79 to 2.62, κ_a from 20.25 to 18.75, κ_{bc} from 10.5 to 9.25. Almost all of the lapse variation comes from 13 decisions in 2017–19 (2014–15 has two; the 2010 and 2012 lapses precede the panel), so the comparison is in effect five lapse years against seven years in law, and a shock to investment in the lapse years would load on the lapse value. I have not bootstrapped these intervals.

Eight of the 16 refinery conversions of 2017–23 were built able to produce SAF, when SAF earned nothing over RD. The margin index cannot explain this choice, which looks in the data like a nearly costless design option. Rather than estimate it, I set future SAF capability at build to the historical proportion; SAF value enters investment only through conversions. The SAF conversion rule predicts 4.5 conversions in 2023–25 against 5 observed.

With margins from observed prices, the rule reproduces observed capacity (RMSE 0.47–0.72 billion gallons across versions). In a fully coupled backtest, where capacity feeds back on margins, capacity rents spike in 2018–19 and the rule overbuilds; the main backtest therefore uses observed capacity. The logit rule also implies a positive announcement probability at zero expected profit, so idle and SAF-capable capacity grow slowly even when margins are nil. I bound its welfare cost by a run with capacity fixed at the main path (lower bound) and the main run with endogenous capacity (upper bound).

D.8 Other estimated blocks

The share response c in equation (36) is identified from how the fat share of renewable diesel and biodiesel moved in California, Washington and the rest of the country over 2016–25 (about 39 state-fuel-year observations), as relative feedstock margins changed. I estimate it by simulating the historical model for a grid of values and minimizing the squared errors of the fat shares, with

a bootstrap over years. Quarterly panel regressions of the fat share on the margin gap are not informative. Estimated separately, the response of biodiesel is precise ($c_{BD} = 0.446$, 95% interval [0.33, 0.76]) and that of renewable diesel is not identified (0.094, [0.012, 1.28]), because California's history favors a small response and Washington's a large one. The main specification uses one response for both fuels, estimated jointly, with the renewable diesel response at the two ends of its separate interval as sensitivities. Two limits: the response is identified from margins, so quantity shocks such as the 2026 drop in imported used cooking oil and tallow under OBBBA's origin rule and tariffs are not captured by it; and within each group the feedstock composition is fixed at 2025.

The light-duty price coefficient β_1, the network coefficients $\beta_2 = \beta_3/2$ and the station entry elasticities $\upsilon_{L2}, \upsilon_{DC}$ are estimated jointly by simulated method of moments on 2021–25, at observed fuel, electricity and credit prices. The moments are the logit sales share, the demeaned growth of the electric stock, and the log Level-2 and DC fast station counts, by region and year. Market stations exclude Tesla, Electrify America and NEVI/CFI stations, which are exogenous. Charger costs in each year use the law known in that year (the 2021 credit, its retroactive restoration in 2022, the Inflation Reduction Act's version from 2023), and next year's station size is this year's.

The estimates are $\beta_1 = -4.918$ (0.353) and $\beta_3 = 0.313$ (0.095), with entry elasticities $\upsilon_{L2} = 0.281$ (0.019) and $\upsilon_{DC} = 0.443$ (0.034) in the LCFS states and 0.724 (0.048) elsewhere. For the aggregate price elasticity of light-duty electric demand, the literature anchor of -2.8 is a sensitivity. The fast-charging network coefficient is close to the local estimate of Springel (2021) (0.42). For trucks the price elasticities are taken from the literature (medium-duty -2.5, the approximate median of light-duty estimates in Cole et al., 2023, p. 318 and fn. 2, applied to medium-duty trucks; heavy-duty -2.65, the midpoint of the range of -2.5 to -2.8 in Weber, 2024, p. 16, an own-price elasticity for class 7–8 diesel trucks transferred to electric trucks) and the intercepts are set to 2025 sales. The truck block is conservative: intercepts are fixed, there is no diffusion or model-availability term, and there is no Advanced Clean Trucks mandate. Elasticities of -6 and -10 are sensitivities.

E Fit to History

The backtest solves the model year by year over 2016–2025 with the policies then in force. In the version reported in the main text, hydroprocessing capacity is set to its observed level; RIN banking follows its observed path; and small-refinery exemptions and RFS volumes are as finally determined.

Table 16 reports the replay from a single start in 2016 under the main specification, with the three shift terms in the history years. California's error is 38.4 over 2016–25 and 40.4 over 2017–25, against 46.9 for a random walk, with an upward bias of about \$24; the difference is not significant (Diebold–Mariano $p = 0.59$). Of all the comparisons with the random walk, only California's

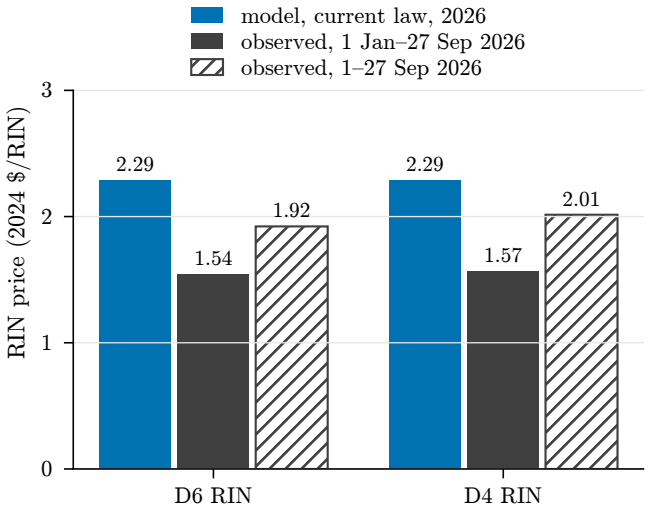


Figure 15: RIN prices in 2026: model and observed (2024 $ per RIN)

Notes: Model: current law, 2026. Observed: volume-weighted EPA EMTS transfer prices for 1 January–27 September and 1–27 September 2026, excluding transfers at $0.05 or less, deflated with CPI-U. Sources: EPA EMTS and model runs.

Table 16: Annual replay of the main specification, 2016–2025: root mean squared error

	Model		Random	DM p	2016	AEO 2016	AEO each
	2016–25	2017–25	walk	vs RW	held	path	year
CA LCFS price ($/t)	38.4 (0.91)	40.4	46.9	0.59	73.0	—	—
OR LCFS price ($/t)	38.5 (0.44)	40.3	36.3	0.58	63.9	—	—
WA LCFS price, 2023–25 ($/t)	34.9 (0.91)	34.9	36.8	—		—	—
D4 / D6 RIN ($)	0.39 / 0.28	0.33 / 0.22	0.46 / 0.46	0.24 / 0.06	0.42 / 0.47	—	—
CA net credits (Mt)	1.46	1.54	4.63	0.24	4.74		
CA renewable diesel (bn gal)	0.181	0.191	0.335	0.04	1.23		
U.S. renewable diesel / biodiesel (bn gal)	0.121 / 0.103	0.122 / 0.100	0.51 / 0.35	0.08 / 0.25	1.36 / 0.49	0.88 / 0.36	0.47 / 0.47

Notes: Main specification with the three shift terms in the history years, observed hydroprocessing capacity. Correlation of model and observed prices in parentheses. The random walk needs the previous year's value, so it and the Diebold–Mariano tests cover 2017–25; Washington's covers 2024–25 and allows no test. The rows were computed on the previous model version; on the current version the California replay error is 38.3. Before AEO 2023, renewable diesel is proxied by AEO's "other biomass-derived liquids" and biodiesel is consumption, while the observed series are EMTS generation; the AEO columns are indicative only.

renewable diesel volume is significant at 5% ($p = 0.04$), with the D6 price and U.S. renewable diesel at 10%. The replay in which capacity follows the investment rule does not converge. Neither replay is the test reported in the main text. The test in Section 5.4 runs the rule as the forward runs use it, without shift terms, from four start years (2017, 2019, 2021 and 2023), with the starting point inferred from the price in the starting quarter. There California's pooled error is 29.6 against 43.9 for a random walk, with a bias of +4.2 ($p = 0.15$, clustered by year), and Oregon's is 43.0 against 34.3. Adding the 2018 shift term in its own quarter only gives 27.4 against 46.9 for California, and the price still reaches the cap in 2019 ($261 against $235 observed). Carrying the event terms into later years and starting from a different point, two choices the forward runs do not make, gives 50 against 47. With γ_{CA} converted to annual units by matching the long-run rather than the first-quarter response (0.94 instead of 0.75), the backtest from four start years gives 26.6 against 43.9 for California, and Oregon's error, 37.1, falls within the criterion of 1.25 times the random walk (42.8) though it remains above the random walk itself; in the single replay California's error is 30.5 against 46.9, and its price still reaches the cap in 2019. Washington's price is evaluated over 2024–25 rather than 2023–25, because the random walk needs the previous year's price and Washington's program began in 2023.

The model tracks California's credit balance and fuel quantities with errors a third to a quarter of the naive benchmarks', and its fuel quantities better than the AEO. Oregon is the exception: the model's Oregon price errors exceed those of a random walk, and its prices are only weakly correlated with the observed ones (0.44). Oregon's credit pool is small, so its price-expectation rule turns small surprises into large revisions of the expected exhaustion year, and its price path is sensitive to the starting point. I treat Oregon's forward price path as the least reliable of the four. The backtest validates the fuel-market block and the forecasting performance of the price-expectation rule, not a model of how holders form beliefs. The backtest is not a fully independent test. RIN banking is set to its observed path, 2016–18 capacity is observed, the gain in the RIN price fit is partly mechanical, and the specification was chosen after earlier backtests had been seen. It does not test the coupling of RIN storage with investment: with capacity from the investment rule, the coupled backtest does not converge.

Table 17 compares the model's 2026 with the observed values that were not targets. Six of the eight test variables are within tolerance: the three programs' net credits, California's electricity credits, U.S. biodiesel and U.S. SAF. Two are outside: the January–August average RIN price, which narrowed after EPA's exemption decisions of 31 August but remained outside the tolerance in September (EPA EMTS: D6 1.92, D4 2.01, 2024 dollars), and the fat share of California's renewable diesel. The fat share fell in 2026 because imported used cooking oil and tallow left the market after OBBBA's origin rule and new tariffs, while the relative price moved toward fat; a model in which shares respond to margins predicts a rise. The earlier SAF shortfall had two causes, both corrected: a coding error that gave HEFA-SAF the reference feedstock mix of renewable diesel,

and a comparison with an EMTS-based observed value that is an upper bound on SAF use. The 2026 D6 price depends on the end-2025 RIN bank, which EPA has not published: the observed price corresponds to a bank of about 6 billion RINs, against 3.21 in the main specification.

Table 17: 2026 out of sample: model against observed

	Observed	Tolerance	Model	Within
D6 / D4 RIN ($)	1.49 / 1.51	±0.25	2.29 / 2.29	no (September: still no)
CA net credits at pinned price (Mt)	−10.7	±4	−9.60	yes
OR / WA net credits (Mt)	+0.44 / +0.10	±1.5 / ±1.0	+0.07 / +0.20	yes
Fat share of CA renewable diesel	0.577	±0.08	0.787	no
U.S. biodiesel production (bn gal)	1.50	±0.21	1.56	yes
U.S. SAF (bn gal)	0.29	±0.15	0.17	yes
CA EV credits, LDV and MDV (Mt)	7.26	±15%	8.18	yes
NM LCFS price ($/t)	none	$0 < P < \bar{P}$	0	boundary
U.S. HEFA output (bn gal)	3.57 (STEO)	report only	3.97	

Notes: Observed: January–August 2026 unless noted. California's fat share is 2025Q3–2026Q1. Biodiesel is production (consumption plus exports), seasonally adjusted, January–August (range 1.43–1.55); the earlier test value of 1.39 (January–June consumption) is superseded. RIN prices are EMTS volume-weighted January–August averages in 2024 dollars (July–August: D6 2.02, D4 2.00); September (EPA EMTS): D6 1.92, D4 2.01 (2024 $), still outside the tolerance. The three programs' 2026 LCFS prices are pinned at their January–August averages and are calibration, not tests.

Before pinning the 2026 prices, I predicted them from the end-2025 state with the estimated rules. This is the most direct test of how the price-expectation rule responds to a large deficit, and it is not used in choosing the specification. Table 18 reports it.

Table 18: One-step prediction of 2026 LCFS prices from the end-2025 state

$/t	CA	OR	WA	NM
Observed (January–August)	63.3	145.8	45.6	—
Rule B, $K = 2$ (main)	68.0	143.5	22.9	—

Notes: Rule B: the estimated rule ($K = 2$, $r = 3.38\%$) stepped once from the end-2025 state; the California entry is the $68.0 of Section 5.2.

In the 2026–30 runs the aggregate RIN bank stays below 10% of the next year's obligation, so the 20% carry-over limit does not bind in aggregate; deficits are repaid the next year. In the final run the RIN bank stays below 5.1% of the next year's obligation, and the deficits of 2027 and 2030 are repaid the next year. California's acceleration mechanism does not trigger: in 2027 its bank ratio is 2.29 (trigger above 3) and credits are 0.83 of deficits (trigger above 1). No program defers in 2026–30. New Mexico's forecast margin is negative in 2029 and 2030 (−0.85 and −0.20 Mt), but its deferral also requires the expected price to reach the cap, and the expected prices are $155 and $211 against a cap of $252.

The coupled forward model, with all blocks endogenous, cannot be tested on history. The backtest validates the fuel market and the credit-holder rules at observed capacity; the investment rule is validated on the build-out given historical margins (estimated on 2014–19 and tested on

2020–25, Section 5.2); the vehicle block is validated separately at observed prices. The 2026 test is one year.

F Additional Results and Proofs for Section 3

F.1 The static model in figures

Figure 16 shows the two markets that matter for national support. Panel (a) is the allocation of national biofuel between the regions, drawn as a box whose width is national output: the line is the extra value of moving one more unit to S, $(c_0 - c_1)\lambda - \tau(q_S)$, and the outlet requires $q_S \geq q - \bar{q}_N$. Panels (b) and (c) are the state credit market of Lemma 1, with credit supply $(\sigma - c_1)q_S + \Gamma$ and deficits $(c_0 - \sigma)(Q_S - q_S)$ drawn separately; net generation b is the horizontal distance between them.

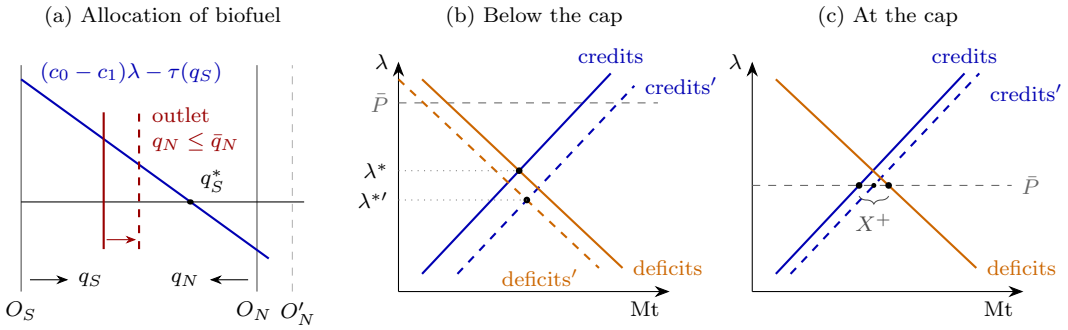


Figure 16: Where the extra biofuel goes, and what it does to the state credit market

Notes: (a) The box has the width of national biofuel output, from O_S to O_N; q_S is measured from the left and q_N from the right. The line is the extra value of a unit sold in S rather than N. Biofuel goes to S up to q_S^*, unless the outlet in N is full (the vertical line), which forces more into S. A subsidy widens the box (O_N to O'_N) and moves the outlet line right; while it stays left of q_S^*, all the extra biofuel goes to N (Proposition 1(iii)). (b) Once the outlet binds, the extra biofuel in S shifts the credit supply curve out and the deficit curve in, and the credit price falls from λ^* to $\lambda^{*\prime}$ (case (iv)). (c) At the cap, the price stays at $\bar{P}$ and the shortfall the state issues, X^+, shrinks (case (v)). Static market without a bank; Section 3.4 adds the bank.

F.2 When the transfer arrives under each description of expectations

Suppose that from year t_0 a permanent subsidy raises the state's net generation by Δb a year at given prices (case (iv) of Proposition 1, the outlet binds below the cap).

Proposition 2 (When the transfer arrives)**.** *At given prices:*

(a) Perfect foresight. *The expected exhaustion year moves out by*

$$\frac{dT}{d\Delta b} = \frac{T - t_0 + 1}{-b_T(\bar{P}) + \ln(1+r)\sum_{t=t_0}^{T} \lambda_t \, \partial b_t / \partial \lambda} > 0, \tag{52}$$

and the credit price falls by the factor $(1+r)^{-dT}$ *in every year from* t_0*: the transfer to the state's fuel buyers starts in the first year.*

(b) Adaptive rule, the subsidy is a surprise. *Under the last-year rule,* T *moves out by* $\gamma\Delta b$ *once, in* t_0*; from* t_0+1 *the subsidy is in the baseline and causes no further surprise. Under the three-year-average rule the revision is spread over three years and totals* $2\gamma\Delta b$*. The price falls from* t_0 *by the factor* $(1+r)^{-\gamma\Delta b}$ *(the last-year rule).*

(c) Adaptive rule, the subsidy is anticipated *(*Δb *is in* Δ^{known}*). There is no surprise;* T *and the price do not change, and the extra credits accumulate in the bank. The price falls only when the larger bank matters: through the bank-years term if* $K>0$*, whose weight rises as exhaustion approaches, and in any case in the years around* T*, when the bank that would have run out does not.*

Part (c) is a property of the form of the rule and does not come from the data. Holding the policy fixed, the same accounting can be used to compare any two descriptions of expectations.

Figure 17 illustrates the three cases with a numerical example. Under perfect foresight the subsidy lowers the credit price from the first year; under the adaptive rule, the same subsidy lowers it by less if it is a surprise, and not at all for many years if it was announced, because holders who already expected it see nothing new in the credit balance. Under the adaptive rule, the part of the subsidy that holders do not treat as news first accumulates in the bank and lowers the price only as the bank runs down. The timing of the transfer therefore depends on what holders count as news, and for this reason the paper reports results under alternative descriptions of expectations.

F.3 Proofs

The equilibrium conditions are (3)–(6). Write $a=(c_0-\sigma)/(c_0-c_1)$. When the standard binds below the cap, $b=0$ in (2) gives $q_S = a\,Q_S - \Gamma/(c_0-c_1)$.

Proof of Lemma 1. With $\nu=0$, (7) gives $q_S=\tau^{-1}((c_0-c_1)\lambda)$, and $Q_S=D_S(m+\pi\rho+(c_0-\sigma)\lambda)$; neither involves s. Differentiating (2) gives (8). Substituting q_S into $Z_S=c_0Q_S-(c_0-c_1)q_S$ gives (9).

Proof of Proposition 1. (i) With $q=H$ and $\rho=\nu=0$, (3) gives $\psi=m+s-C'(H)$, so $d\psi/ds=1$, and nothing else depends on s.

(ii) With $q=M$ and $p_N=m+\pi\rho$, (3) gives $C'(M)=m+(1+\pi)\rho+s$, so $d\rho/ds=-1/(1+\pi)$ and fuel prices fall by $\pi/(1+\pi)$ per unit. Fuel buyers gain $\frac{\pi}{1+\pi}\,ds$ per unit on Q^F+M units with $Q^F=M/\pi$, that is $M\,ds$. Effects on the state market, with the outlet slack: $\tau(q_S)=\lambda(c_0-c_1)$ gives $d\lambda=\tau' dq_S/(c_0-c_1)$; with $dQ_S=D'_S[\pi\,d\rho+(c_0-\sigma)d\lambda]$, $dq_S\big[1-aD'_S(c_0-\sigma)\tau'/(c_0-c_1)\big]=$

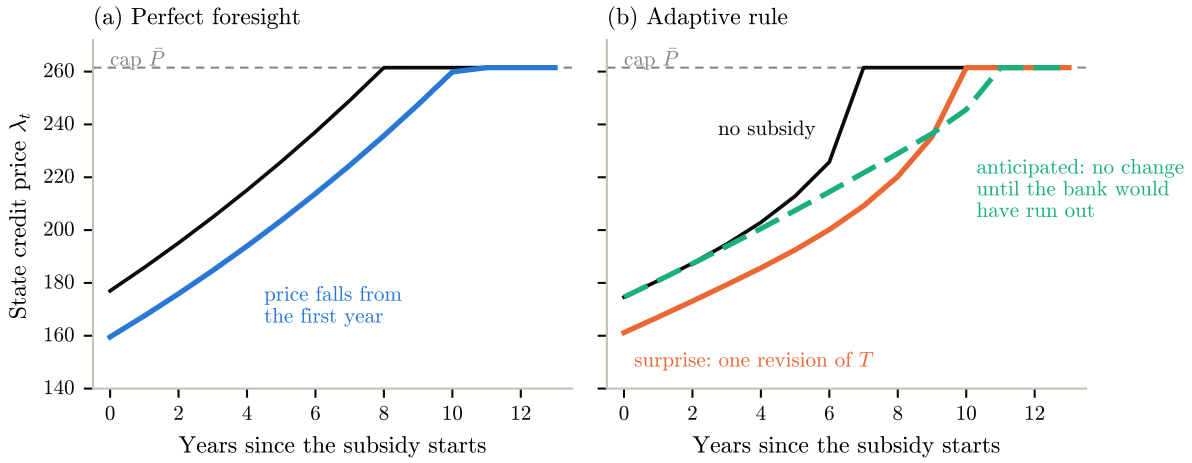


Figure 17: When the transfer arrives: a numerical example

Notes: Credit price paths with and without a permanent subsidy that raises net credit generation by 3 Mt a year from year 0. (a) Perfect foresight, equation (14). (b) Adaptive rule, equation (15) with the last-year rule, $\gamma = 0.75$ years per Mt and $K = 2$; "surprise" and "anticipated" differ only in whether the subsidy is in Δ^{known}. Prices include the within-year response of generation to the price. Illustrative parameters of California's order of magnitude: $b_t(\lambda) = -14 - t + 0.06\lambda$ Mt, initial bank 40 Mt, $\bar{P} = \$261.5$/t, $r = 5\%$.

$aD'_S\pi\, d\rho \geq 0$, and the bracket is positive, so $dq_S \geq 0$, $d\lambda \geq 0$ and $dq_N = -dq_S$. The price in S changes by $\pi\, d\rho/(1+x) < 0$ with $x = -a^2 D'_S \tau' \geq 0$, so Q_S and, by (9), Z_S rise; in N, p_N falls and q_N falls, so emissions rise. These effects are of the order of the demand response to the RIN pass-through and are small when demand is inelastic.

(iii) With $\rho = \psi = \nu = 0$, (3) gives $dq/ds = 1/C''$. By Lemma 1, λ and q_S do not depend on s, so $dq_N = dq$; N's demand is unchanged because $p_N = m$, so emissions fall by $(c_0 - c_1)/C''$. At the cap the same holds: $q_S = \tau^{-1}((c_0 - c_1)\bar{P})$ is fixed, and so is X^+.

(iv) With $q_N = \bar{q}_N$ and $\pi\rho = 0$, (4) gives $(C'' + \tau')\, dq_S = (c_0 - c_1)\, d\lambda + ds$, and $b = 0$ gives $dq_S = aD'_S(c_0 - \sigma)\, d\lambda = -\varpi\, d\lambda$. Solving gives (10); $dZ_S/ds = \sigma(c_0 - \sigma)D'_S\, d\lambda/ds > 0$ by (9), and N does not change.

(v) With $\lambda = \bar{P}$, p_S and Q_S are fixed and (4) gives $(C'' + \tau')\, dq_S = ds$; fossil use in S falls one for one, so $dZ_S = -(c_0 - c_1)\, dq_S$, and differentiating $b + X^+ = 0$ gives $dX^+ = -(c_0 - c_1)\, dq_S$.

Proof of Proposition 2. (a) Let $\Phi(T, \Delta b) = B_{t_0-1} + \sum_{t=t_0}^{T} \big[b_t(\lambda_t(T)) + \Delta b\big]$ with $\lambda_t(T) = \bar{P}(1+r)^{-(T-t)}$, treating T as continuous. Then $\partial\Phi/\partial\Delta b = T - t_0 + 1$ and $\partial\Phi/\partial T = b_T(\bar{P}) - \ln(1+r)\sum_t \lambda_t\, \partial b_t/\partial\lambda < 0$, and the implicit function theorem gives (52). (b) Under the last-year rule, $\bar{b}_{t-1} = b_{t-1}$, which includes Δb from $t_0 + 1$; under the three-year-average rule, $\bar{b}_{t-1}$ includes $\frac{1}{3}\Delta b$, $\frac{2}{3}\Delta b$ and Δb in $t_0 + 1$, $t_0 + 2$ and $t_0 + 3$, so the surprises are Δb, $\frac{2}{3}\Delta b$, $\frac{1}{3}\Delta b$ and zero after, which sum to $2\Delta b$. (c) With Δb in $\Delta^{known}_{t_0}$ (and, under the three-year-average rule, the part of Δb not yet in the average in $\Delta^{known}_{t_0+1}$ and $\Delta^{known}_{t_0+2}$), the surprise is zero in every year. The bank is larger

by $(t - t_0 + 1)\Delta b$ in year t, which raises Y_t and lowers the weight $e^{-Y_t/K}$ on the bank-years term. Within-year feedback: the price that clears a year satisfies (13) with T_t from (15), and a lower price lowers b_t and hence the surprise, so the revision is smaller than $\gamma\Delta b$ but of the same sign.

Proof of Corollary 1. Before either bank runs out no credits are issued at the cap, so $B_t = B_{t_0-1} + \sum_{k=t_0}^{t} b_k(\lambda_k)$. Net generation is increasing in the price: by Lemma 1 when the outlet is slack, and with the outlet binding because then $\partial b/\partial\lambda = (c_0 - c_1)^2/(C'' + \tau') - (c_0 - \sigma)^2 D'_S > 0$. For $t < \min(T^e, T^{e'})$, $\lambda^e_k \le \lambda^{e'}_k$ for every $k \le t$, so $B^e_t \le B^{e'}_t$. If $T^e < T^{e'}$ there is nothing to prove. Otherwise $T^{e'} - 1 < \min(T^e, T^{e'})$, so $B^e_{T^{e'}-1} \le B^{e'}_{T^{e'}-1}$, and at $T^{e'}$ the bank under e' cannot cover the shortfall at the cap, $B^{e'}_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) < 0$. Then $B^e_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) \le B^{e'}_{T^{e'}-1} + b_{T^{e'}}(\bar{P}) < 0$, so the bank under e cannot cover it either, and $T^e \le T^{e'}$.

F.4 Notes on the numerical example

Panel (b) of Figure 17 has two further features that come from the adaptive rule. Without the subsidy, the price rises from 226 to the cap in one year (year 6 to 7), much faster than the 5% a year at which holders are willing to store credits; holders with perfect foresight would have bought earlier. The bank-years term raises the price only as the bank runs low, and the last step comes when the bank is overdrawn and the market clears at the cap. Prices under the rule therefore do not follow Hotelling's rule near exhaustion. In addition, the surprise path reaches the cap a year before the anticipated path. The surprise lowers the price at once, so less credit is generated and the bank runs down sooner; when the subsidy is anticipated, the price and credit generation do not change and the extra credits stay in the bank.

Part (c) follows from the form of the rule, not from data: an announced change enters expected net credits and so never moves T. It illustrates one end of a range rather than a finding. A more general rule lets an announcement move T by a share θ of its effect on the perfect-foresight exhaustion year, $T_t = T_{t-1} + \gamma(b_t - E_{t-1}[b_t]) + \theta\,\Delta T^{PF}_t$. Part (c) is the case $\theta = 0$, and with $\theta = 1$ an announced subsidy moves the price at once, as in (a). Appendix D.5 estimates θ close to zero but imprecisely; the main specification keeps $\theta = 0$, and Section 7 reports the other end.

The state rules that respond to the bank work through the same channel: a dollar that adds to California's bank makes its acceleration mechanism more likely to tighten the standard, and a dollar that adds to the other states' banks makes their deferral rules less likely to relax it. Perino (2024) shows that a supply rule that responds to the bank, the EU's Market Stability Reserve, makes prices more sensitive to expectations that depart from fundamentals.

G Additional Results

G.1 The current-law path

Under current law the emissions avoided relative to no fuel policy are 44.5 Mt in 2026, 83.5 Mt in 2030 and 112.5 Mt in 2045 (Figure 6a).

For medium- and heavy-duty trucks in the clean fuel states, the electric share of new sales peaks in 2032–34 and then falls, as each kilowatt-hour earns fewer credits. Washington's path depends on whether Ecology lifts the statutory 20% limit. The main specification keeps the limit; lifting it to 45% is described below.

The D6 price is $2.29–2.45 in 2026–28 and falls to about $1.27 in 2029,when hydroprocessing utilization completes its ramp to the long-run reference rate and about 0.5 billion gallons of capacity move to the flat part of the utilization-cost curve; the ramp is applied as a single step, so I report 2028–29 as an average ($1.86). It then drifts down to about $0.95 in 2045; capacity earns a rent above the annualized conversion cost only in 2026–28, and in every year some hydroprocessing capacity is idle.

SAF output is 0.14–0.29 billion gallons in 2026–32.Almost all of it is made outside the clean fuel states, with Illinois's SAF credit. That credit ends in 2032, and SAF output falls to almost nothing. SAF earns about the same credits per gallon as renewable diesel. But it sells at the price of jet fuel, which carries no clean fuel standard obligation, and it costs $0.4–0.6 a gallon more to make. In California the gap between the diesel and jet prices widens from $0.5 to $3.2 a gallon, as the credit price passes into diesel. A gallon of feedstock therefore earns more as road fuel in every year (Wu et al., 2026).

California's shortfall at the cap grows as the electric share of the light-duty stock rises. Electricity credits peak in the mid-2030s and then fall, because each kilowatt-hour earns less as the standard approaches zero. CARB's own projection for the 2024 amendments (California Air Resources Board, 2024) has similar deficits until the mid-2030s and lower ones after, because it assumes the Advanced Clean Cars II rule, whose federal waiver Congress revoked in 2025, and credits from direct air capture, hydrogen and other projects after 2040, which the model does not include.

The model closes each year by issuing the shortfall as credits at the cap, the safety valve a regulator would have to adopt. The statute instead carries an unmet obligation forward with 5% interest and requires it to be retired within five years (17 CCR §95485(c)(5)), which cannot happen when the cap binds every year. Carrying the deficit forward instead of issuing changed the resource cost within 2045 very little in the previous version of the model (Appendix G.5). With issuance the state receives the money. Under the statute fuel sellers collect it in fuel prices and owe it as an unpaid obligation, which becomes a windfall if it is never enforced.

New Mexico reaches its cap in 2032; its deferral then holds it close to balance, and its standard reaches only about 30% by 2045.

From 2030 the RIN bank sits at its floor. Because the one-year deficit carry-forward applies to the industry as a whole in the model, the market can borrow only every other year, and RIN prices alternate between years; firms' cycles need not coincide, so I report RIN prices from 2030 as two-year averages, with a version in which obligated parties are grouped as a sensitivity.

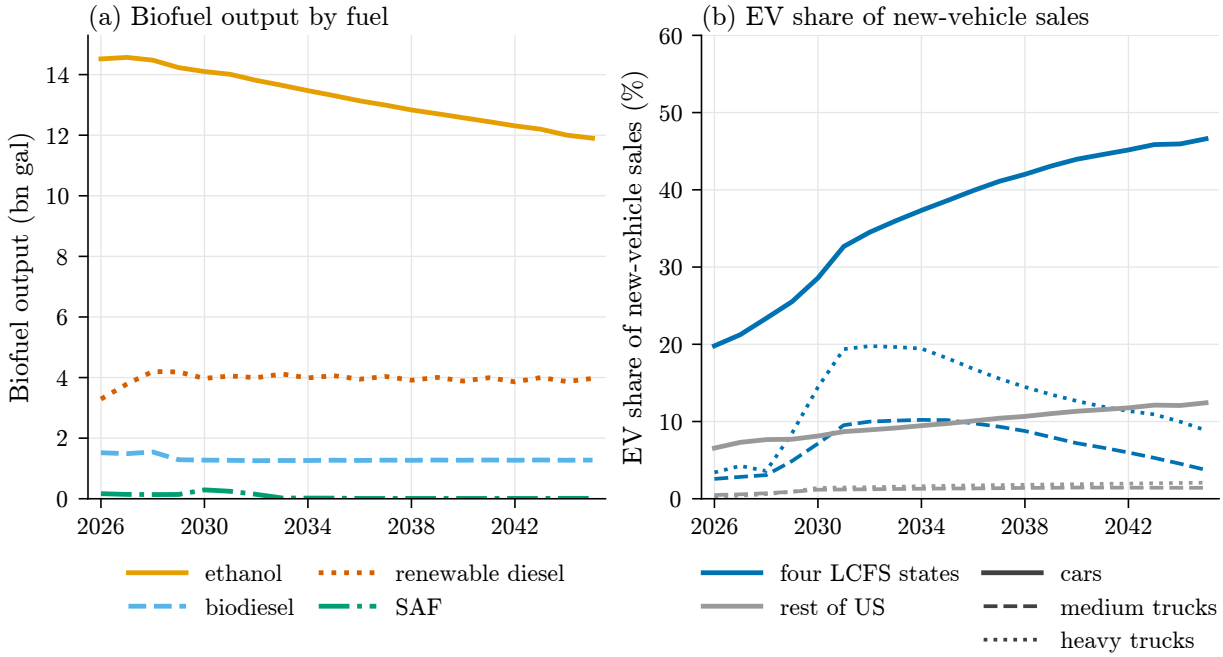


Figure 18: Current law, 2026–2045: biofuel output and electric vehicle sales

Notes: Biofuel output by fuel, and the electric share of new car, medium- and heavy-truck sales in the four clean fuel states and in the rest of the country. Main run. The truck shares in the four states peak in 2032–34 and then fall: California reaches its cap in 2032, after which the credit price can no longer rise while each unit of charging earns fewer credits as the standard tightens.

Washington's path depends on whether Ecology lifts the statutory 20% limit. Under the law as it stood in October 2026 (the main specification), its standard stops tightening in 2030 while electricity and renewable fuel credits keep growing, so its bank grows from 3 Mt in 2032 to 22 Mt in 2045 and its price falls from about $50 to $6. This is the consequence of a standard frozen at 20% and continued growth in credit supply, not a forecast. If Ecology makes the demonstration the statute requires, which its 2027 rulemaking may do, the standard follows the path to 45% and Washington's price reaches its cap. I report Washington's results under both, as a range set by that one administrative step.

Direct air capture does not change this under current law. CARB's own market model lets DAC credits set the price once their cost net of the federal 45Q credit falls below the cap (California Air Resources Board, 2024), but 45Q pays only for facilities whose construction begins before 2033 (26 U.S.C. 45Q(d)(1)), and on CARB's cost path DAC costs more than the cap in every year to 2045

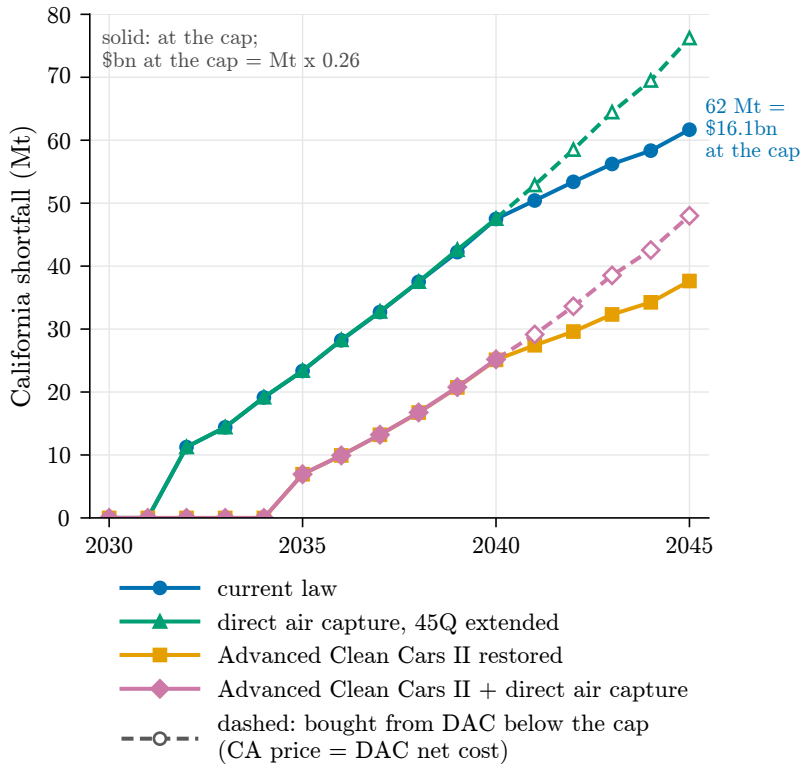


Figure 19: From 2032 California cannot meet its standard at the cap

Notes: Shortfall at the cap, met by credits issued at the cap. Under current law the shortfall in 2045 is 62 Mt, $16.1 billion at the cap (536 Mt over 2026–45). With Advanced Clean Cars II restored, California first reaches the cap in 2035 and the shortfall is 38 Mt in 2045 (254 Mt over 2026–45). With direct air capture under an extended 45Q, California is at its cap from 2032 to 2040; from 2041 DAC's cost net of 45Q ($230 a tonne, falling to $102 in 2045) is below the cap and sets the price. Solid lines show the shortfall at the cap (257 Mt over 2032–40 with DAC; 93 Mt over 2035–40 with Advanced Clean Cars II and DAC). Dashed lines show credits bought from DAC below the cap, where California's price equals DAC's net cost (322 Mt and 192 Mt over 2041–45).

with or without it ($544 a tonne in 2035, $273 in 2045, 2024 dollars).[21] If 45Q were extended to later facilities, DAC would take California off its cap from 2041, with a price of $102 in 2045 and 322 Mt removed by 2045 (Figure 19).

G.2 Details of the main-text figures

Figure 4 (which constraint binds). 45Z is in force through 2029 as legislated. "def" marks a year in which the state defers a step of its standard under its codified rule (Oregon, New Mexico; Washington's rule never triggers), with the price state on the second line. 2026 is observed (prices pinned; New Mexico's first program year). California's bank runs out in 2031 and its price reaches the cap in 2032 and stays there through 2045. New Mexico reaches its cap in 2032; Oregon and New Mexico defer from 2033; Washington stays below its cap with a positive bank throughout.

[21] CARB's cost path is taken to be in 2021 dollars; its source does not state the year.

New Mexico 2031 is a knife edge: the bank left at the end of the year is 0.003 Mt. In 2037–45 New Mexico's price sits at the cap but no credits are issued at the cap (issuance 0.2–0.6 Mt a year in 2032–36 only): deferral alone closes the market. Oregon's price is within $1 of its cap in the deferral years labelled "bank" or "no bank".

Section 2.1: horizons and bounds. At a 3.38% holding cost, the real carry measured from forward quotes, the 2019–21 horizon was 2.5–5.6 years, close to, and if anything shorter than, the fundamentals projection of 3–7 years. The bank reached 37 Mt in 2024. In 2024–25 the fundamentals horizon is 6 years on the model's current-law path, in which the bank runs out in 2031, and 15–25 years if the recent growth in net credits is extrapolated. From 2024 to 2025 the horizon fundamentals imply fell from about 10 years to about 7 (from 25 to 15 with the growth extrapolated), while the price-implied horizon rose from 43 to 45 years (Appendix Table 14). The price averaged $64 from 1 January to 18 September 2026 although deficits exceeded credits by about 10.7 Mt at an annual rate in 2025Q3–2026Q1 (CARB data, restated at the 2026 standard; 9.6 Mt in the model's 2026), against an end-2025 bank of 39.7 Mt. The regulator expected scarcity sooner than the market did. In 2018 CARB projected a bank of about 2.2 Mt in 2020 and 1.0 Mt in 2021–22, against 8.3 Mt observed in 2020 and 9.8 and 15.4 Mt in 2021–22 (California Air Resources Board, 2017). A projection-free bound makes the same point: at the observed 2025Q3–2026Q1 deficit the bank lasts fewer than four years (about four at the model's 2026 deficit), so a price near $60 for decades requires about 10 Mt a year of extra net credits, more than replacing all of California's fossil diesel with renewable diesel would generate (8.2 Mt). At a 5% holding cost the 2024–25 horizon is 29–31 years; at 2.8%, a market interest rate, 51–54.

Figure 2 (California's credit market). In panel (a) the model path is the estimated expectation rule under current law (annual; the bank runs out in 2031 and the price is at the cap from 2032), and the bar for CARB's 2024 market-model projection (California Air Resources Board, 2024) is drawn at the right edge, not at dates. The bank rose far above CARB's 2018 projections (California Air Resources Board, 2017); the 2024 amendments were approved on 8 November 2024 and took effect on 1 July 2025. In panel (b) the quarters are grouped by period of the 2024 amendments; the diamonds are $P/\bar{P} = (1+r)^{-T}$, with T from Appendix Table 14 and the main $r = 3.38\%$; in 2023Q2–Q4 T is censored at 40 years, so those diamonds are upper bounds.

G.3 The cost of current law and who pays

At a 3% discount rate with climate valued on EPA's 2.5% path, current law gains $5.8 billion net of climate damages, and the share of the first-best gain is 2%. At 3% with the Interagency Working Group's 2021 interim 3% values (Interagency Working Group on Social Cost of Greenhouse Gases, 2021), current law loses $130.8 billion. No share of the first-best gain is reported on that basis: the carbon-price run is priced at EPA's 2% values, 3.7 times the interagency value, and both net

Table 19: Who gains and who pays: all rows, 2026–2045 ($ billion, present value at 2%)

	State standards		Current law (whole stack)		45Z extension	
	$bn	share of W	$bn	share of W	$bn	per $1 fed.
Fuel users, the four states (road)	−338.7	2.29	−374.9	1.43	4.6	0.08
Fuel users, rest of US (road + non-road diesel RIN cost)	12.3	−0.08	−218.2	0.83	42.5	0.77
Aviation fuel users (national; not split by state)	0.0	0.00	0.0	0.00	0.0	0.00
Vehicle buyers, the four states	18.9	−0.13	18.4	−0.07	0.0	0.00
Vehicle buyers, rest of US	0.3	0.00	0.6	0.00	0.0	0.00
Feedstock suppliers, US	−11.7	0.08	104.8	−0.40	8.7	0.16
Feedstock suppliers, foreign (cane ethanol)	0.0	0.00	0.0	0.00	0.0	0.00
Biofuel producers (location not identified)	75.7	−0.51	118.3	−0.45	2.2	0.04
Credit holders and other credit generators	33.2	−0.22	54.4	−0.21	−1.9	−0.03
Charging station owners	0.3	0.00	0.5	0.00	0.0	0.00
Other private	0.0	0.00	0.0	0.00	0.0	0.00
Federal budget	−4.1	0.03	−18.7	0.07	−54.9	−1.00
State budgets, the four states: excise and other lines	−17.9	0.12	−15.9	0.06	0.2	0.00
State budgets, the four states: credits issued at the price caps	104.4	−0.71	104.4	−0.40	0.9	0.02
State budgets, CA / WA: lost cap-and-trade allowance revenue (E+A)	−19.9	0.13	−28.0	0.11	−0.1	0.00
State budgets, other states	−0.5	0.00	−7.2	0.03	1.2	0.02
Total = economic welfare W (= - net cost)	−147.6	1.00	−261.6	1.00	3.4	
memo: of which US farmers (soybean, corn, livestock)	−13.2		86.0		6.9	
memo: of which CA in-state RD freight rent (in biofuel producers)	8.2		8.2		0.0	
Climate damages avoided (EPA SCC, 2 %)	127.3		432.2		−16.3	
Net welfare incl. climate	−20.4		170.6		−12.9	
Abatement (Mt CO2e, undiscounted; + = abated)	520.8		1,755.6		−66.6	
$ per t abated, + tax wedge (E+A) (45Z: resource saving per extra t)	284		149		51	

Notes: Each column compares two runs: state standards = current law vs current law without the state standards; current law = current law vs no fuel policy; 45Z extension = extension vs current law. Positive: the group gains. Share: the row divided by the column total (economic welfare W, = minus the net cost incl. forgone fuel-tax and CA / WA allowance revenue); the shares of the groups sum to one. For the 45Z extension, whose total is a small net gain from a $54.9 billion federal budget cost ($54.0 billion of credit outlays, the denominator in Table 2), the column gives the amount per $1 of federal budget cost. "Four states" = CA, OR, WA, NM. Rest-of-US fuel users include non-road diesel buyers outside the model, who pay the RIN cost. Aviation fuel users' surplus is not split by region; it does not change in these runs. Biofuel producers cannot be split by location: the model indexes biofuel rows by consumption state, not the plant; the only in-state producer item is California's in-state renewable-diesel freight rent ($8.2 billion, inside the producer row). Credits issued at the price caps are state revenue (an unpaid obligation of fuel sellers under the statute). Climate damages at the EPA SCC (2%). The cap-and-trade programs are assumed to offset none of the standards' covered abatement.

welfare terms are negative.

The duties on imported fat treat the whole gap between the import and domestic price of fat as tariff, which the duties in force in 2026 exceed. Counting only the most-favored-nation tariff and the Section 301 duty, without the temporary Section 122 surcharge of February–July 2026, raised the resource cost slightly in the previous version of the model. The model holds that gap at its 2026 value and does not follow later changes in tariffs.

Capital costs are a range: endogenous capacity (main run) and capacity fixed at observed levels.

I allocate changes in federal spending and revenue by each quintile's share of federal taxes in 2022: 0.3, 3.9, 8.6, 16.8 and 70.2% from the lowest quintile to the highest (Congressional Budget Office, 2026, supplemental Table 12). The credits issued at the caps are state revenue that I do not allocate to households; under the statute fuel sellers would hold the same amount: for the state standards the credits issued at the caps are worth $104.4 billion (Table 1), and the resource cost is the same either way.

The fuel-price figures leave out the credit revenue that returns to households in the four states. Credits from residential charging pass through utilities, which spend them on vehicle rebates, charger grants and administration; credits from non-residential charging go to station owners (Appendix Table 31).

G.4 The blenders' credit and the RIN price

Proposition 1(ii) implies that a credit on top of a binding mandate is capitalized one for one into the RIN price. I test this on the $1 blenders' credit for biomass-based diesel, which lapsed and was restored retroactively several times. The regression is monthly, January 2014 to July 2021, of the D4 price on the credit in force per RIN (the credit divided by 1.5 RINs per gallon) and the gap between the biodiesel and diesel prices per RIN, with Newey–West standard errors, following Irwin et al. (2020), extending their sample to 2021. Full capitalization gives a coefficient of −1 on the credit. Daily event windows around each lapse, restoration and extension from 2010 to 2026 complete the evidence. Figure 20a shows the credit as in law at the time. It lapsed in January–November 2014, January–November 2015 and January 2017–November 2019, and each lapse was later made retroactive, which the series does not show.

Table 20: Capitalization of the blenders' credit into the D4 price

Sample	Coefficient (s.e.)	Months	Implied q [95% interval]
All months, 2014–21	−0.39 (0.12)	89	0.61 [0.37, 0.84]
Months in which the mandate binds	−0.41 (0.13)	86	0.59 [0.34, 0.84]
2014–16	−0.10 (0.11)	33	0.90 [0.69, 1.12]
2017–21	−0.59 (0.13)	53	0.41 [0.15, 0.67]
Months with D4 above D6 by $0.05 or more	−0.53 (0.11)	82	0.47

Notes: q: the probability of retroactive restoration that would make the estimate consistent with full capitalization, $-(1-q)$; it is a reading of the estimate, not a test. With the D4–D6 spread as the dependent variable the coefficient is not significant in the samples above; it is −0.28 (s.e. 0.10) in 2014–19. In the event windows, the retroactive restoration of December 2019 moved the spread by −0.04 to −0.07 dollars per RIN over ten days, against a standard deviation of 0.076 in non-event windows; neither the 2022 extension nor the 2025 extension of 45Z moved it detectably.

Over 2014–21, when the $1 blenders' credit lapsed and was restored, a dollar of credit moved the D4 price by −0.41 (s.e. 0.13, 86 months in which the mandate bound), well short of the −1 that full capitalization implies. Two readings fit this number. Every lapse was later reversed retroactively, so traders may have treated a lapse as only partly a loss of the credit; if they expected restoration with probability q, full capitalization gives $-(1-q)$, and the estimate implies $q = 0.59$. Or capitalization is partial. Two other measures of q point to the second reading. The decomposition of the D4 price used for the model gives a median of 0.18 over the lapse months, and investors' own decisions imply 0 ([0, 0.31]; Section 5.2). At q between 0 and 0.18 the regression implies that only 40–50% of the credit was capitalized. A third measure points the other way. In EPA's weekly transaction prices,

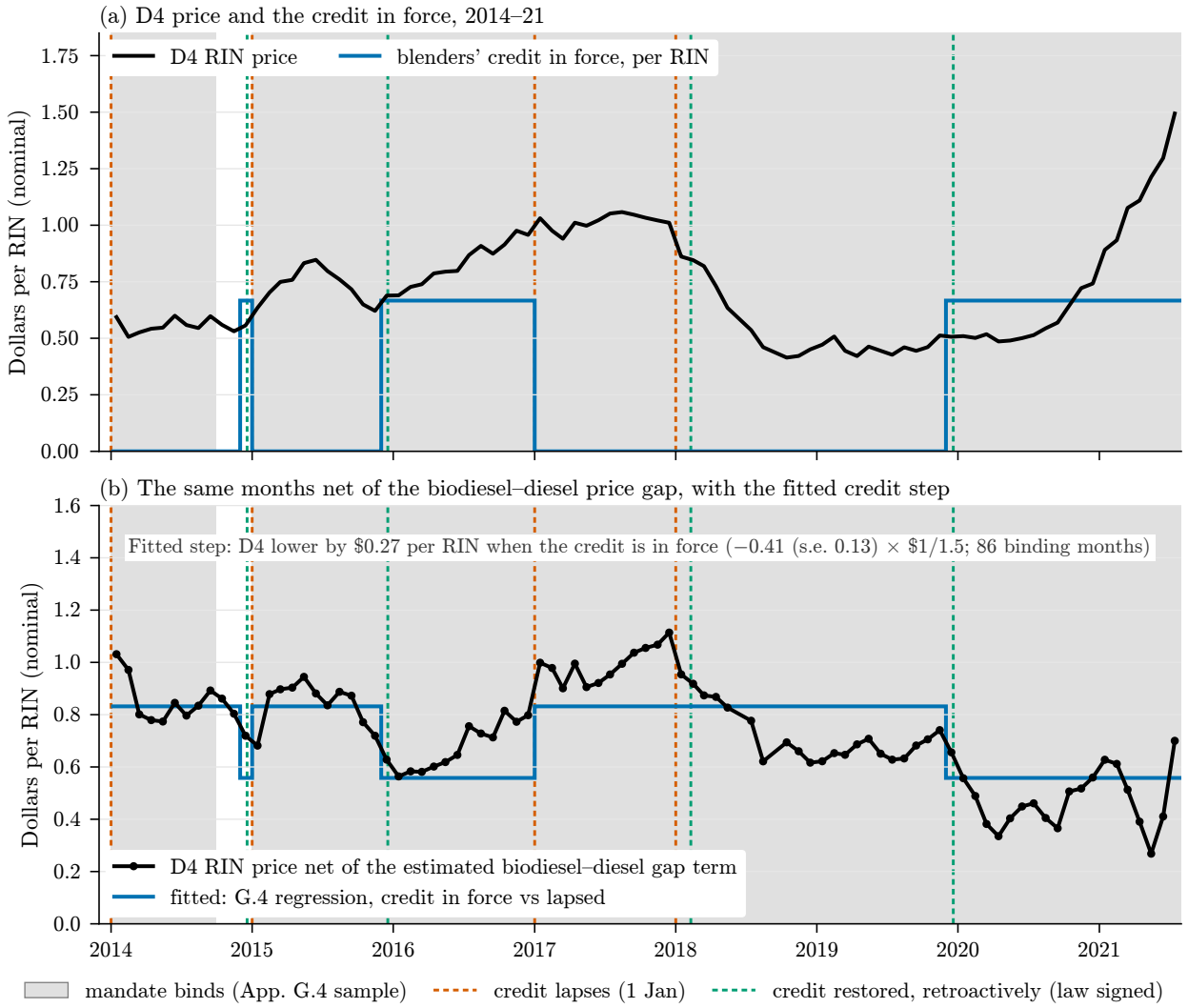


Figure 20: The blenders' credit and the D4 RIN price, 2014–21

Notes: Panel (a) shows the D4 price and the credit in force, and panel (b) shows the same months net of the biodiesel–diesel price gap, with the fitted credit step.

current-vintage D4 RINs barely move in the weeks around each lapse and restoration, which under full capitalization implies $q = 0.98$ ([0.79, 1.18], nine events). The design is weak: the lapses on 1 January were scheduled, and the RIN vintage changes in the same weeks. Taken together, RIN traders appear to have treated restoration as nearly certain over a few weeks, investors did not, and the monthly regression lies between. Other evidence does not separate the two readings: the coefficient is −0.10 in 2014–16 and −0.59 in 2017–21, the D4–D6 spread responds significantly only in 2014–19, and neither the December 2019 restoration nor the 2022 and 2025 extensions moved it detectably in the following ten trading days (Appendix G.4).

Table 2 and Figure 21 report the 45Z extension when only part of the credit is priced.

G.5 The 45Z extension

Larger small-refinery exemptions. In a pair of runs with small-refinery exemptions of 13.38 billion gallons in 2026–27 (both converged), the model's 2026 D6 price is $1.71, the closest to the observed. The mandate still binds, and the 45Z extension sends 0.756 of each federal dollar to fuel buyers and raises emissions by 67.5 Mt, against 0.751 and 66.6 Mt in the main run.

Timing of the transfer. Expectations also affect when the extension's transfer reaches fuel buyers. Under the rule the share of each year's federal cost reaching fuel buyers is 0.74 in 2030 and 0.68 in 2031. In 2030 the advanced-biofuel requirement adds a premium above the D6 price, and the extension lowers the D4 price slightly more than the D6 price. In 2031 the extension brings California to its cap a year early in the main specification; with the higher effective cap that the statutory carry-forward implies, both runs reach the cap in 2032 and the lower share in 2031 remains.

Where the RIN value goes. The RIN value that fuel buyers in the model pay falls by 1.15 times the federal cost, and another 0.12 reaches buyers of non-road diesel outside the model. The RIN value lost exceeds the federal cost partly because fuel made from imported fat earns no 45Z after 2025, so foreign fat suppliers bear that credit through a lower import price (6.3% of the federal cost). Cheaper fuel also adds obligated gallons at the lower RIN price, which works the other way (−1.2%). The rest of the excess comes from differences in the credit per RIN across fuels, which I do not attribute pathway by pathway.

EPA's response. The result holds the RFS percentages fixed, and EPA has changed them in both directions (Appendix A.2); Irwin et al. (2020) document that EPA takes the blenders' tax credit into account in its rulemakings. If EPA cut the requirement by 1.5% from 2030, the extension and the cut together would send 95 cents of each federal dollar to fuel buyers and raise emissions by 99 Mt (both 2030–45). Figure 8 shows the other direction. Emissions stop rising when EPA raises the requirement by about 2.65%. At that point fuel buyers receive 40% of the federal cost and feedstock suppliers 29%. Returning the average 2030–45 D6 price to its current-law level takes an increase of 5.64%; then fuel buyers receive 1% and feedstock suppliers and producers most of the rest. The pair lowers welfare at every offset: by $12.9 billion with none, $8.6 billion at 2.65% and $2.0 billion at the full offset. The extension's own effect, against the same requirement without it, stays between −$12.9 and −$13.5 billion, so the smaller loss of the pair at larger increases comes from the higher requirement, not from the extension. A higher requirement alone reaches about 100 Mt at $175 a tonne, while the extension paired with the full offset costs $272 a tonne (Table 3). The shares at each increase are in Figure 8: the federal cost rises from $54.0 billion to $57.9 billion

across the range; the pair's net welfare is −$12.9 billion with no offset and −$2.0 billion at the full offset; the dotted line shows the size of adjustments EPA has made, not a prediction.

Table 21: Incidence of a 45Z extension: all rows, 2026–2045 (shares of the federal cost; present value at 2%)

	Capitalized share of the credit, χ				$p = 0.5$	Bootstrap
	1 (main)	0.75	0.50	0.083	($\chi = 1$)	s.e.
RIN transfer to fuel buyers in the model	1.150	0.861	0.566	0.112	1.148	
Fuel buyers, other (biofuel prices and quantities)	−0.391	−0.290	−0.183	−0.042	−0.388	
Transfer through the state credit prices	−0.007	−0.006	−0.004	−0.002	−0.007	
Share to fuel buyers in the model	**0.751**	**0.565**	**0.379**	**0.068**	**0.753**	(0.0041)
RIN transfer to non-road diesel buyers	0.122	0.091	0.060	0.012	0.121	
Feedstock suppliers	0.161	0.126	0.087	0.014	0.161	(0.0007)
Biofuel producers (net of capital costs)	0.042	0.031	0.017	−0.003	0.031	(0.0030)
Biofuel producers: unpriced 45Z credit (windfall)	0.000	0.250	0.500	0.917	0.000	
State revenue from credits issued at the caps	0.016	0.012	0.007	−0.002	0.016	
Other	−0.026	−0.020	−0.013	−0.002	−0.026	
Net resource cost	−0.066	−0.055	−0.037	−0.004	−0.057	(0.0051)
Federal cost F ($ billion)	54.0	53.4	52.8	51.6	54.0	(0.01)
F at the priced rate, F - windfall ($ billion)	54.0	40.1	26.4	4.3	54.0	
Emissions change (Mt)	+66.6	+49.7	+33.2	+6.1	+66.7	(0.6)
of which outside the four states (Mt)	+61.8	+45.9	+30.8	+5.3	+62.0	
Lost CA / WA allowance revenue ($ billion)	0.134	0.106	0.086	0.014	0.135	
Resource saving per extra t, + tax wedge (E+A) ($)	**51.5**	**56.7**	**56.7**	**31.6**	**44.1**	(3.8)
same, excise only (E) ($)	53.5	58.8	59.3	33.9	46.1	(3.8)
Net welfare incl. climate (E+A) ($ billion)	−12.9	−9.4	−6.3	−1.3	−13.4	(0.2)

Notes: Extension of 45Z from 2030 to 2045 against current law; window 2026–45, present values at 2%. Producers, the RIN market and investors value the extended credit at the share χ of its rate; the Treasury pays it in full, and the part not priced is a windfall to producers. $\chi = 0.083$ is the change in the D4 price per dollar of credit change at the two dates on which 45Z replaced the blenders' tax credit and its 2026 rules took effect (January 2025 and January 2026; 95% interval −0.43 to 0.49); because both dates were known in advance it is a lower bound. $p = 0.5$: investors' extension probability 0.5 with full capitalization, against current law at the same p. Cost per tonne: resource cost plus lost fuel-excise and California / Washington allowance revenue (E+A). Bootstrap s.e.: 184 of 200 converged draws of the estimated parameters; estimated-parameter uncertainty only. The main estimate of the producers' share, 0.042, lies just below the 5th percentile of the draws (0.043); the main parameters are central in the draw distribution, so the gap reflects a difference between the main run and the draw runs other than the estimated parameters, which I have not resolved. Shares sum to one. Case (v), California at its cap, applies in the state market from 2032.

Under perfect foresight the share reaching fuel buyers is 0.752 (0.752–0.785 over the last rounds of a run that has not converged) and emissions rise by 65.9 Mt; with a probability of extension of $p = 0.5$ they are 0.753 and 66.7 Mt (Tables 2 and 4).

The result does not depend on the biomass-based diesel requirement binding on its own: under current law it never does. The D4 price exceeds the D6 price only in the few years in which the advanced requirement carries a premium, so the fall in the RIN price reaches gasoline as well as diesel.

Oregon, Washington and New Mexico defer their standards when credits run short, and California tightens its standard when the bank is large. These responses could offset part of the extension. They did not in the previous version of the model, which is the only one in which the responses were switched off. Once the federal mandate binds, the extension barely moves the state banks, so

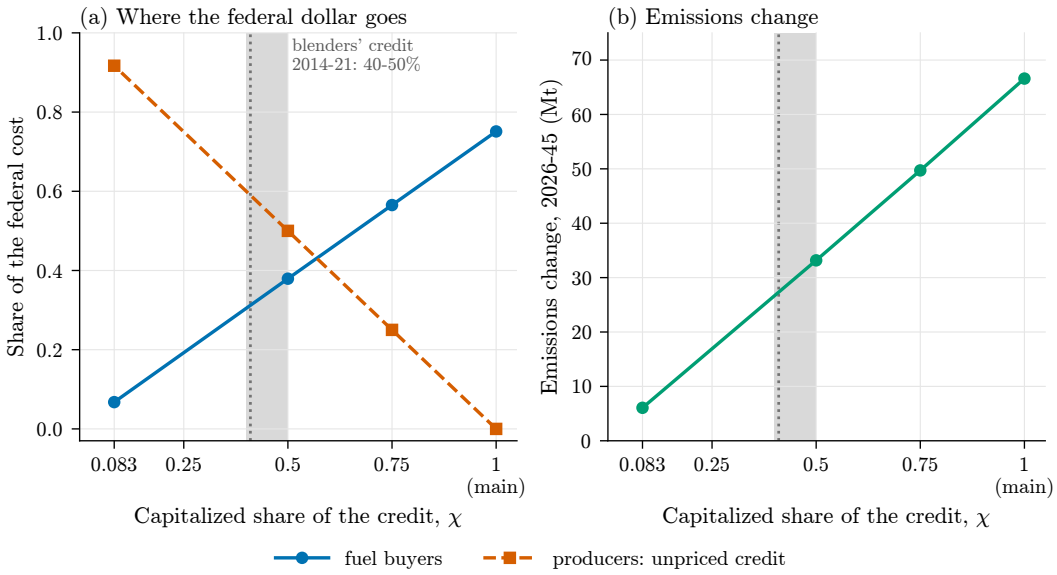


Figure 21: The 45Z extension against the priced share of the credit, 2026–2045

Notes: Each point: the 45Z extension from 2030 to 2045 against current law, with producers, the RIN market and investors pricing the share χ of the credit. (a) Shares of the federal cost reaching fuel buyers in the model and kept by producers as an unpriced credit. (b) Change in emissions, including indirect land-use change, 2026–45. Shaded: 40–50% capitalization, the range implied by the 2014–21 blenders' tax credit; two of three measures imply 40–50%, the third, from weekly transaction prices, is consistent with full capitalization. Dotted line: the regression coefficient 0.41 read as capitalization.

the rules have nothing to respond to (Appendix Figure 22).

Table 22 splits the full offset into the extension and the higher requirement. The full offset raises the requirement by 5.64%. Measured against the run with that requirement and no extension, the extension sends 73% of its cost to fuel buyers and raises emissions by 72 Mt, close to its effect under current law. With the higher requirement the extension costs the Treasury $3.7 billion more, because more biofuel earns the credit.

Table 22: Decomposition of the full EPA offset, 2026–2045 (present value at 2%)

	Federal cost ($ billion)	Share to fuel buyers	Emissions (Mt)	Net welfare incl. climate ($ billion)
45Z extension alone, against current law	54.0	0.751	+66.6	−12.9
45Z extension given RFS +5.64%	57.7	0.729	+71.5	−13.5
Both together, against current law	57.9	0.011	−78.8	−2.0

Notes: "Given RFS +5.64%": the extension against the run with the requirement raised by 5.64% and no extension. Emissions: positive is an increase. Net welfare counts lost fuel-excise and California / Washington allowance revenue (E+A).

The states have written part of their response to the bank into law. A dollar of national support that adds to Oregon's, Washington's or New Mexico's bank makes their forecast deferral less likely, so their standards stay tighter and part of the dollar becomes abatement in those states rather than a lower credit price. California's acceleration mechanism does not trigger in any run. Figure 22

compares the 45Z extension with the codified responses on and off.

Switching them off makes little difference to who receives the federal dollar or to emissions (0.751 against 0.743; 66.6 against 66.1 Mt); the resource saving per additional tonne differs by about \$4 (\$51.5 against \$55.1). Once the mandate binds, the extension barely moves the state banks, so the deferral rules have nothing to respond to. In the rule's version the codified response is the only state response; California's discretionary tightenings of 2018 and 2024 both followed large banks, so this offset is a lower bound on the states' response. The same figure reports two state reforms that move credit value from renewable diesel toward SAF (SAF credits multiplied by 1.5, or renewable diesel credits by 0.5), the treatment of the shortfall at California's cap as an accumulated deficit carried forward at 5% rather than as issuance and Washington's 20% limit against its statutory path to 45%.

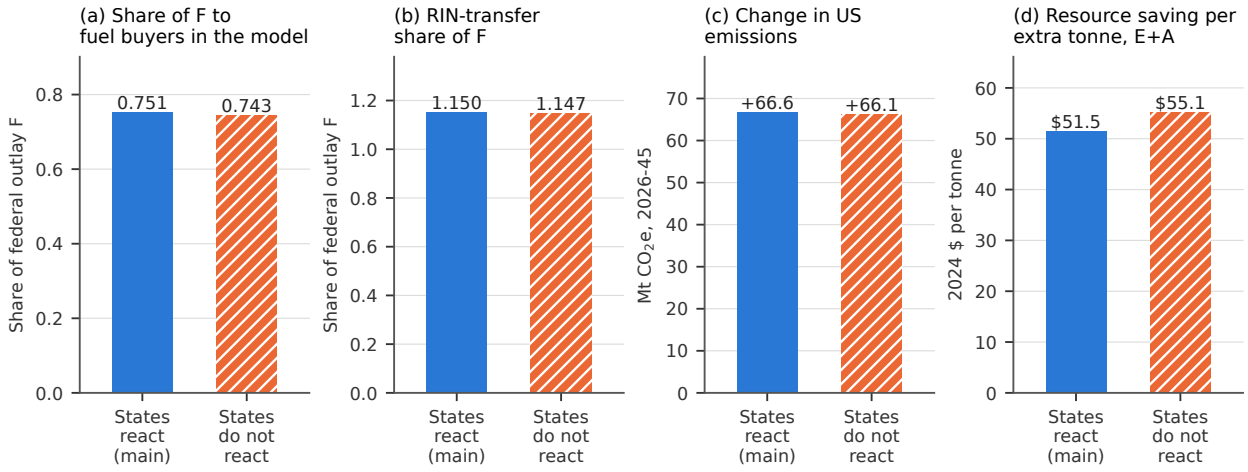


Figure 22: The 45Z extension with the states' codified responses on and off, 2026–2045

Notes: Codified responses: California's automatic acceleration mechanism and the deferral rules of Oregon, Washington and New Mexico. Each bar compares the 45Z extension with current law under the same state rules; with the responses off, both runs are without them. F: federal cost of the extension. Panel (d): resource saving per extra tonne emitted, including lost California / Washington allowance revenue (E+A). Present values at 2% to 2025, 2024 dollars.

G.6 Exemptions and the RIN price

At the exemptions in EPA's final rule (7.55 billion gallons, about 1.11 billion RINs), the model's 2026 D6 price is \$2.58, against a volume-weighted D6 price of \$1.92 in September 2026 (2024 dollars; EPA EMTS), after EPA's decision on 2025 exemptions on 31 August 2026 (\$2.02 in July–August, \$1.49 over January–August) (Section 5.4). The main specification uses the 2023–25 average exempted volume, 10.07 billion gallons, which gives \$2.29. Holding the 2025 bank at its observed level, the model reproduces the market price at a market-implied exemption volume close to EPA's decision EPA projected 1.11 billion exempted RINs for 2026–27; for 2025 EPA projected 0.78–0.99 billion exempted RINs and on 31 August 2026 granted 1.76 billion RINs on 29 of 34 petitions. This

market-implied volume is a calibration, not an estimate: it absorbs any other 2026 misfit of the RIN block, so I report it next to the 2026 D4 fit.

G.7 A national low carbon fuel standard

This appendix asks what a single national standard would do in place of the current stack. Congress announces it in 2028 and it takes effect in 2030. It replaces the four state programs and repeals the RFS. The state programs run as before in 2028–29, and their banks, 17.77 Mt in all, merge into the national market at the start of 2030. The national standard starts from the 2027 pool carbon intensity under current law and declines at California's scheduled rate, to 87% below that level by 2045. It has one price cap, California's $261.52 a tonne. Credits issued at the cap are federal revenue. 45Z ends after 2029, as under current law; 45Q, the state SAF credits and the vehicle policies are unchanged. Electric vehicles earn credits in every state. I report two versions. In the first (E1a), jet fuel carries no obligation and SAF earns credits against the diesel standard, as in California's program. In the second (E1b), fossil jet fuel is obligated against a national jet standard.

Both versions bind at the cap. The merged bank is used up in 2030, and from 2031 the national market issues credits at the cap in every year. In E1a the shortfall grows to 1.16 Gt in 2045, and federal revenue from issuance from $42 billion to $302 billion a year. In E1b the shortfall grows to 1.38 Gt, and revenue from $51 billion to $360 billion a year. A national standard on California's path therefore works as a carbon price of $262 a tonne on transportation fuel, extended to jet fuel in E1b. Its effect comes from the cap price, not from meeting the standard.

Table 23 compares the two versions with current law. Over 2026–45, against no fuel policy, current law abates 1,756 Mt at $149 a tonne. A national standard abates about three times as much at about two thirds of the cost per tonne: 5,395 Mt at $98.8 in E1a, and 5,910 Mt at $95.1 in E1b. Measured against current law, moving to E1a abates another 3,640 Mt at $74.3 a tonne, and moving to E1b another 4,155 Mt at $72.1. Net welfare including climate damages is $803 billion (E) for E1a and $899 billion (E) for E1b, against $199 billion (E) for current law. Governments gain as well, because the federal government collects the issuance revenue: against current law the fiscal gain is $1,948 billion in E1a and $2,351 billion in E1b, in present value. Because the national standard starts in 2030, the 2026–45 comparison includes 2026–29 under the current state stack; Table 23 also gives 2030–45 on the excise-only definition.

The comparison makes the point of Section 6.4 at scale. The current stack pays for fuel inside a binding federal mandate and leaves California's cap to do the pricing in one state. One national market at the same cap prices carbon in every state, and the cost per tonne falls.

Table 23: A national low carbon fuel standard from 2030

row	Mt, 2030-45	net cost -W ($bn), 2030-45	$/t (E), 2030-45	net welfare incl. climate, E ($bn), 2030-45	Mt, 2026-45	$/t (E+A), 2026-45	net welfare incl. climate, E ($bn), 2026-45
Current law	1,511.9	198.5	131.3	+171.9	1,755.6	149.0	+198.7
National standard (E1a)	5,151.6	479.6	93.1	+775.5	5,395.3	98.8	+802.9
National standard with jet (E1b)	5,666.5	508.7	89.8	+871.8	5,910.3	95.1	+899.2
E1a against current law	3,639.7	280.0	76.9	+604.7	3,639.7	74.3	+605.3
E1b against current law	4,154.7	309.1	74.4	+701.1	4,154.6	72.1	+701.6

Notes: Rows 1-3 against the run without fuel policy; rows 4-5 against current law. Both national-standard runs, without and with jet fuel obligated, converged (14 and 13 rounds) on the main specification. The national standard starts in 2030, so 2026-29 is under the current state stack: the 2026-45 columns include the state stack's 2026-29 abatement and cost relative to no fuel policy, identical in both national-standard runs and current law (against current law the two windows differ by < 0.1 Mt). $/t (E) = (resource cost + lost fuel excise) per undiscounted tonne; (E+A) also counts lost CA / WA allowance revenue at 2025 prices (negative against current law: CA / WA covered fossil use is higher under a national standard). PV 2 % to 2025, 2024 $. The 2026–45 net welfare column is on the excise-only convention (E); on the E+A convention used in Table 1 and Section 6.4, current law is +170.6, the national standard +784.0 and the standard against current law +614.3.

G.8 Provenance of the headline numbers

Table 24 lists, for each headline number, the parameters that drive it, whether they are estimated, calibrated or a convention, and how far the number moves across the alternatives reported in the paper.

Table 24: Where each headline number comes from

headline number (2026-45)	driving parameter(s)	estimated / calibrated / convention	range across the reported alternatives
Stack abates 1,756 Mt over 2026-45 at $149/t (E+A)	fuel-demand elasticities; oil price path; biofuel utilization response	calibrated (literature: Coglianese 2017, Dahl 2012; AEO 2025); utilization curves estimated	demand x0.5 / x1.5: 1,481 / 2,045 Mt, $160.9 / $140.2; AEO high / low oil: 898 / 3,176 Mt, $186.9 / $140.0; utilization nearly fixed 2,338 Mt at $183.2
$/t definition: + tax wedge (E+A); allowance prices held at 2025 ($27.34 CA, $59.36 WA)	what counts as cost: lost fuel excise and lost CA / WA allowance revenue	convention	resource cost only $118.3; (E) excise only $133.0; (E+A) $149.0
Net welfare of current law about one eighth of that of a carbon price at the SCC (E+A)	EPA social cost of carbon path; discount rate 2 %	convention (EPA SCC, 2 % path)	E 0.138 (198.7 / 1,434.8); E+A 0.121 (170.6 / 1,412.9); at 3%: E 0.100, E+A 0.020
About three quarters of a 45Z-extension dollar reaches fuel buyers (0.751)	capitalized share of the tax credit χ (main 1); imported-fat supply elasticity (5); soy / fat elasticities	χ: convention / assumption (lower bound 0.083 estimated at the two 45Z switch dates); imported-fat elasticity calibrated; soy / fat elasticities set (out-of-sample price fit as the check)	χ 0.75 / 0.50 / 0.083: 0.565 / 0.379 / 0.068; fat elasticity 2 / 10: 0.822 / 0.694; soy/fat +50%: 0.790 (2026–45); bootstrap 5th-95th 0.741-0.754
45Z extension raises emissions by 66.6 Mt and lowers net welfare by $12.9bn (E+A)	fuel-demand elasticities (gasoline price response to the lower D6 price); SCC for the welfare figure	calibrated (literature) / convention (SCC)	demand x0.5 / x1.5: +38.4 / +94.3 Mt; χ 0.083: +6.1 Mt; bootstrap +65.4 to +67.3 Mt
Resource saving $51.5 per extra tonne emitted (E+A)	imported-fat supply elasticity; investors' extension probability p	calibrated; p estimated (at the bound 1)	fat elasticity 2 / 10: $82.4 / $35.4; p 0 / 0.5: $37.4 / $44.1
45Z extension adds almost no biofuel (+0.11 %); hydroprocessing capacity +0.064 bn gal/yr (+0.8 %)	investors' extension probability p (main 1); entry costs	estimated (p at the bound; entry costs on 46 decisions)	p 0.5: +0.113 bn gal (1.7 %); p 0: +0.204 (3.0 %)
California's bank runs out in 2031 and its price reaches the cap in 2032	credit-holder expectations: exhaustion horizon K, response to balance surprises γ, holding cost r	estimated (γ_{CA} 1.108, r 3.38%, K fixed at 2 inside its 95% interval [1.07, 2.28])	bootstrap cap year 2031-32; perfect foresight 2033 (last round of a non-converged run); price-level bound 2031; storage + estimated regulator: never
A carbon price abates 100 Mt at about a fifth of the $/t of a higher RFS requirement	tax-wedge convention; imported-fat elasticity	convention (wedge) / calibrated	E+A 0.207 ($36.2 vs $175.0); E 0.218; resource only 0.048; fat elasticity 10: $35.0 vs $171.3, no flip
A national standard abates 5,395 Mt at $98.8/t (E+A), about three times current law at two thirds of its $/t	statutory design of the standard (start 2030, 87 % CI decline by 2045, cap issuance)	convention (policy design)	with jet fuel obligated: 5,910 Mt at $95.1; 2030–45 (E): 5,151.6 Mt at $93.1
Model 2026 D6 $2.29 vs observed $1.54 (2024 $, 1 Jan-27 Sep)	small-refinery exemption volume; RIN-holder storage and end-2025 RIN bank	calibrated / bracketed	small-refinery exemptions at the market-implied 11.6 bn gal: D6 2.11 (current-law run not converged; the converged extension run gives the same 2026 price); end-2025 RIN bank of 6.0 bn: D6 1.05; of 2.27 bn: 2.51

Notes: Estimated: by the author on data (credit-price rule, investment, utilization, RD delivery, EV demand). Set: soy / fat supply elasticities, with an out-of-sample price fit as the check. Calibrated: taken from the literature, statute or one-year data (fuel-demand elasticities, oil path, imported-fat elasticity, small-refinery exemption volume). Convention: accounting choices (SCC, discount rate, $/t definition, window, $\chi = 1$ in the main). Ranges are values for the 45Z extension or for current law at the ends of the input ranges in Figure 9, 2026-45, (E+A) unless stated; empty ends (not converged) are named. Bootstrap: 184 of 200 converged draws (2030–45), estimated parameters only.

G.9 Results under the five descriptions of expectations: full table

Table 25 is the full version of Table 4, with the ranges, the excise-only cost and the status of each run.

Table 25: Results under five descriptions of expectations, 2026–2045: full version

expectations	stack Mt	stack $/t (E+A)	stack $/t (E)	45Z share to fuel buyers	45Z +Mt	CA bank runs out	CA first year at cap	OR / NM first deferral	status
Estimated rule (main)	1,756	149.0	133.0	0.751	+66.6	2031	2032	2033 / 2033	converged
Storage model, statutory regulator	1,754	148.8	132.8	0.749†	+66.8†	2031	2031	2033 / 2033	converged
Storage model, estimated regulator (200 paths; mean [5th, 95th])	1,471 [1,429, 1,492]	125.5 [121.9, 127.3]	109.7 [106.6, 111.3]	0.710 [0.707, 0.724]†	63.4 [62.8, 66.4]†	never (all paths)	never (all paths)	2035 / 2035 (all paths)	200 of 200 paths converged
Perfect foresight (round 6 of a non-converged run; rounds 2-6 in brackets)	1,784	152.8	137.0	0.752 [0.752, 0.785]	+65.9 [65.9, 68.0]	2033 (rounds 2-6 same)	2033 (rounds 2-6 same)	2036 / 2036 (round 6 only)	not converged (last round)
Price-level bound (holders expect a 10 Mt/yr relaxation from 2027)	1,747	148.6	132.7	0.745	+66.1	2030	2031	2033 / 2032	converged

Notes: Each current-law column against the no-fuel-policy run, 2026–45; 45Z columns: extension vs current law under the same expectations, 2026–45. $/t (E+A) = (resource cost + lost fuel excise + lost CA / WA allowance revenue at 2025 prices) per undiscounted tonne, PV 2% to 2025, 2024 $; (E) = excise only. n/c: not computed. † 2030–45 (the storage column was aggregated on 2030–45; the main-column share is 0.751 on both windows). Perfect foresight: round 6 of a run that has not converged; its 45Z values compare a converged extension run with round 6 of the current-law run, with the range over rounds 2–6; no round range exists for the current-law rows; Oregon / New Mexico deferral years exist for round 6 only. Storage model: California holders price with a storage model (Appendix D), Oregon, Washington and New Mexico with the rule; the estimated regulator relaxes California's standard by 2029 in every path, so California never reaches its cap. Residuals: Appendix Table 27. CA cap year under current law (main): 2032.

G.10 Sensitivity

This subsection points to the sensitivity results. Figure 9 and Table 24 report the headline quantities with each input at the end of its range; the delivery slopes are varied from half to twice their estimated values, a range that contains the 95% confidence interval of California's estimate. The holding cost in the main specification is 3.38%, the real carry measured from forward quotes. Table 4 reports the descriptions of expectations, and Figure 19 Advanced Clean Cars II and direct air capture with 45Q extended. Across the delivery slopes the relocated share of the state standards' abatement stays between 49% and 53%. Figures 23 and 24 show every input, including those left out of Figure 9.

The price-expectation rule projects net credits from the last three years. When a standard is amended, past net credits should be restated to the amended standard; otherwise holders learn the same tightening twice, once in the starting price and again as a surprise. The backtest restates them. The forward runs start in 2026 without the restatement of the 2024 net credits to the 2025 amendments. Before running the correction I set five criteria for changing the main specification: the cost per tonne of current law moves by more than $5, its abatement by more than 1%, the



Figure 23: Every input at the end of its range: the 45Z extension, 2026–2045 (I)

Notes: As Figure 9, for every input. The 45Z extension against current law under the same input: shares of its federal cost reaching (a) fuel buyers, (b) feedstock suppliers and (c) producers; (d) extra emissions (Mt, incl. indirect land-use change). All bars 2026–45. Orange: low end or alternative setting; green: high end; hollow: one end only. MISSING: the end did not solve; n/a: not computed by design; not run: the SAF-capable share of new conversions at 0.5, because the SAF retrofit investment parameters are estimated at 0.2. The utilization high end is not paired: its current-law row converged only on a later code version. The oil price bar moves the no-fuel-policy baseline itself and is shown last. In Figure 9 the bootstrap bar spans 184 of 200 converged draws, and a row with one marker had the other end fail to solve (Appendix Table 27).



Figure 24: Every input at the end of its range: the extension's cost and the current fuel policies, 2026–2045 (II)

Notes: (e) Resource saving per extra tonne emitted by the 45Z extension (E+A); (f) federal cost of the extension ($ billion, present value); (g) abatement and (h) cost per tonne (E+A) of the current fuel policies against no fuel policy under the same input. All bars 2026–45. Markers as in Figure 23. At cane elasticity 0.50 the no-fuel-policy row did not solve, so panels (g) and (h) have no value at that end. Present values at 2% to 2025, 2024 dollars.

share of a 45Z extension reaching fuel buyers by more than 0.01, the extension's extra emissions by more than 2 Mt, or California's bank-out or cap year changes. None is met.

Table 26: Forward runs with and without the restatement of past net credits

	Main specification	With restatement
Cost per tonne, current law, 2026–45 ($, E+A)	148.99	149.10
Abatement, current law, 2026–45 (Mt)	1,755.6	1,755.5
Share of a 45Z extension reaching fuel buyers	0.751	0.745
Extra emissions from the extension (Mt)	66.6	65.8
California: bank runs out / first year at the cap	2031 / 2032	2031 / 2032
Oregon / New Mexico: first year of deferral	2033 / 2033	2033 / 2033
California credit price, 2027 / 2028 ($/t)	77.0 / 115.1	82.8 / 118.4
California shortfall at the cap, 2045 (Mt)	61.7	61.7

Notes: The restatement raises California's price in 2027–28, when the expected net credits are lower, and leaves later years almost unchanged ($159 in 2029 in both runs).

H Further Results and Convergence

H.1 Convergence

Every row used in the paper is solved by iterating between the yearly market equilibrium and the dynamic blocks until the target gap is at most 0.06, the change in the state credit prices between rounds at most $0.5 a tonne, the change in rack prices at most $0.005 a gallon, and no regime flag is still moving. Table 27 lists the rows and their last-round residuals. Rows that do not meet the rule are reported as not converged, with their residuals, or with the range over the last saved rounds. Every failure sits where a layer of policy switches between binding and slack: the year the RFS stops binding, the ethanol blend kink, cap and deferral cycles in the small state markets (mainly Oregon), a feedstock corner, or the high end of electric vehicle adoption, where electricity credits push Washington's price to its floor. None fails where all markets are interior.

Results from runs that have not converged. The main text reports none of these. A perfect-foresight path is reported here. Its solver has not converged, but California's cap year is 2033 in every round. Under perfect foresight (round 6 of a run that has not converged) California's bank runs out and its price reaches the cap in 2033, the same year in rounds 2–6. Holders who foresee the scarcity price it in from the start, so California's price in 2028 is $221 under perfect foresight against $115 under the rule; the higher price draws more renewable diesel into the state and lowers fuel use, and the bank lasts longer. In the last round the cost of current law is $152.8 a tonne and the share of the 45Z extension reaching fuel buyers is 0.752 (0.752–0.785 over rounds 2–6); California's bank-exhaustion year, its cap year and its 2028 price are the same in rounds 2–6. Appendix Table 25 gives the row and Appendix Figure 28 the price paths. The perfect-foresight

one-step forecast of 2026 rests on the same run and is not shown in Figure 5. In the policy menu, the D2 SAF category with jet fuel obligated (round 40 of a run that has not converged) abates 235.7 Mt at $75 a tonne, with a fiscal cost of −$62.2 billion, federal revenue from D2 waiver credits sold by EPA. For the fast-charging credit, with the network effect of Cole et al. (2023) the credit also ranks second, in a run whose reference has not converged; with the strongest specification the runs do not converge.

Tornado ends not drawn in Figure 9. The following ends did not solve or were not run: learning rates of zero (current-law row not converged); the high end of the utilization response (current-law row not used, because its rerun is on a different code version without a documented parity check); a SAF-capable share of new conversions of 0.5 (not run, because the SAF retrofit investment parameters are estimated at 0.2); soybean-oil and fat supply elasticities −50% (current-law row not converged); the AEO high and low oil-price paths (extension rows not converged; appendix bar only); the 2026 transitional terms kept permanently (current-law row not converged); high electric-vehicle adoption drift (current-law, extension and charging-credit rows not converged); and a sugarcane supply elasticity of 0.50 (no-fuel-policy row not solved; panel (c) only).

Table 27: Convergence of the model rows used in the paper

row	used in	status	last-round residuals
Current law (main run)	every main-text number	converged	9 rounds; dP_LCFS 0.192; dRack 0.0018; gap 0.0591; rin 0.03
45Z extension (main run)	Table 2, abstract, intro	converged	10 rounds; dP_LCFS 0.103; dRack 0.0019; gap 0.0529; rin 0.0051
No fuel policy (reference)	stack, 6.1, national standard	converged	13 rounds; dP_LCFS 0; dRack 0; gap 0.045
Carbon price at the SCC	welfare share	converged	12 rounds; dP_LCFS 0; dRack 0; gap 0.0414
No RFS	RFS layer	converged	13 rounds; dP_LCFS 0.0122; dRack 0.00021; gap 0.0233
No state standards	state layer	converged	14 rounds; dP_LCFS 0; dRack 0.0033; gap 0.051; rin 0.031
45Z extension, no state standards	state-price channel	converged	14 rounds; dP_LCFS 0; dRack 3e-05; gap 0.0509; rin 0.02
No 45Z in 2026-29	45Z as legislated	converged	9 rounds; dP_LCFS 0.156; dRack 0.002; gap 0.0334; rin 0.0099
45Z extension, chi 0.75	capitalization	converged	13 rounds; dP_LCFS 0.374; dRack 0.0018; gap 0.0243; rin 0.034
45Z extension, chi 0.50	capitalization	converged	14 rounds; dP_LCFS 0.0808; dRack 0.00019; gap 0.046; rin 0.011
45Z extension, chi 0.083	capitalization	converged	19 rounds; dP_LCFS 0.00194; dRack 0.00031; gap 0.0568; rin 0.034
No fuel policy with E10 floor	stack footnote	converged	13 rounds; dP_LCFS 0; dRack 0; gap 0.045
45Z extension + EPA offset (+5.64 %)	Table 2 EPA column, tab:menu	converged	14 rounds; dP_LCFS 0.137; dRack 0.0022; gap 0.057; rin 0.047
RFS +5.64 % alone	EPA-offset decomposition	converged	12 rounds; dP_LCFS 0.111; dRack 0.00066; gap 0.0575; rin 0.018

row	used in	status	last-round residuals
Higher RFS requirement (+3.94 %, 100 Mt)	tab:menu	converged	9 rounds; dP_LCFS 0.278; dRack 0.0024; gap 0.053; rin 0.011
Carbon price (100 Mt)	tab:menu	converged	11 rounds; dP_LCFS 0.103; dRack 0.00053; gap 0.0426; rin 0.04
Fast-charging credit (B1)	tab:menu	converged	10 rounds; dP_LCFS 0.104; dRack 0.0022; gap 0.057; rin 0.0048
Purchase credit (B4)	tab:menu	converged	11 rounds; dP_LCFS 0.135; dRack 0.00037; gap 0.0514; rin 0.027
45Z extension + RFS +2 %	EPA-offset figure	converged	10 rounds; dP_LCFS 0.0159; dRack 0.0017; gap 0.0348; rin 0.025
RFS +2 %	EPA-offset figure	converged	9 rounds; dP_LCFS 0.161; dRack 0.0018; gap 0.0459; rin 0.0095
45Z extension + RFS +2.65 %	EPA-offset figure	converged	18 rounds; dP_LCFS 0.000459; dRack 4.1e-05; gap 0.0247; rin 0.024
RFS +2.65 %	EPA-offset figure	converged	11 rounds; dP_LCFS 0.0553; dRack 0.00045; gap 0.0404; rin 0.0064
45Z extension + RFS +4 %	EPA-offset figure	converged	11 rounds; dP_LCFS 0.0338; dRack 0.00046; gap 0.0339; rin 0.032
RFS +4 %	EPA-offset figure	converged	12 rounds; dP_LCFS 0.325; dRack 0.0035; gap 0.0538; rin 0.01
EPA cuts RFS 1.5 % from 2030 (current law)	Table 2 file	converged	14 rounds; dP_LCFS 0.0143; dRack 0.00012; gap 0.0502; rin 0.044
EPA cuts RFS 1.5 % from 2030 (45Z extension)	Table 2 file	converged	9 rounds; dP_LCFS 0.144; dRack 0.0019; gap 0.0475; rin 0.039
EPA cuts RFS 1.5 % from 2035 (current law)	Table 2 file	converged	12 rounds; dP_LCFS 0.0518; dRack 0.0015; gap 0.0575; rin 0.017
Investor p = 0.5 (current law)	Table 2 p = 0.5, capacity	converged	11 rounds; dP_LCFS 0.053; dRack 0.00059; gap 0.0498; rin 0.0018
Investor p = 0.5 (45Z extension)	Table 2 p = 0.5, capacity	converged	13 rounds; dP_LCFS 0.0901; dRack 0.00051; gap 0.0274; rin 0.027
National standard (E1a)	6.5, tab:natlcfs	converged	14 rounds; dP_LCFS 0.000986; dRack 1.9e-05; gap 0.0384; rin 0.039
National standard with jet (E1b)	6.5, tab:natlcfs	converged	13 rounds; dP_LCFS 0.00197; dRack 6.1e-05; gap 0.0542; rin 0.039
Price-level bound (current law)	Sec. 7, T5	converged	14 rounds; dP_LCFS 0.0149; dRack 0.00012; gap 0.0433; rin 0.0032
Price-level bound (45Z extension)	Sec. 7, T5	converged	14 rounds; dP_LCFS 0.235; dRack 0.00061; gap 0.0335; rin 0.027
Perfect foresight (45Z extension)	Table 2 PF, T5	converged	15 rounds; dP_LCFS 0.0767; dRack 0.00032; gap 0.0389; rin 0.013
Perfect foresight (current law), round 6	Table 2 PF, T5, fig:ca-4lines	NOT converged (band over rounds 2-6 reported)	6 rounds; dP_LCFS 8.97; dRack 0.2; gap 0.0962; rin 0.031; moving: _lcfs_pf_moving, _dfr_moving
Perfect foresight, holders' rate 8 % (current law)	appendix PF r rows	NOT converged (band over rounds 2-6 reported)	6 rounds; dP_LCFS 2.86; dRack 0.045; gap 0.0614; rin 0.037; moving: _lcfs_pf_moving, _dfr_moving
Perfect foresight, holders' rate 10 % (current law)	appendix PF r rows	NOT converged (band over rounds 2-6 reported)	6 rounds; dP_LCFS 10.5; dRack 0.31; gap 0.184; rin 0.024; moving: _dfr_moving

row	used in	status	last-round residuals
C2: nested D2 SAF category, jet obligated (round 40)	tab:menu C2 row	NOT converged; complete outputs at round 40 (values labelled)	20 continuation rounds (40 overall); dP_LCFS 0.486; dRack 0.0013; gap 0.0811; rin 8.4; moving: pf_moving
Storage model, statutory regulator (current law / 45Z)	T5	converged	M0 29 rounds (20 + 9 continuation), Z45 14 rounds
Storage model, estimated regulator (200 paths x 2)	T5	converged	200 / 200 paths in both blocks, 11-17 rounds (uniform rule)
EPA bound x 13.1/15.8 from 2030 (current law / 45Z)	Table 2 file (not in the .tex)	not converged	M0 pf_moving, gap 0.0645; Z45 pf_moving, gap 0.028 (40 rounds); last saved round, approximate
High network: B1_nethigh (main-text pair)	state layer under the high network	not converged	40 rounds: dP_LCFS 0.556, dRack 0.165, gap 0.082, _dfr_moving; nothing computed
High network: M0_nethigh (reference)	network ranking	not converged	40 rounds: dP_LCFS 0.377, dRack 0.0502, gap 0.0639, _dfr_moving
Netcole reference (M0_ netcole)	network ranking	not converged	20 rounds dRack 0.033, _dfr_moving; damped rerun not converged at 40
Low oil price, 45Z extension	tornado oil-low end	not converged	40 rounds (damped continuation): gap 0.1657, dRack 0.0239, dP_LCFS 0.344, _dfr_ moving; end MISSING
Parameter bootstrap (200 draws of the extension and current law)	Table 2 band (0.741-0.754)	converged (184 / 200)	16 draws failed; IPW leaves medians and intervals unchanged

Convergence: uniform rule: target gap ≤ 0.06, dP_LCFS ≤ 0.5, dRack ≤ 0.005, no moving flag. Residuals are those of the last round of each run. "Band reported": the cells are the last round of a non-converged run with the range over the saved rounds.

H.2 The cost per tonne on other definitions

Table 28 reports the policy menu on the excise-only definition; Tables 29, 30 and 31 give the forgone allowance revenue under four price assumptions, the fiscal effects by government and where the state credit value goes. In Table 3, the fiscal cost counts the federal and state budgets, including the lost allowance revenue and excluding the credits issued at the state caps. Including those credits as state revenue, the fiscal cost (excise only) is −$62.6 billion for current law, $16.4 billion for the charging credit, $114.0 billion for the purchase credit and $2.3 billion for the higher requirement. The 45Z credit of 2026–29 is evaluated over 2026–45; the run without fuel policy has no ethanol blend floor; the cap-and-trade programs are assumed to offset none of the standards' covered abatement.

A carbon price at the social cost of carbon in place of current fuel policy abates 11,640 Mt over 2026–45, 6.6 times current law, at $125.1 a tonne; its net welfare including climate damages is $1,413 billion.

Table 28: The policy menu on the excise-only definition, 2026–2045

Policy (vs current law; first row vs no fuel policy)	Abatement (Mt)	$/t resource	$/t + tax wedge (E)	$/t + other ext. (E)	Fiscal cost ($bn)	Federal outlay F ($bn)	Share to fuel buyers
Current law against no fuel policy	1,755.6	118	133	111	41.8	17.0	—
45Z credit as legislated, 2026-29	−14.9	costs 8	saves 33	costs 42	16.9	16.9	0.71
45Z extension	−66.6	saves 23	saves 53	costs 12	53.6	54.0	0.75
45Z extension with EPA offset (RFS +5.64%)	78.8	263	255	267	58.2	57.9	0.01
Fast-charging credit (credits 2028-32)	171.8	24	59	60	14.1	8.2	0.15
Purchase credit (credits 2028-32)	434.9	60	100	102	99.3	105.3	0.07
Higher RFS requirement (+3.94%), 100 Mt target	103.5	155	165	141	0.5	0.1	—
Carbon price, 100 Mt target	100.4	7	36	−29	−54.9	0.0	—
D2 SAF category with jet obligated (C2; round 40, NOT converged)	235.7	56	82	20	−60.5	−0.2	—

Notes: As Table 3, but forgone allowance revenue is not counted; columns: resource cost, + tax wedge (E), + other external costs (E, central), fiscal cost (E).

Table 29: Forgone allowance revenue under four allowance-price assumptions, 2026–2045

	(b) main	(a) excise only	(c) floor path	(d) ceiling path
Lost CA / WA allowance revenue, current law vs no fuel policy ($ billion)	28.0	0.00	40.4	144.1
Lost CA / WA allowance revenue, state-standards layer ($ billion)	19.9	0.00	29.5	106.3
Current law vs no fuel policy: $ per t, + tax wedge	149.0	133.0	156.0	215.1
State-standards layer: net welfare incl. climate at zero cap offset ($ billion)	−20.4	−0.44	−30.0	−106.7
Fiscal: federal, change in revenue ($ billion)	−18.7	−18.7	−18.7	−18.7
Fiscal: state excl. cap issuance, change in revenue ($ billion)	−51.2	−23.2	−63.6	−167.3
Fiscal cost excl. cap issuance ($ billion)	69.8	41.8	82.2	186.0
Government net revenue incl. credits issued at the state caps ($ billion)	34.6	62.6	22.2	−81.6
Cap offset: break-even share s*, current law vs no fuel policy	0.68	0.71	0.66	0.40
Cap offset: break-even share s*, state-standards layer	none	none	none	none
Cap offset share above which the state standards raise national emissions	0.63	0.63	0.63	0.63

Notes: (b), the main-text value, holds the observed 2025 average auction price in real terms (CA $27.34, WA $59.36 per tonne, 2024 $). (a), (c) and (d) give the range; (d), every covered tonne at the legal price ceiling, is an upper bound. (b) is not a lower bound. (c) and (d) re-price the same tonnes; the runs are not re-solved.

Table 30: Fiscal effects by government, 2026–2045 ($ billion, present value at 2%)

	State standards	Current law (whole stack)	RFS layer	45Z extension
Federal: 45Z / 45Q credit outlays	0.12	−16.95	−4.85	−54.04
Federal: other credit outlays (SAF, EV, chargers, investment)	−0.02	−0.02	−0.01	0.00
Federal: fuel excise revenue	−5.13	−6.20	−1.07	0.66
Federal: other lines (duties, tariffs, D2 waiver credits)	0.88	4.53	2.92	−1.57
Federal total	−4.15	−18.65	−3.01	−54.95
Four LCFS states: fuel excise revenue	−17.88	−15.93	1.84	0.16
Four LCFS states: incentive outlays and other lines	0.00	−0.00	−0.00	0.00
Four LCFS states: lost CA / WA allowance revenue (E+A)	−19.91	−28.02	−4.20	−0.13
Four LCFS states total excl. cap issuance (E+A)	−37.79	−43.96	−2.35	0.03
Other states: fuel excise revenue	−0.52	−3.73	−3.04	1.22
Other states: other lines	−0.01	−3.49	−1.20	−0.00
Other states total	−0.53	−7.22	−4.25	1.22
Fiscal cost excl. cap issuance (E+A) = -(federal + state)	42.47	69.84	9.61	53.70
Fiscal cost excl. cap issuance (E)	22.55	41.81	5.41	53.57
memo: credits issued at the state caps (state revenue)	104.40	104.40	−21.72	0.89
Government net revenue incl. cap issuance (E+A)	61.94	34.56	−31.33	−52.82
Federal share of the fiscal cost excl. cap issuance (E+A)	0.10	0.27	0.31	1.02

Notes: Positive: revenue gained. Four LCFS states: CA, OR, WA, NM. Fiscal cost excludes the credits issued at the state caps (memo row) and includes the lost CA / WA allowance revenue (main definition); the excise-only value is given beside.

Table 31: Where the LCFS credit value goes, by state, current law, 2026–2045 ($ billion)

State	Biofuel	EV charging	Other gen.	Imports	Bank holders	State (cap)	DAC	Total (PV)	Total (undisc.)
CA	−4.4	48.9	30.9	0.0	4.8	104.0	0.0	184.2	234.1
WA	0.9	1.4	0.1	0.0	−0.1	0.0	0.0	2.3	2.7
OR	8.4	8.3	0.7	0.1	0.2	0.0	0.0	17.7	22.3
NM	6.2	1.6	0.0	0.0	0.2	0.4	0.0	8.4	10.6
All four	11.1	60.2	31.7	0.1	5.1	104.4	0.0	212.6	269.8

Notes: Credit value: each year's credits times the state's LCFS price, booked to the recipient; recipients sum to the payments of the obligated parties. Bank holders: net sales out of the bank. State (cap): credits issued at the price cap. California's net biofuel credit value is negative: from the 2030s several biofuels' carbon intensity is above the standard, and their deficits exceed the low-carbon fuels' credits. Present value at 2% to 2025; undiscounted total in the last column.

H.3 Sources of abatement

Table 32 splits the RFS's abatement by source, and Table 33 gives the renewable diesel and biodiesel drawn into each of the four states.

Table 32: Abatement from the RFS by source, current law against no RFS (Mt CO_2e)

	2026–45			2030–45		
Source	CA, OR, WA, NM	Rest of US	National	CA, OR, WA, NM	Rest of US	National
RD and BD relocated between states	−125.4	125.4	0.0	−90.8	90.8	0.0
Other RD and BD (volume and CI)	339.0	19.5	358.5	272.6	7.4	280.0
EVs: fuel displaced	−31.3	73.0	41.6	−31.1	67.2	36.1
EVs: charging emissions	1.2	−9.8	−8.6	1.2	−8.3	−7.1
Price-induced fall in fossil fuel demand	−1.0	175.6	174.5	5.7	133.0	138.7
Ethanol	2.2	50.1	52.3	1.7	47.6	49.3
SAF	2.3	6.2	8.5	2.3	4.1	6.4
Fossil CI change (residual)	0.0	0.0	0.0	0.0	0.0	0.0
Total	**187.0**	**440.0**	**626.9**	**161.5**	**341.9**	**503.4**

Notes: Current law against the run without the RFS from 2026 (no RIN market, advanced tier or non-road volume; the state standards, 45Z, 45Q, state SAF programmes and the EV block as under current law; neither run has a hard E10 floor, so without the RFS ethanol falls below 10% of gasoline outside the four states in 2030–33 and 2041–45). Positive = abatement. Relocation is valued at the national pooled net factor and sums to zero nationally. Same caveats as Figure 7.

Table 33: Renewable diesel and biodiesel drawn into the four states, cumulative 2026–45

	Renewable diesel (bn gal)			Biodiesel (bn gal)			Relocated
State	Current law	No state LCFS	Relocated	Current law	No state LCFS	Relocated	(Mt CO_2e)
California	55.53	13.46	43.90	4.90	0.14	4.75	421.7
Oregon	7.62	0.09	7.54	1.82	0.55	1.28	78.6
Washington	7.89	2.94	5.29	0.82	0.01	0.81	47.0
New Mexico	5.00	0.08	4.93	1.46	0.41	1.07	53.1
Four states	76.04	16.57	61.66	9.00	1.11	7.92	600.5
Rest of US	3.08	63.88	−61.66	17.17	24.64	−7.92	−600.5
United States	79.13	80.45	0.00	26.17	25.75	0.00	0.0

Notes: Current law against the run without the four state standards. Consumption by state, billion physical gallons. "Relocated" = the change in a state's consumption less its no-standard share of the national change, by feedstock; valued at the national pooled net factor (Mt CO_2e) it gives the relocation term of Figure 7.

H.4 Pump prices and feedstock markets

Table 34 and Figure 25 give the pump-price effect of each layer by state and fuel; Table 35 and Figure 26 give the upstream incidence and the feedstock markets.

Table 34: Pump-price effect of each policy layer by state and fuel, cents per gallon (2024 dollars)

		State standards			45Z extension			Current law vs no fuel policy		
State	Fuel	2030	2026–45	vol.-wt.	2030	2026–45	vol.-wt.	2030	2026–45	vol.-wt.
California	gasoline	66.9	156.3	140.4	−1.1	−1.1	−1.0	78.3	164.1	148.3
	diesel	82.9	191.1	188.3	−2.8	−2.7	−2.6	102.7	210.3	207.7
Oregon	gasoline	36.2	81.1	75.2	−1.4	−1.7	−1.5	47.6	89.4	83.4
	diesel	52.2	108.5	107.7	−3.2	−3.3	−3.2	72.0	127.7	127.0
Washington	gasoline	10.1	5.2	5.4	−1.2	−1.3	−1.3	21.5	13.5	13.7
	diesel	11.3	5.9	6.0	−3.0	−2.9	−2.9	31.1	25.2	25.3
New Mexico	gasoline	34.4	48.9	47.5	−1.6	−2.3	−2.2	45.5	55.0	53.6
	diesel	39.5	56.9	56.6	−3.3	−4.0	−4.0	59.2	76.1	75.9
Rest of US	gasoline	0.1	−0.3	−0.3	−1.6	−1.2	−1.2	5.6	3.3	3.3
	diesel	−0.9	−0.9	−0.9	−3.4	−2.8	−2.8	18.8	18.4	18.4

Notes: Scenario minus reference change in the price of blended gasoline (E10-type) and blended diesel. State standards: current law vs current law without the four state programs; 45Z extension: extension vs current law (0 in 2026–29, included in the mean); current law vs no fuel policy. 2026–45 = simple mean over years (main); vol.-wt. = weighted by the average of the two runs' volumes. The retail wedge (taxes, distribution, and the California and Washington allowance cost, held at its 2025 level) is the same in both runs, so the retail change equals the blended-price change. Outside the four states prices move only through the national RIN price. Jet fuel: 0 in every state and layer (jet carries no RFS obligation or state deficit in the model).

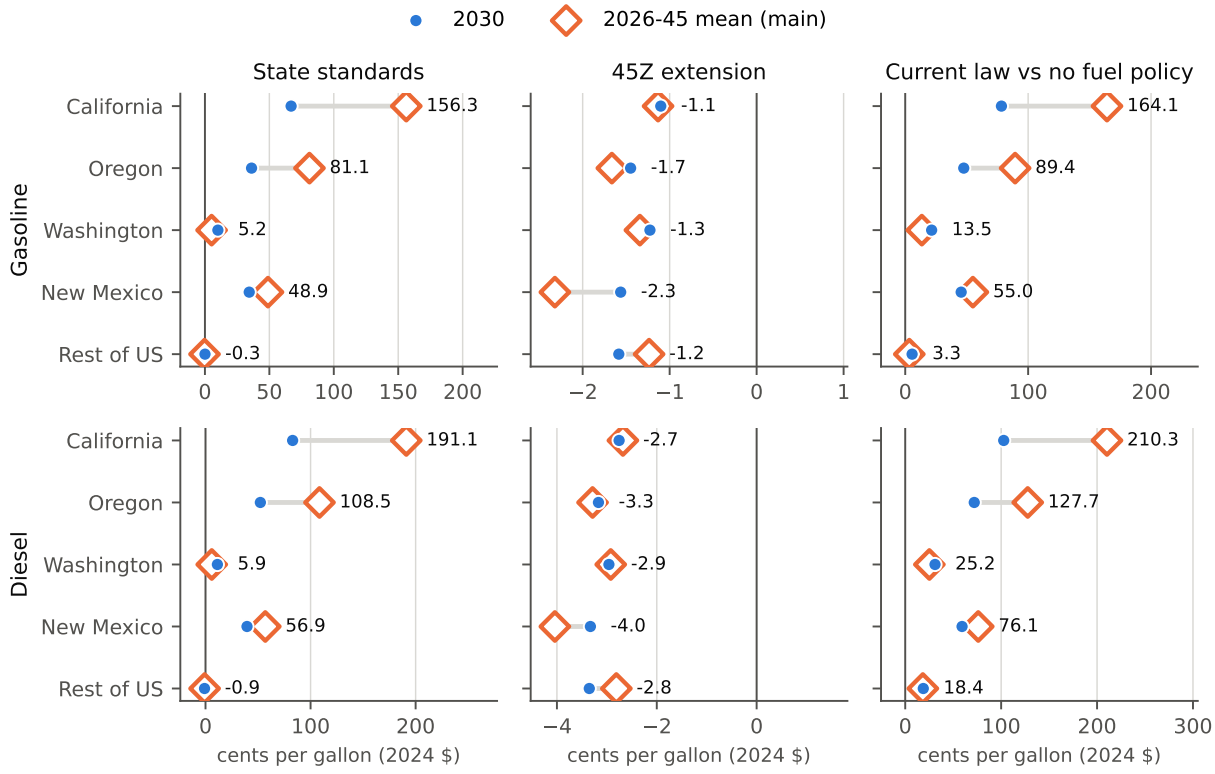


Figure 25: Pump-price effect by state: state standards, 45Z extension and current law vs no fuel policy

Notes: Circles: 2030; diamonds: 2026–45 mean (main), labelled. Cents per gallon, 2024 $; note as Table 34. Panels have their own horizontal scales (the 45Z extension panels are in single cents).

Table 35: Upstream incidence and feedstock markets by policy layer

A. Producer surplus by layer, $bn PV at 2%, 2026–45

	RFS	State standards	45Z extension	Current law vs no fuel policy
U.S. farmers (soybean oil, NA fat, corn)	65.1	−13.2	6.9	86.0
soybean oil	32.7	−12.8	2.6	36.3
North-American fat	14.2	3.5	4.5	26.1
corn	18.1	−3.9	−0.2	23.6
of which Midwest	53.1	−11.8	5.1	69.0
Distillers corn oil (ethanol plants)	6.9	1.5	1.8	18.8
U.S. feedstock suppliers, total	72.0	−11.7	8.7	104.8
Memo: foreign fat / UCO suppliers (not in U.S. welfare)	5.2	1.6	−2.9	7.4

B. Feedstock prices (2024 $) and uses: current law, and change caused by each layer

	Current law		State standards		45Z extension		Stack vs NFP		RFS	
	2030	2026–45	2030	2026–45	2030	2026–45	2030	2026–45	2030	2026–45
Soybean oil price ($/lb)	0.482	0.470	−0.032	−0.034	0.010	0.007	0.132	0.118	0.132	0.103
North-American fat price ($/lb)	0.564	0.556	0.017	0.020	0.029	0.023	0.284	0.272	0.104	0.089
Imported fat / UCO price ($/lb)	0.514	0.489	0.017	0.020	−0.050	−0.041	0.284	0.256	0.104	0.089
Corn price ($/bu)	4.22	3.71	−0.04	−0.06	0.00	0.00	0.77	0.37	0.77	0.29
Crop oil to fuel (bn lb)	20.8	20.2	−2.3	−2.6	0.7	0.6	20.8	20.0	9.9	9.1
North-American fat to fuel (bn lb)	12.1	12.0	1.1	1.2	2.0	1.5	10.6	10.4	5.5	4.9
Imported fat / UCO to fuel (bn lb)	6.9	5.6	1.1	1.1	−2.8	−2.0	6.8	5.4	4.7	3.5
Distillers corn oil to fuel (bn lb)	4.8	4.8	0.1	0.1	0.1	0.1	1.4	1.4	0.5	0.4
Corn for ethanol, gross (bn bu)	4.66	4.33	−0.04	−0.08	0.00	0.00	0.92	0.47	0.92	0.36

Notes: Panel A: producer surplus, $bn PV at 2%, 2026–45. U.S. farmers = surplus of the North-American soybean-oil, fat and corn supply curves (the fat curve covers the U.S., Canada and Mexico, including yellow grease and used cooking oil); distillers corn oil surplus goes to ethanol plants; foreign fat and UCO suppliers are outside U.S. welfare (memo). Panel B: feedstock prices (2024 $) and quantities used for fuel, current-law level and change caused by each layer, 2030 and 2026–45 mean. Without the RFS or any fuel policy the soybean-oil price is at its floor ($0.35/lb) in most years and the fat price at $0.28/lb.



Figure 26: Feedstock prices and uses under current law and without each layer, 2026–45

Notes: Soybean (crop) oil includes canola (one crop-oil pool). Fuel use = gallons × conversion per pathway. North-American fat = U.S., Canada and Mexico incl. yellow grease and used cooking oil. Prices in 2024 $. The soybean-oil price falls over time under current law because crushing capacity grows (USDA baseline) and the 2026 crush shortfall unwinds by 2028, while crop-oil use for fuel rises. Floors: $0.35/lb soybean oil, $0.28/lb fat.

H.5 Electric vehicle paths

Figure 27 and Tables 36 and 37 give the electric share of new-vehicle sales by class and region under five policy cases.

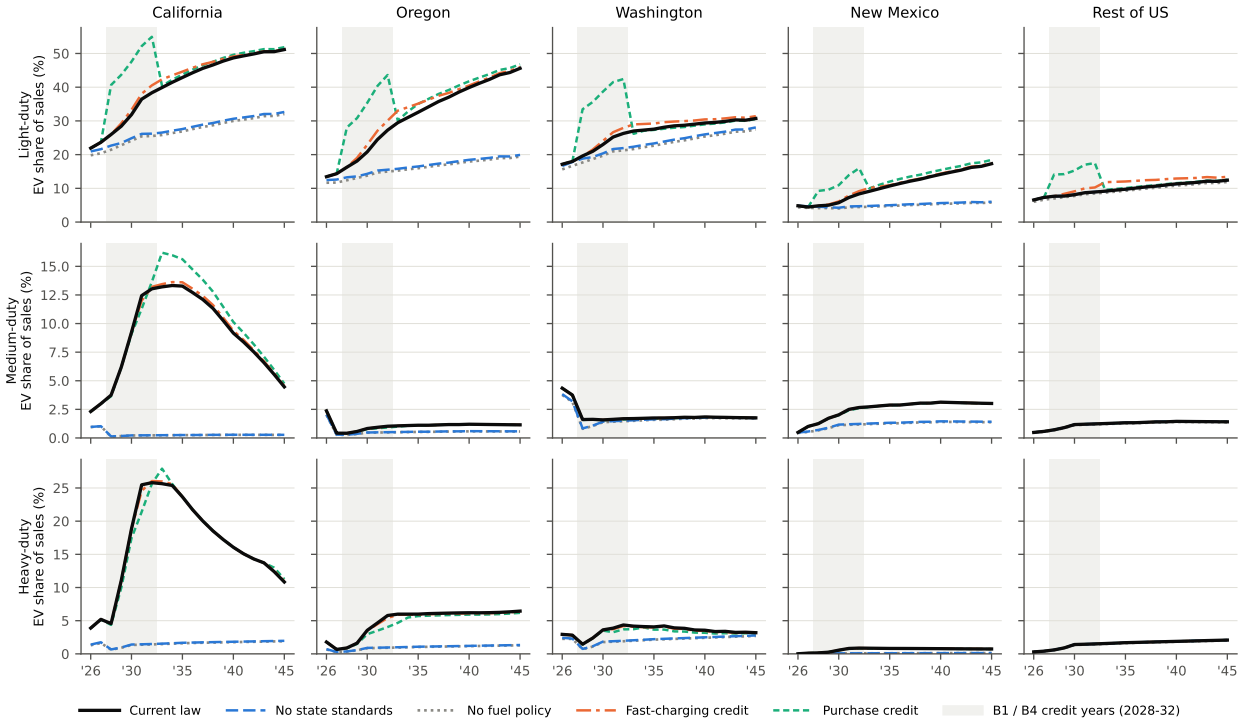


Figure 27: Electric share of new-vehicle sales by class and region, five policy cases, 2026–45

Notes: Rows: light-, medium- and heavy-duty vehicles; columns: California, Oregon, Washington, New Mexico and the rest of the United States. Lines: current law (main run), no state standards, no fuel policy, and current law with a federal fast-charging credit or a federal EV purchase credit paid in 2028–32 (shaded). Share = EV sales / all new sales of the class. The heavy- and medium-duty peaks in California come from the state standard: the credit price rises to its cap by 2032, after which each MWh of charging earns fewer credits as the standard tightens (Appendix Table 37). Source: model runs.

Table 36: Electric share of new-vehicle sales (%), 2026–45: light-duty vehicles

Region	Scenario	2026	2028	2030	2032	2035	2040	2045
Panel A: light-duty vehicles								
California	Current law	21.9	26.0	31.9	38.3	42.9	48.7	51.2
	No state standards	20.9	22.7	24.9	26.2	27.6	30.6	32.7
	No fuel policy	19.8	21.4	24.1	25.5	26.9	30.0	32.0
	Fast-charging credit	21.9	26.0	33.3	40.5	44.6	49.1	51.3
	Purchase credit	21.9	40.7	47.6	54.9	43.8	49.6	51.9
Oregon	Current law	13.4	16.3	20.8	27.3	32.6	39.9	45.6
	No state standards	12.5	13.3	14.3	15.5	16.5	18.5	19.9
	No fuel policy	11.7	12.4	13.8	15.0	16.0	18.0	19.4
	Fast-charging credit	13.4	16.3	22.9	30.1	35.1	40.5	46.0
	Purchase credit	13.4	28.0	35.3	43.6	35.0	41.8	46.8
Washington	Current law	17.1	19.6	23.0	26.3	27.6	29.4	30.7
	No state standards	16.6	18.8	20.4	21.9	23.3	26.0	28.1
	No fuel policy	15.6	17.7	19.7	21.2	22.6	25.4	27.5
	Fast-charging credit	17.1	19.5	24.1	27.9	29.2	30.4	31.4
	Purchase credit	17.1	33.5	38.7	42.4	27.3	29.0	30.6
New Mexico	Current law	4.8	4.8	5.7	8.4	10.7	14.1	17.3
	No state standards	4.8	4.6	4.2	4.7	5.0	5.6	6.0
	No fuel policy	4.4	4.1	4.0	4.5	4.8	5.4	5.8
	Fast-charging credit	4.8	4.7	6.1	9.2	10.9	14.4	17.5
	Purchase credit	4.8	9.2	10.9	16.0	12.0	15.4	18.4
Rest of US	Current law	6.6	7.7	8.1	8.9	9.7	11.3	12.4
	No state standards	6.6	7.7	8.1	8.9	9.7	11.3	12.3
	No fuel policy	6.0	7.0	7.7	8.5	9.3	10.8	11.9
	Fast-charging credit	6.6	7.7	9.1	10.3	12.0	12.9	13.4
	Purchase credit	6.6	14.1	15.3	17.5	10.1	11.5	12.5

Note: share of electric vehicles in new sales of each class. Current law is the main run; "no state standards removes the four state programs; "no fuel policy removes the RFS, 45Z and the state programs; the fast-charging and purchase credits are paid in 2028–32 on top of current law. Rest of US is the model region outside the four clean fuel states. Source: model runs (post-processing of the converged runs).

Table 37: Electric share of new-vehicle sales (%), 2026–45: medium- and heavy-duty vehicles

Region	Scenario	2026	2028	2030	2032	2035	2040	2045
Panel B: medium-duty vehicles								
California	Current law	2.3	3.7	9.2	13.0	13.3	9.2	4.5
	No state standards	1.0	0.1	0.2	0.2	0.3	0.3	0.3
	No fuel policy	0.9	0.1	0.2	0.2	0.2	0.3	0.3
	Fast-charging credit	2.3	3.7	9.2	13.2	13.6	9.4	4.5
	Purchase credit	2.3	3.7	9.1	13.6	15.6	10.1	4.8
Oregon	Current law	2.4	0.4	0.8	1.0	1.1	1.2	1.1
	No state standards	2.0	0.3	0.5	0.5	0.5	0.6	0.6
	No fuel policy	2.0	0.3	0.5	0.5	0.5	0.6	0.6
	Fast-charging credit	2.4	0.4	0.8	1.0	1.1	1.2	1.1
	Purchase credit	2.4	0.4	0.8	0.9	1.1	1.2	1.1
Washington	Current law	4.4	1.6	1.6	1.7	1.7	1.8	1.8
	No state standards	3.8	0.8	1.4	1.5	1.6	1.8	1.7
	No fuel policy	3.8	0.8	1.4	1.5	1.6	1.7	1.7
	Fast-charging credit	4.4	1.6	1.6	1.7	1.7	1.8	1.8
	Purchase credit	4.4	1.6	1.6	1.6	1.7	1.8	1.7
New Mexico	Current law	0.5	1.3	2.0	2.7	2.9	3.1	3.0
	No state standards	0.5	0.7	1.2	1.2	1.3	1.5	1.4
	No fuel policy	0.4	0.7	1.1	1.2	1.3	1.4	1.4
	Fast-charging credit	0.5	1.3	2.0	2.7	2.9	3.1	3.0
	Purchase credit	0.5	1.3	2.0	2.6	2.9	3.1	3.0
Rest of US	Current law	0.5	0.7	1.2	1.2	1.3	1.4	1.4
	No state standards	0.5	0.7	1.2	1.2	1.3	1.5	1.4
	No fuel policy	0.4	0.7	1.1	1.2	1.3	1.4	1.4
	Fast-charging credit	0.5	0.7	1.2	1.2	1.3	1.4	1.4
	Purchase credit	0.5	0.7	1.2	1.2	1.3	1.4	1.4
Panel C: heavy-duty vehicles								
California	Current law	3.9	4.6	18.8	25.8	23.7	16.1	10.8
	No state standards	1.4	0.7	1.4	1.5	1.7	1.8	2.0
	No fuel policy	1.3	0.6	1.3	1.4	1.6	1.8	1.9
	Fast-charging credit	3.9	4.5	18.8	26.0	23.7	16.1	10.9
	Purchase credit	3.9	4.4	17.4	25.6	23.7	16.1	11.3
Oregon	Current law	1.8	0.9	3.5	5.8	6.0	6.2	6.4
	No state standards	0.7	0.4	0.9	1.0	1.1	1.2	1.3
	No fuel policy	0.7	0.3	0.9	0.9	1.0	1.2	1.3
	Fast-charging credit	1.8	0.9	3.5	5.4	5.9	6.1	6.4
	Purchase credit	1.8	0.9	3.0	4.0	5.7	5.9	6.2
Washington	Current law	2.9	1.5	3.6	4.4	4.1	3.6	3.2
	No state standards	2.4	0.7	1.8	2.0	2.2	2.5	2.8
	No fuel policy	2.3	0.7	1.8	1.9	2.1	2.4	2.7
	Fast-charging credit	2.9	1.5	3.6	4.2	4.0	3.5	3.2
	Purchase credit	2.9	1.4	3.5	3.7	3.6	3.2	3.1
New Mexico	Current law	0.0	0.1	0.6	0.9	0.8	0.8	0.7
	No state standards	0.0	0.0	0.1	0.1	0.1	0.1	0.1
	No fuel policy	0.0	0.0	0.1	0.1	0.1	0.1	0.1
	Fast-charging credit	0.0	0.1	0.6	0.9	0.8	0.8	0.7
	Purchase credit	0.0	0.2	0.5	0.8	0.8	0.8	0.7
Rest of US	Current law	0.3	0.6	1.4	1.5	1.7	1.9	2.1
	No state standards	0.3	0.6	1.4	1.5	1.7	1.9	2.1
	No fuel policy	0.3	0.6	1.3	1.4	1.6	1.8	2.0
	Fast-charging credit	0.3	0.6	1.4	1.5	1.7	1.9	2.1
	Purchase credit	0.3	0.6	1.4	1.5	1.7	1.9	2.1

Note: share of electric vehicles in new sales of each class. Current law is the main run; "no state standards removes the four state programs; "no fuel policy removes the RFS, 45Z and the state programs; the fast-charging and purchase credits are paid in 2028–32 on top of current law. Rest of US is the model region outside the four clean fuel states. Source: model runs (post-processing of the converged runs).

Fast-charging credit: floor, phase-down and depreciation. Each row is the credit against current law under the same assumption, 2026–45, E+A. Main (10% floor, the station stock does not depreciate (the entry condition's user cost includes depreciation at 10.72% a year)): 172 Mt, $59 a tonne, federal outlay $8.2 billion. Floor of 5%: 303 Mt, $103, $14.7 billion; the rush of entry in 2032 is 2.6 times as large. Credit phased down to 35% in 2031 and 15% in 2032: 150 Mt, $36, $6.3 billion; the rush moves to 2030, the last year at 50%. Stations depreciating at 10.7% a year, the rate in the light-duty model's inputs: 121 Mt, $12, $9.0 billion. The cost per tonne falls with depreciation not because the credit buys more abatement (it buys 30% less, for more federal money) but because the main run's resource cost includes $17.9 billion of losses on the stations built in 2032, which earn less than their cost in every later year; with depreciation those losses are $4.4 billion. A floor of 20% is not reported, because it also removes the rush of entry before the 30C credit expires in 2026 in current law itself.

H.6 California's price under the rule and under perfect foresight

Figure 28 shows California's credit price under the estimated rule and under perfect foresight, with and without the 45Z extension.

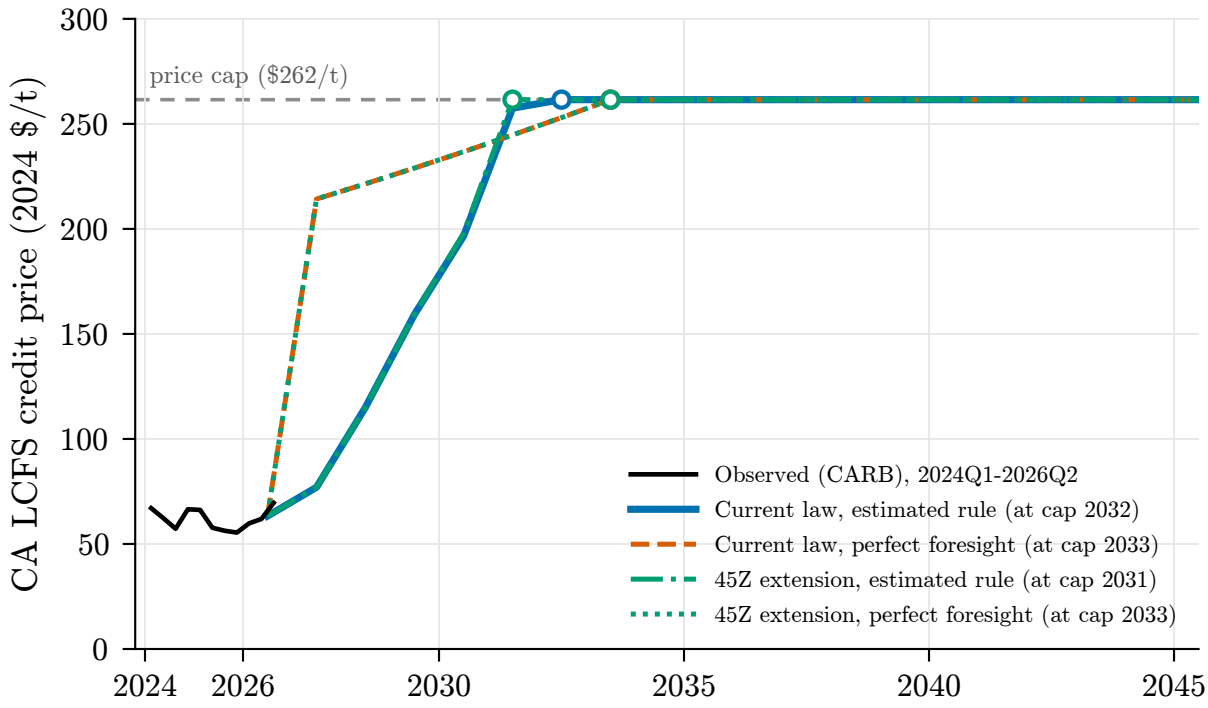


Figure 28: California's credit price and the timing of the 45Z transfer under the estimated rule and under perfect foresight

Notes: California's credit price under current law and the 45Z extension, each under the estimated price-expectation rule and under perfect foresight; observed price 2024–26. 2024 dollars. The perfect-foresight paths are the last round (round 6) of a run that has not converged; California's path is the same in rounds 2–6.

I Welfare Accounting

The cost of current law on other definitions. On the excise-only definition, which does not count forgone allowance revenue, the cost is $133 a tonne; counting the resource cost alone, it is $118. With tonnes discounted at the same rate (levelized) it is $186. Excluding indirect land-use change, current law abates 1,992 Mt at $117 a tonne on the excise-only definition. Counting the change in congestion, accident and local pollution costs as well, it is $127 (range $109–139). If the whole wedge between the delivered price and the absorption cost of renewable diesel into California is a resource cost rather than a rent (below), it is $176; if the 10% ethanol blend floor is kept in the reference runs and the loss it imposes on blenders is counted, it is $152. With entry at engineering cost it is $156 on the excise-only definition (Section 5.2).

This appendix states how welfare is computed. Every result compares a scenario S with a reference R, year by year, and reports present values at 2% to 2025 over the stated window. The building blocks are standard partial-equilibrium surplus measures (Just et al., 2004): consumer surplus under each demand curve, producer surplus above each supply curve, government revenue, and climate damages. Transfers between groups cancel in the total.

Economic welfare is

$$W = \sum_r \Delta CS_r^{fuel} + \Delta CS^{veh} + \Delta PS^{feed} + \Delta \Pi^{bio} + \Delta \Pi^{sta} + \Delta T + \Delta G^{fed} + \sum_r \Delta G_r^{state}, \quad (53)$$

where T collects the credit-market transfers to holders and other generators. The net cost of a policy is $-W$. Net welfare adds avoided climate damages, $W - \sum_t SCC_t \, \Delta E_t$, with EPA's social cost of carbon at a 2% discount rate (U.S. Environmental Protection Agency, 2023) and lifecycle emissions E_t including indirect land-use change. The cost per tonne is the present value of $-W$ over undiscounted tonnes abated. An accounting identity checks the total: the credit ledgers (RIN and each LCFS program) close in every state and year, and the run stops if they do not.

Road and jet fuel demand is isoelastic in the retail price, $Q = A(P + w)^{\beta}$, where w is a fixed retail wedge that includes excise taxes. The change in consumer surplus is the exact integral

$$\Delta CS^{fuel} = -\frac{\bar{A}}{\beta + 1}\Big[(P_S + w)^{\beta+1} - (P_R + w)^{\beta+1}\Big], \qquad \bar{A} = \tfrac{1}{2}\Big[\frac{Q_S}{(P_S + w)^{\beta}} + \frac{Q_R}{(P_R + w)^{\beta}}\Big], \quad (54)$$

by region and fuel. P is the pool price per gasoline- or diesel-equivalent gallon, so buyers bear the lower energy content of ethanol. The price carries every pass-through: RIN and LCFS compliance costs and the cap-and-trade charge. Averaging the demand level A over the two runs lets the curve shift with the electric share of the fleet; the shift itself is counted with vehicle buyers.

The electric share of new sales is logit. The change in buyers' surplus is the Small and Rosen (1981) log-sum, converted to dollars with the reference run's marginal utility of money. The

internal-combustion alternative is held at the reference run's cost of ownership, so fuel-price changes are counted once, with fuel buyers. Purchase credits, LCFS-funded rebates and the charging network enter through the log-sum.

Each feedstock has an isoelastic supply, $Q = \alpha P^{\beta}$, and producer surplus $\alpha P^{\beta+1}/(\beta+1)$, for soybean oil, North American animal fat, corn oil and corn. The surplus of foreign suppliers of imported fat and cane ethanol is reported but not counted in W.

Producers' surplus is revenue less feedstock, operating, utilization and capital costs. Three rules decide what is a resource cost and what is a rent.

- *Rising costs.* Where a cost rises with volume, the area under the marginal-cost curve is the resource cost and the rest of the payment is a rent to producers. This applies to the utilization cost of hydroprocessing capacity and to the cost of delivering renewable diesel into a clean fuel state, $\tau(q)$, on both segments of California's curve: the resource cost is $\int_0^q \tau(x)\,dx$ and the rent is $\tau(q)\,q - \int_0^q \tau(x)\,dx$. Because the curve is calibrated from credit values (Section 5.3), the split between cost and rent is an assumption. Counting the whole payment $\tau(q)\,q$ as a resource cost gives an upper bound, reported with the main results.

- *Capital.* New capacity is charged its capital cost, annualized over a 20-year life and counted only within 2025–45. The part of the estimated entry cost above the engineering capital cost is not a resource cost. Idle capacity carries its fixed operating cost.

- *Freight.* Freight into the clean fuel states is a resource cost.

Changes in credit banks, valued at the year's price, are transfers between deficit payers and holders. End-2045 banks are valued at the 2045 price; the counterparty lies after the horizon, and results are also reported without them. The federal line collects excise taxes, duties on imported fat and carbon revenue, less outlays on 45Z, 45Q and the other credits. State lines collect excise taxes and, when a program is short at its cap, the credits issued at the cap, less state outlays.

Six choices are conventions rather than measurements; the text reports each alternative.

1. *Fuel taxes.* The change in excise revenue is a cost of using less fuel. The cost per tonne is also reported with that revenue treated as a transfer, and with the change in congestion, accident and local pollution costs added (Section 4.5).

2. *Credits issued at the cap* are state revenue. Under the statute they are an unpaid obligation of fuel sellers; prices, quantities and the resource cost are the same either way.

3. *Cap-and-trade* is a fixed charge at 2025 allowance prices, and allowance revenue is not counted. Covered abatement is counted in full (Section 8).

4. *The ethanol blend floor.* The reference runs without fuel policy and without the RFS let the market choose the ethanol share. In those runs a cane-ethanol price floor at 1% of its 2025 price keeps the solver away from a zero price; cane ethanol is foreign and not counted in W. Keeping a 10% floor in both reference runs and counting blenders' loss from it is a sensitivity: current law then costs $143 a tonne over 2026–45, against $134, and the blenders' loss is $20.6 billion over 2030–45 in the run without the RFS.

5. *Price changes are not rounded.* Every price change between two runs enters surplus, however small.

6. *Foreign surplus* is excluded; the terms-of-trade gain on imported fat is counted as a saving of resources.

Social welfare is the sum of the surpluses of fuel users (by state), vehicle buyers, biofuel producers, feedstock suppliers, the federal government, the state governments and owners of capital, less climate damages. Feedstock supplier surplus is the area above the upward-sloping supply curves of North American soybean oil, fat, corn oil and corn; the surplus of foreign suppliers of imported feedstock is reported separately. Transfers between groups (credit payments, tax credits, rents) cancel in the total, which is checked by an accounting identity. Climate damages use EPA's 2023 social cost of carbon at a 2% discount rate, with 1.5% and 2.5%, the interagency value and a domestic value as sensitivities; electricity used by electric vehicles is valued at the long-run marginal emission rates of NREL's Cambium 2024 Mid-case (Gagnon et al., 2025). Costs are resource costs: capital at its capital cost, not the investment hurdle (the part of κ_ℓ above capital cost is a sensitivity), fixed operating cost on idle capacity, the feedstock-mix cost (36) and the state diesel differentials. The shadow price of the utilization curve is not a social cost. Two capital ledgers are reported side by side: all SAF conversion capital, and only the SAF-capable capacity that is used; the difference comes from the logit rule's positive announcement probability at zero expected profit. When a program is at its cap, issuance at the cap is state revenue; the statutory deficit is reported as an unpaid liability, not added to that revenue. The first-best benchmark is a carbon price at the social cost of carbon on all fuels and vehicles in the model, so each policy can be scored by the share of the first-best welfare gain it achieves; a carbon price that delivers the same cumulative abatement is the cost-effectiveness benchmark. Both are first best within the model's scope: the model is partial equilibrium, and world feedstock markets, indirect land use and foreign emissions are not endogenous. I do not compute marginal values of public funds, because the LCFS's cost does not pass through a government budget; for 45Z I report the outcome per dollar of federal cost.

Other external costs of fuel use. A mile of car travel costs 5.0 cents in congestion and 3.0 cents in accidents (2005 dollars, Parry et al., 2006), and 0.54 cents in local air pollution for a new gasoline car (Holland et al., 2016, 2014 dollars), with the older-fleet value of 2.0 cents as the upper bound; I convert at 21 miles per gallon. A mile of truck travel adds congestion, accident, noise and pavement

costs of 95 cents per gallon (2000 dollars, Parry, 2008). Gallons displaced by electric vehicles cause no change in miles; in the upper bound they add the local damages of charging on the 2014 grid (Holland et al., 2016). All values are converted to 2024 dollars with the CPI-U. The other external costs of fuel use are discounted to 2026 rather than 2025, which overstates them by about 2%.